VERSION SEPTEMBER 10, 2026
Preprint typeset using LaTeX style openjournal v. 09/06/15

A SEARCH FOR TIME-VARIABLE AND DRIFTING TECHNOSIGNATURES TOWARD ALMA-OBSERVED STARS WITHIN 40 PC

GLENN J. WHITE[1,2] AND ROBIN DEY[3]
Version September 10, 2026

ABSTRACT

We find no credible technosignature in 431 spectral windows searched toward 82 independent nearby stellar systems. The sample is archive-defined: every one of the 168 stars within 40 pc with usable public ALMA coverage, which is the intersection of a volume-limited catalogue with a strongly selected archive and not a volume-limited stellar sample. At the epoch of this release, 107 of 208 planned star/band datasets across 88 stars have been searched to completion, spanning 1.30–39.63 pc and 93.1 GHz of unique sky frequency.

That null applies to a specific signal class. The per-channel temporal-median subtraction that rejects stationary ground-based interference also suppresses phase-steady carriers, so the experiment constrains carriers time-variable on the timescale of an observing track and/or drifting in frequency, and says nothing about a stable beacon. Each window's stellar position is compared with 512 control positions carried through an identical search, calibrating the false-alarm rate empirically rather than assuming Gaussian statistics. Four windows contain a control-ring exceedance: three are circumstellar or foreground CO (two toward β Pictoris, one toward HD 48370) and one, toward CP$-$72 2713, is a marginal crossing consistent with the 0.84 expected under the noise null. None survives vetting, so this release reports 0 candidates.

Nominal 5σ trigger thresholds span 1.6×10^{13}–2.8×10^{17} W in EIRP (median 1.4×10^{15} W). Because most windows are channelised at 15.625 MHz, these bound carriers unresolved at the native ALMA channel width and are not Hz-resolution sensitivities. A 1200-trial injection campaign into real calibrated visibilities measures 42^{+16}_{-14} per cent recovery at that threshold *for the drifting class on fine-channel windows only* (118 of 431 windows); recovery of the time-variable morphology that defines the coarse-channel majority (313 windows) is not yet measured at all. These thresholds are therefore not 95-per-cent completeness limits.

Conditionally – and the condition does the work – a transmitter must occupy the particular frequencies already observed toward its own host and emit throughout the archival epochs for this survey to have seen it. Under that condition the zero-detection result bounds the in-band fraction of systems hosting a transmitter of EIRP $\geq 10^{16}$ W at < 6.2 per cent, weakening to 12.3 per cent at duty cycle 0.5 and 62 per cent at 0.1. Under any transmitter-frequency prior spread across ALMA's tuning range the searched islands cover too little spectrum to exclude anything at all. The number is a pilot-grade worked example of how these data will constrain a population once completeness is fully calibrated, and it is not a prevalence measurement.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

A technological civilisation, signalling deliberately or simply going about its affairs, may leave a detectable imprint on the electromagnetic spectrum: a *technosignature* (Sheikh 2020; Vidal et al. 2026). Six decades of radio searches now cover many thousands of stars yet, as Wright et al. (2018) quantified, sample only an infinitesimal fraction of the multi-dimensional space of transmitter frequency, bandwidth, power and duty cycle. Three gaps persist: the millimetre and submillimetre regime above ∼50 GHz is essentially unsearched; most surveys target interest-selected classes of star, not the local stellar population; and few systematically reprocess *existing* interferometric archives.

This paper addresses all three. We census every star within 40 pc of the Sun with public ALMA archival coverage, irrespective of spectral type, disc-hosting status or known planets, and mine it for narrowband technosignature candidates and anomalous millimetre continuum, correcting for stellar proper motion and for Doppler drift rates plausible for a transmitter on an orbiting platform. The processing sets the search class – suppressing phase-steady carriers – so this paper constrains *time-variable and/or sufficiently frequency-drifting narrowband emission* only (Appendix E measures both sides of that definition). Four archival-specific analyses supplement the classical search: closure-phase vetting, a continuum anomaly screen, a chirp and periodicity re-analysis, and a cross-target frequency-occupancy check. Per-target tables are in online-only Appendix I. We publish at 51 per cent completion because the citable content is methodological – the census definition, the empirically calibrated false-alarm machinery, the measured completeness

[1] School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk
[2] RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK
[3] VBRL Holdings Inc.

– with a first set of per-target limits, every number tied to a frozen pipeline and named snapshot that later work will supersede row by row without invalidating this one.

2. BACKGROUND

These six decades divide into wide-field commensal surveys and targeted programmes on preselected stars, latterly dominated by Breakthrough Listen, whose GBT, Parkes and MeerKAT surveys published multi-thousand-star limits from 1.1 to 8 GHz (Isaacson et al. 2017; Enriquez et al. 2017; Price et al. 2020; Czech et al. 2026), with parallel efforts on FAST, LOFAR, NenuFAR, the Sardinia Radio Telescope and commensal VLA, MeerKAT and MWA back-ends (Zhang et al. 2020; Johnson et al. 2023; Valente et al. 2025; Tremblay et al. 2024); machine-learning rejection of RFI-dominated hit lists is now central (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026) – a capability our classical threshold search lacks. As Margot et al. (2023) note, a non-detection bounds the transmitter fraction only if the sample represents the underlying population, which interest-selected samples do not. Essentially all this activity lies below $\sim$10 GHz; above it the mm/submm regime has been searched once, also with ALMA (Mason et al. 2024). Sixteen programmes are tabulated in Appendix D.

Mason et al. (2024) searched archival Band 3 windows toward 28 "stellar bycatch" stars, placing $\mathrm{EIRP_{min}} > 7\times10^{17}$ W for the nearest and identifying high-frequency work's challenges: drift rates scale with frequency, smearing erodes sensitivity without a drift search, and the molecular line forest confuses. Our pilot (White 2026; under review, citation provisional) extended that methodology to TRAPPIST-1 across Bands 3, 6 and 7 ($\mathrm{EIRP_{min}} \sim 5$–$10 \times 10^{13}$ W, no detection) and built the ranked 100-star list this survey grew from. New here are the census selection (§3), the 512-position control-ring false-alarm calibration, closure-phase vetting, the spectral-type-normalised continuum screen, the chirp and periodicity extension and the cross-target frequency-occupancy check – each new, to our knowledge, *in an ALMA archival technosignature survey*, though we claim no wider priority: several have centimetre-wave counterparts, and we have not made the exhaustive search a general claim would require. No execution block is analysed in both papers: the pilot's 100-star search products are superseded, not re-used, here, and every window here was re-reduced and re-searched from the raw archive data with the pipeline of §4.

Figure 1 places this release's nominal EIRP thresholds among published searches in *total power*. Earlier versions divided each threshold by its channel width, in $\mathrm{W\,Hz^{-1}}$; that was wrong for the transmitter class considered here, and we withdraw it. $\mathrm{EIRP}_{5\sigma}/\Delta\nu_{\mathrm{ch}}$ is a channel-averaged equivalent spectral luminosity, correct only for emission *filling* the channel, not an unresolved carrier's spectral power density: a 1 Hz carrier in our deepest window, 1.6×10^{13} W in a 15.625 MHz channel, still needs all 1.6×10^{13} W to be detected, so its intrinsic spectral power is of order $10^{13}\,\mathrm{W\,Hz^{-1}}$, not the $10^{6}\,\mathrm{W\,Hz^{-1}}$ the division returns – which, beside Hz-scale backends, flatters ALMA by some six orders of magnitude against exactly the signal class sought. The linewidth convention is in §4.

A non-detection must become a quantitative exclusion statement. Wright et al. (2018) formalised the "Cosmic Haystack" search volume, Margot et al. (2023) its conversion into an upper bound on the transmitter-hosting fraction, and Sheikh et al. (2025a) asked where Earth's technosignatures would be detectable – a calibration point for any survey's sensitivity.

Hence the case for a census over interest-selected lists. Most surveys select by proximity, spectral type, planets or Earth Transit Zone membership – reasonable per unit time – while the ALMA-specific form, disc selection, skews the sample young and dynamically active, ALMA-detected discs being orders of magnitude brighter than the Solar System's zodiacal cloud at interstellar distances. We instead take every star within 40 pc in the usable field of at least one public ALMA observation (§3): a bycatch census of the ALMA-observed subset of the local stellar population in a fixed volume, not of that population itself. This removes technosignature-specific but not archival astrophysical selection; our 168 stars are $\sim$1 per cent of the catalogued population within 40 pc, and that residual coverage selection is characterised as a completeness function (§3, Table 3). The SKA (Siemion et al. 2015; Tremblay et al. 2026) and ngVLA (Ng et al. 2022) will widen the coming decade's search with wide bandwidth and commensal architectures below $\sim$50 GHz, but neither matches ALMA's resolution and collecting area above it near-term, so a systematic mm/submm census stays ALMA-specific for some years.

3. SAMPLE SELECTION

Our sample is every star within 40 pc of the Sun (Gaia DR3 parallaxes, distances direct inversions $d = 1/\pi$; every entry $\pi \geq 25$ mas with $\sigma_\pi/\pi \leq 0.8$ per cent) whose proper-motion-corrected position at the observation epoch falls in the field of view of at least one public ALMA execution block, not necessarily at the pointing centre: mosaics and wide pointings serendipitously cover more than one catalogued star. Disc, planet and spectral type are descriptive metadata here, not selection criteria. This gives 168 unique target stars, processed nearest-first – a sensitivity-ranked order, not an unbiased draw, on which the prevalence statements of §6 are conditional.

Composition follows from the archival selection: 76 of the 168 stars are M dwarfs (63 per cent), then 12 K, 11 G and 8 F dwarfs, one A-type star (β Pic) and six white dwarfs; six faint companions lack tabulated types. Eighteen of the 88 processed stars are confirmed exoplanet hosts (§6); the debris-disc hosts (β Pic, AU Mic, ϵ Eri, τ Cet) contribute dusty continua, kept from masquerading as anomalies by the continuum lane's spectral-type-normalised screen (§4).

The 168 stars are complete with respect to, and conditional only on, the 40-parsec reference catalogue intersected with the public-archive snapshot (2026 September 8) defining this release. Against that Gaia DR3 solar-neighbourhood catalogue (Figure 13, online-only appendix; 17 566 stars within 40 pc passing the parallax-quality cut of Table 16) they are $\sim$1%; that list is itself incomplete at both ends, so the denominator is a convention and the fraction says only that coverage is small. Table 16 (online-only appendix) leaves the selection function inspectable: strong enrichment of the

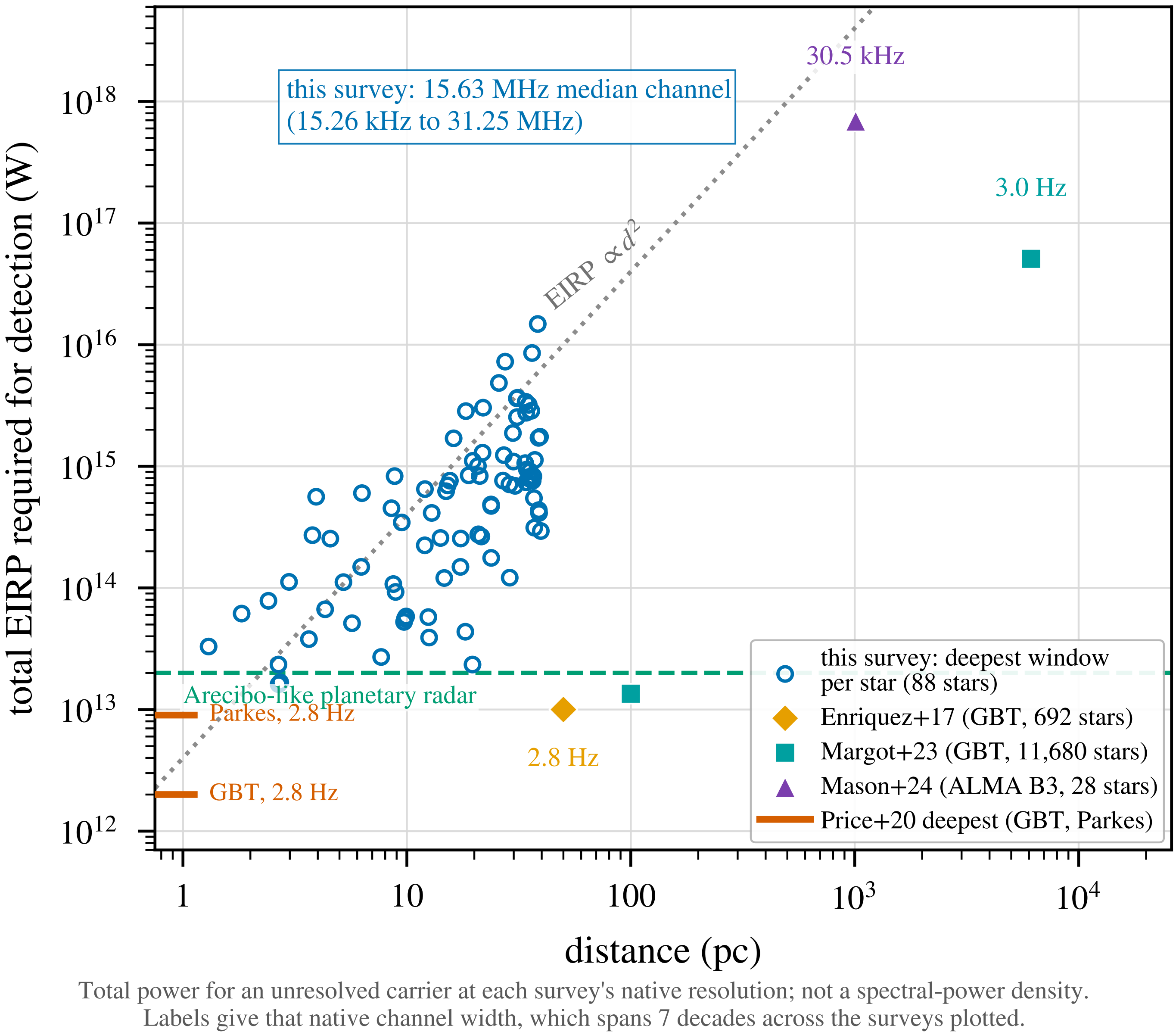

Fig. 1.— Literature context of this release's thresholds, in total power (see text; the earlier W Hz^{-1} panel is withdrawn). Blue: this survey, deepest window per star (88 stars, 1.30–39.63 pc), 1.6×10^{13}–1.5×10^{16} W, median 5.6×10^{14} W. Orange diamond: Enriquez et al. (2017), $\sim 10^{13}$ W within 50 pc. Aqua squares: Margot et al. (2023), 1.35×10^{13} W within 100 pc and 5.08×10^{16} W within 6135 pc. Violet triangle: Mason et al. (2024), archival ALMA Band 3, 6.91×10^{17} W for their closest star at 1.01 kpc. Left-edge ticks: deepest per-target limits of Price et al. (2020), 2×10^{12} W (GBT) and 9×10^{12} W (Parkes), no single sample distance. Grey dotted: EIRP $\propto d^2$ at constant received flux, normalised to 4×10^{12} W at 1 pc, a guide not a fit. Green dashed: an Arecibo-like planetary radar (2×10^{13} W), purely a technological power scale; only our G 272-61, in its two resolved components, falls below it. Thresholds, drift coverage and confidence conventions differ among these surveys: context, not a ranking.

nearest systems falling towards parity with distance, mild M-dwarf depletion with F/G enrichment, planet-hosting in 20 of the 168 census entries. Age and disc-hosting flags are not uniformly available and are documented qualitatively; a machine-readable selection-function table ships with the release.

Stellar counts throughout are catalogue-entry counts: the 168 census entries comprise 82 independent systems among those searched (seven designation-linked component pairs),[4] and since the 44 narrowband-covered entries include the unresolved UV Ceti and AT Mic pairs, the 107 star/band datasets, 431 windows and 102 execution blocks derive from 82 systems – the paired rows being bookkeeping, not independent trials.

"Usable public ALMA coverage" is a five-criterion decision tree, frozen for this release absent an erratum: (i) at least one public execution block (EB) whose QA2-calibrated visibilities cover the star's epoch-propagated position in the primary beam, with no minimum integration, sensitivity, antenna count, mosaic-tile count or array configuration beyond QA2 itself; (ii) that position within one primary-beam FWHM of the field centre (largest accepted offset $\sim$0.7

[4] Designations follow SIMBAD; in the per-target file (Appendix I) each entry has its own row and distance, and spatially unresolved pairs (e.g. the UV Ceti entries at 2.7 pc, with blended fluxes) are marked in the notes column.

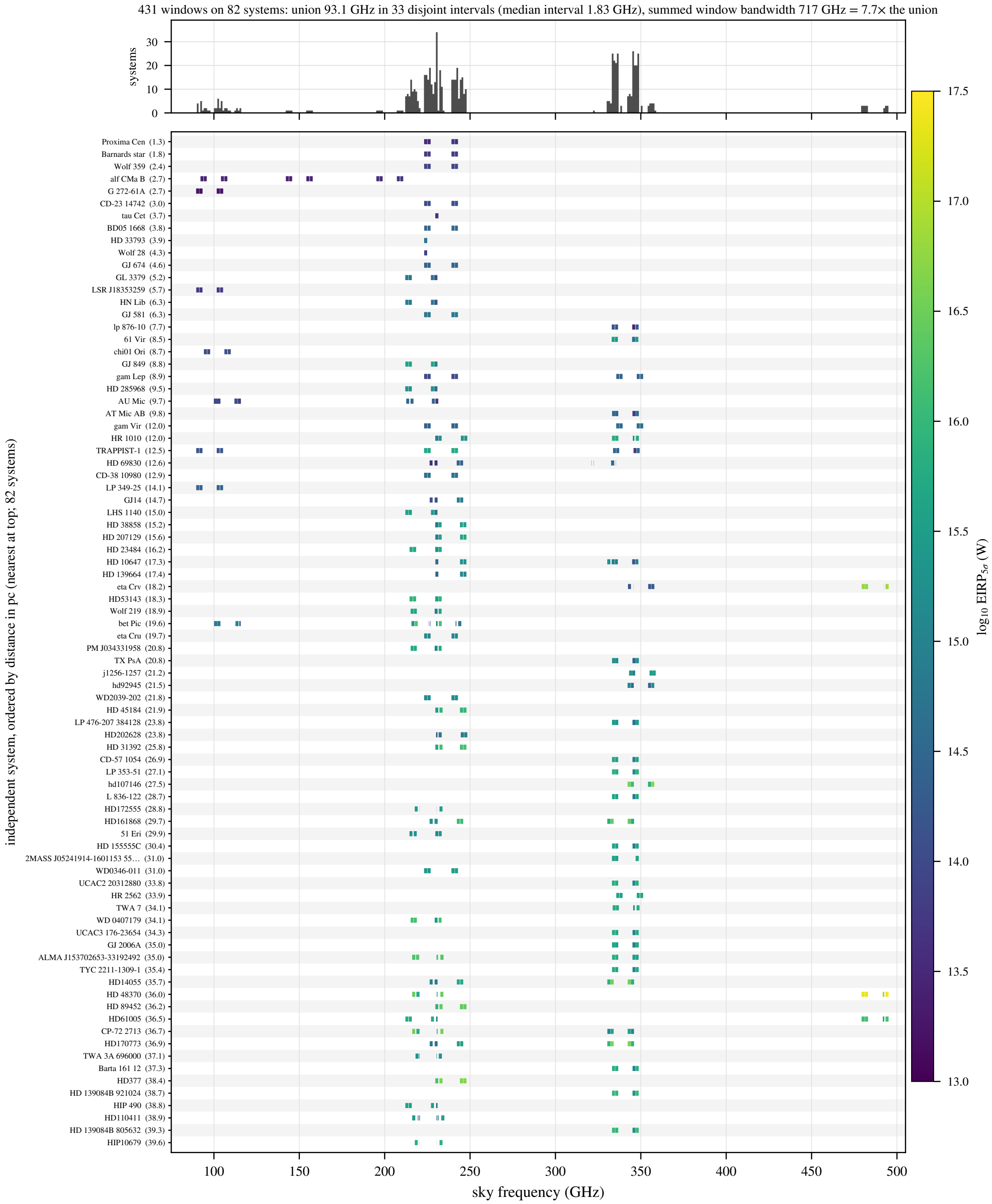

Fig. 2.— What this survey sampled in frequency. Each searched window is a horizontal segment on its host system's row, rows ordered by distance, coloured by nominal $\mathrm{EIRP}_{5\sigma}$; the upper panel counts systems per gigahertz of sky frequency. The 93.1 GHz frequency union comprises 33 disjoint islands of median width 1.8 GHz, concentrated near 230 and 345 GHz, and the summed window bandwidth exceeds the union by a factor 7.6 – coverage narrow, clustered and repeatedly revisited, not broad. The in-band transmitter fractions of §6 are conditional on these intervals. Thresholds are nominal in the sense of §4.

FWHMs, Sirius B); (iii) each spectral window admitted individually, on the pipeline's quality vetting alone; (iv) no programme-category restriction, science, calibration and observatory projects entering on identical terms; (v) a fixed archive snapshot date (2026 September 8), re-evaluated only at release boundaries. No star or window was excluded by an unlisted criterion, none for short integration (the shortest retained are 1.0 and 1.5 min).

ALMA project metadata quantify pointed versus serendipitous coverage: almost every narrowband-covered star is the pointed science target in at least one execution block, the exceptions a few blocks on another programme's field or calibrator-only, leaving Wolf 219 the sole genuinely serendipitous star. An archival census's residual selection function is the union of ALMA target lists – a contrast with interest-selected samples of degree, but of *quantified* degree: a bycatch fraction of 1 star in 44, 1 EB in 57.

4. DATA AND METHODOLOGY

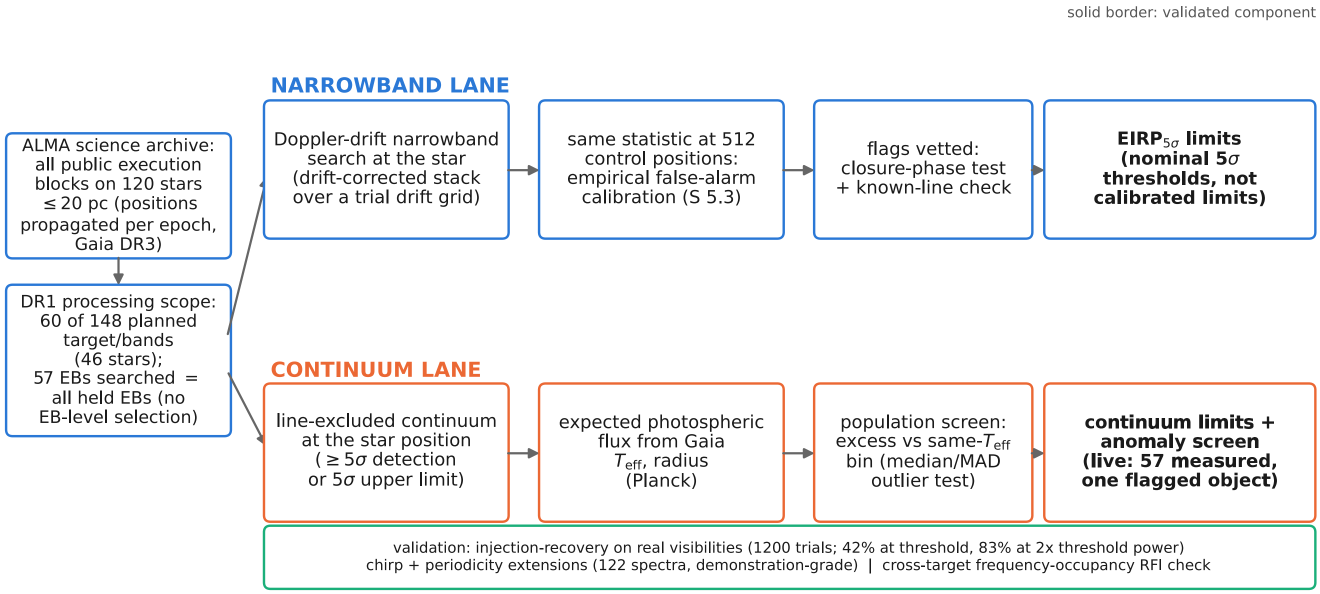

Fig. 3.— Design of the survey. All public execution blocks on the 168 targets are mined from the ALMA archive, positions propagated to each epoch from Gaia DR3, and searched along the two independent lanes described in the text. Solid borders mark validated components; the validation rail summarises the injection-recovery (1200 trials, Appendix E), chirp/periodicity and frequency-occupancy analyses.

Survey vocabulary – target/band dataset, spectral window, execution block, control ring, candidate, molecular-line exclusion mask – is defined in Table 12, Appendix B.

Figure 3 summarises the design: one shared archive-ingestion stage feeds two independent search lanes. For each target we mine, within a bounded budget, all calibrated (or locally recalibrated) archival execution blocks covering the star's Gaia-propagated position, and search their visibilities in (i) the *primary* lane, a narrowband, Doppler-drift-corrected candidate search following Mason et al. (2024) and Enriquez et al. (2017), converting each non-detection into an equivalent isotropic radiated power limit $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\rm min}\Delta\nu$ ($S_{\rm min}$ five times the searched window's per-channel rms, primary-beam corrected; $\Delta\nu$ its native channel width; d the target distance), and (ii) an *ancillary* line-excluded continuum lane (Appendix L). "5σ" throughout is the local instrumental *trigger* threshold of Table 2, separately and empirically calibrated (§5.4). Every limit, disposition, false-alarm calibration and completeness statement derives from the narrowband lane; the chirp/periodicity extension and molecular-line census are in Appendices G and N.

Both lanes exclude catalogued molecular transitions at tolerances suited to their statistics. The continuum lane masks the eight common species (CO, ^{13}CO, C^{18}O, HCN, HCO$^+$, CS, CN, SiO) at native resolution before integrating; the narrowband lane applies a $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame velocity exclusion mask (§§4.2, 5.4) to any threshold crossing, and sets the effective searched bandwidth (≈87 of the 93.1 GHz unique frequency union). Masking reduces the frequency space a candidate can occupy, not the per-channel noise, so no threshold in the per-target file changes. All flux densities and EIRP thresholds are Stokes I, so the limits apply to total transmitted power independent of polarisation geometry; the loss is diagnostic, not sensitivity (no Stokes $V/Q/U$ extraction or circularity test), and a polarisation-sensitive test is a natural future addition (§6.2).

How to read every limit in this paper. *Every EIRP$_{5\sigma}$ value here is a nominal, local instrumental detection threshold: five times the per-channel rms of the searched window, at that window's native channel width, for a transmitter at the star's position matching the searched temporal and spectral morphology – time-variable on track timescales, since phase-steady carriers dwelling in one channel for most of the track are suppressed by the statistic's own continuum subtraction (Appendix E). It is not a calibrated completeness limit: the injection campaign measures a recovery of roughly one half at the threshold itself (EIRP$_{50}$ = 1.10 EIRP$_{5\sigma}$ for the searched class), rising to 83 per cent at twice the threshold power (EIRP$_{90}$ > 2 EIRP$_{5\sigma}$; Appendix E). It is a total-power threshold set by ALMA's coarse native channelisation, not a Hz-resolution narrowband sensitivity. The same reading applies to every subsequent "5σ" and to the per-target file (Appendix I); it is stated once here and not repeated at each occurrence.*

4.1. *The search statistic, and the words used for its output*

The primary statistic is formed on extracted spectra, not images or visibilities. For a sky position $\mathbf{x}$ the per-channel time series is median-subtracted along the time axis, the residual dynamic spectrum stacked along each trial linear

TABLE 1
The statistical data path, in order. Steps 1–8 run identically at the stellar position and all 512 controls; only step 9 onward distinguishes them.

1	calibrated measurement set from the ALMA archive
2	stellar position propagated to the observation epoch (Gaia DR3)
3	spectral extraction at that position, native channelisation
4	per-channel subtraction of the time median
5	stack along each trial linear drift rate
6	robust per-position noise scale $\sigma(\mathbf{x})$ (MAD)
7	$T(\mathbf{x})$ from Eq. 1
8	repeat 3–7 at 512 control positions on a ring
9	threshold: $T_\star \geq 5$ (a hit)
10	rank test: $T_\star > \max_i T_i$ (control-ring exceedance)
11	molecular-line mask, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar frame
12	vetting: recurrence, closure phase, cross-target occupancy

TABLE 2
Nomenclature for search output and the two senses in which a probability is quoted.

Term	Meaning
hit	one channel×drift cell at the stellar position with $T \geq 5$
control-ring exceedance	a window whose $T_\star$ exceeds all 512 control statistics
candidate	a control-ring exceedance surviving every veto; none in this release
5σ trigger threshold	local instrumental trigger level, *not* a completeness limit
completeness	measured recovery of injected signals by the unmodified pipeline
$p_{\mathrm{rank,min}}$	$1/(N_{\mathrm{ctrl}}+1) = 1/513$, the rank *resolution* of the control ensemble
P(false flag)	the survey false-positive probability after thresholding, correlation and non-Gaussianity; not the same quantity

drift rate $\dot\nu$, and the statistic is the largest resulting single-channel amplitude in units of the window's noise,

$$T(\mathbf{x}) = \max_{\dot\nu,\,\nu} \frac{S(\mathbf{x},\nu,\dot\nu)}{\sigma(\mathbf{x})}, \tag{1}$$

where S is the drift-stacked flux density in one native channel and $\sigma(\mathbf{x})$ a single robust scale per position and window, estimated as 1.4826× the median absolute deviation of that position's median-subtracted spectrum over all channels: local in position, global in frequency. $T_\star \equiv T(\mathbf{x}_\star)$ at the proper-motion-propagated stellar position; the same operation, drift grid, channelisation and noise estimator give $\{T_i\}$ at all 512 control positions. Table 1 sets out the data path in order, so the statistic can be reimplemented without the appendices.

Being global in frequency, that scale can misstate the noise *local* to a crossing wherever a window carries bandpass-edge structure, atmospheric features or a line forest. The retained products keep the per-window scale but not the per-channel spectra, so no local estimate around a crossing can be recomputed for this release; a per-crossing local robust scale, computed with the crossing itself excluded, is committed for the next release with trial-level injection logging (§6.2). One indirect check the frozen products do support: the same estimator runs at the star and at all 512 control positions of a window, so a window whose σ were badly misstated would carry a displaced control distribution, and the pseudo-star ranks are uniform in every band and at both channelisations (§5.4) – the one departing stratum is the fine-channel one, and that departure is the four flagged windows themselves.

The output vocabulary nests strictly (Table 2); no object of this release is a candidate.

The searched drift-rate range is the larger of (a) a generic frequency-scaled default (12–$13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$), the Mason et al. (2024) fiducial with a threefold safety margin – a design boundary, not a claim that faster drift is implausible – and (b) for confirmed exoplanet hosts, the maximum line-of-sight orbital acceleration of the innermost known planet, so a transmitter with a short-period planet cannot drift out of the window. Drift rate maps onto line-of-sight acceleration through the first-order Doppler relation

$$a_{\mathrm{los}} \;=\; c\,\frac{\dot\nu}{\nu}, \tag{2}$$

so the ceiling is most compactly quoted in cross-band-comparable form $\dot\nu/\nu$: 12–$13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ is $\dot\nu/\nu = (1.2$–$1.3)\times10^{-8}\,\mathrm{s^{-1}}$, i.e. $|a_{\mathrm{los}}| = 3.6$–$3.9\,\mathrm{m\,s^{-2}}$, independent of observing frequency – the acceleration at $\sim$0.04 au around a solar-type star. Earth-motion contributions ($\lesssim 0.13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$) lie two orders of magnitude below, subsumed by the grid. These are *apparent* topocentric drift rates, including deliberate transmitted chirp and acceleration-induced drift, which the survey does not distinguish; our constraints concern linear apparent drift within the searched range only.

Across the 431 windows maximum searched drift rates span 1.1–$5.9\,\mathrm{kHz\,s^{-1}}$, native channel widths $15.3\,\mathrm{kHz}$–$31.25\,\mathrm{MHz}$ (median $15.625\,\mathrm{MHz}$), trial drift rates per window 2–1944 (median 4). The threefold margin matters: at 1× the release would have been blind to two of its threshold-crossing windows, including the AU Mic crossing; at 5× no disposition changes, transmitters drifting faster than $\sim 13\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ remaining unconstrained. Intra-integration smearing is negligible almost everywhere: at most 1.10 channels, median 0.001; only three windows have $\eta_{\rm smear} < 0.99$ (marked * in Appendix I; Table 13), the worst the β Pic Band 6 fine window (effective threshold 1.74× nominal). Channel widths are effective resolutions, verified against the measurement sets' spectral window tables (Appendix C).

4.2. *Frequency and velocity reference frames*

Every frequency axis here – searched spectra, tabulated ranges, candidate frequencies – is *topocentric*, as delivered by the ALMA correlator, no velocity-frame shift applied to any data product at any stage. Velocities enter only in the vetting stage's stellar-frame line-exclusion mask, whose conversion chain is in Appendix C. The stacking is coherent in frequency and in the drift-rate trial but preserves no electric-field phase coherence between integrations.

The native channel width is also the linewidth convention for every EIRP threshold here – the commonest confusion when comparing technosignature limits across surveys. The quoted $\mathrm{EIRP}_{5\sigma}$ applies to *any* transmitter narrower than the native channel; dividing by it gives the minimum detectable spectral power for wider ones. Channelisation is the dominant price the archive exacts, but we deliberately do not project an Hz-resolution sensitivity: a radiometer-equation scaling $P_{\rm min} \propto \sqrt{\Delta\nu}$ across six orders of magnitude in spectral resolution is a thought experiment, not an achievable ALMA sensitivity, and quoting it invites the misreading this paper avoids. The worked example, sub-channel qualifications, smearing table and channel-width verification are in Appendix C.

The narrowband search covers every spectral window of an observation, capped at 8 per target to bound compute cost; the cap bound exactly one target/band (61 Vir Band 7: four of eight windows searched), whose frequency-space selection function is therefore tabulated. Of the 107 star/band datasets, 53 have every configured window searched; the remaining 3 (τ Cet, Kapteyn's Star, Van Maanen's Star) are represented by their deepest single window, marked in Appendix I, full multi-window coverage awaiting the next release. Each searched window yields its own $\mathrm{EIRP}_{5\sigma}$ row; Figure 8 plots only the deepest per target/band.

Velocity coincidence, not drift, is the operative discriminant when a crossing is checked against the eight masked species (§4): circumstellar emission sits at rest offset by the gas's kinematics, whereas a technosignature may appear at any frequency and drift rate. The searches as executed used a narrower one-channel-width implementation, agreeing with the velocity-space form on every disposition here; its justification and the bandwidth removed are in §5.4.

Both narrowband thresholds and continuum fluxes are corrected for ALMA's primary-beam response at the star's offset from the phase centre, negligibly for most targets (median factor 1.0001, 90th percentile 1.012, none above 2%). Two sit at large offset. ϵ Eri Band 6 lies at ∼0.7 primary-beam FWHM, where the correction reaches 2.9–3.4× and the Gaussian beam form fails; rather than publish provisional values we *withhold* those four windows, which enter no number here but are released, flagged, in the per-target file and return once the holography beam model is applied. Sirius B is retained and quoted as such: corrections 3.6–16 per cent across bands, a bounded systematic rather than an invalid model. Excluding it too would change neither endpoint of the quoted EIRP range, nor the median ($1.3 \to 1.4 \times 10^{15}$ W), nor any conclusion.

4.3. *Reference transmitter benchmarks*

The EIRP thresholds of §5 are compared throughout to two human-transmitter benchmarks, both from Sheikh et al. (2025a): an Arecibo-like planetary radar (Drake 1974) (S-band EIRP 2×10^{13} W, the most powerful directed transmitter an Earth-equivalent civilisation could point at us, purely a technological-power benchmark), and unintentional Earth mobile-network leakage, no signalling intent (EIRP of order 4×10^{9} W).

4.4. *Ancillary and validation analyses*

Four analyses beyond the narrowband search are executed here, each specified in full online (Table 15) and summarised only as far as the main text depends on it.

Injection-recovery validation (Appendix E). Synthetic tones of known amplitude were injected into real visibilities and recovered with the unmodified pipeline (1200 trials across six window configurations, three bands and both resolution classes). Per-channel time-median subtraction absorbs phase-steady carriers, totally so on the coarse (15.625 MHz) windows (0 of 500 coarse injections recovered from 4σ to 10σ), so coarse-window limits bound transmitters time-variable on track timescales. For the drifting class on the fine window recovery is 17, 42, 58, 75 and 83 per cent at 4, 5, 6, 8 and 10σ, giving $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ and, as a bound, $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$ (Fig. 12, Table 18). Tables here stay in nominal $\mathrm{EIRP}_{5\sigma}$; the conditional in-band transmitter fraction of §6 is the one place measured completeness is folded in, Band 8 completeness being transferred by shared construction, not measured.

Drift-ceiling caveat. The fixed $12\,\mathrm{Hz\,s^{-1}\,GHz^{-1}}$ searched ceiling, line-of-sight $|a| = 3.6\,\mathrm{m\,s^{-2}}$, is uniform across every window: two orders of magnitude clear of Earth-rotation and Earth-orbit drifts, but 11 per cent *below* the $4.0\,\mathrm{m\,s^{-2}}$ of a close-in planet on a TRAPPIST-1 b-like orbit. The limits of §6 are therefore conditional on it (Figure 14).

Closure-phase vetting (Appendix F). A one-sided point-source test – evidence against celestial point-source behaviour, never an authentication – unavailable to single-dish and beamformed surveys, needing multiple independent baselines. Implemented and simulation-validated, it has at this survey's flux densities *no discriminating power*: per-baseline S/N never exceeds 5.1 in the positive-control fields, so no flagged window could have failed it whatever its nature. It is a proof-of-principle diagnostic for deeper releases, not an operative filter here, and no window is described as having "survived" it.

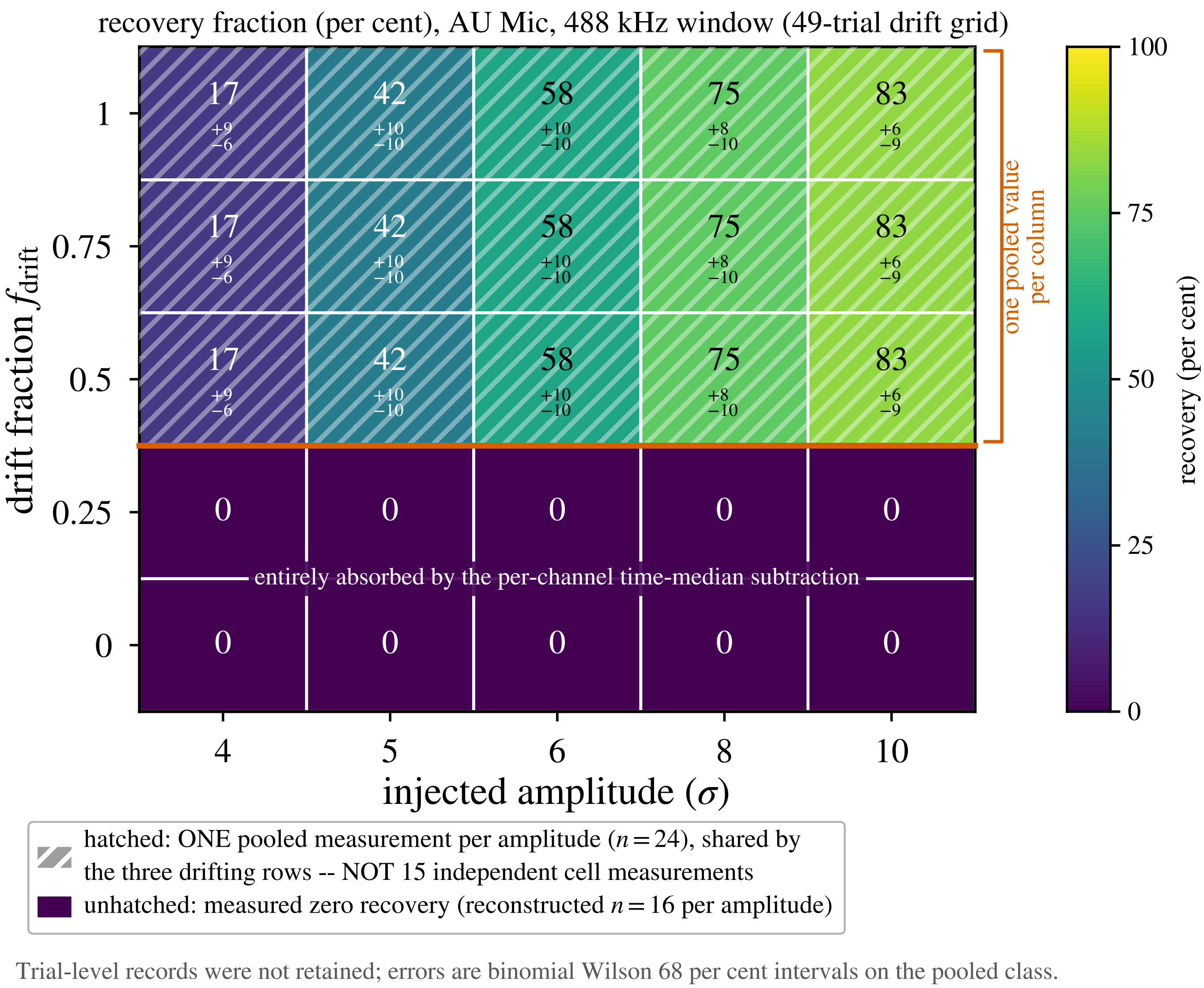

FIG. 4.— What the pipeline recovers, for the best-calibrated fine-channel configuration, against injected amplitude and drift fraction $f_{\rm drift}$ (injected drift as a fraction of the searched ceiling). Carriers with $f_{\rm drift} \leq 0.25$ are absorbed entirely by the time-median subtraction at every amplitude; those with $f_{\rm drift} \geq 0.5$ recover progressively, from 42–83 per cent between 5σ and 10σ. Trial-level records were not retained, so surviving fractions pool grid phase, sub-channel placement and the three drifting rows: the figure is drawn as discrete cells, not an interpolated surface, each column's hatched rows sharing one pooled measurement ($n = 24$ per amplitude) rather than fifteen independent ones. Intervals are binomial Wilson 68 per cent; absorbed rows are measured zeros.

Extended and cross-target screens (Appendices G and H). A chirped (quadratic-drift) extension and per-channel periodicity search ran on the 122 retained spectra as a demonstration, not a survey-grade screen; null behaviour is as expected (star-exceeds-ensemble rates 10.7 and 13.1 per cent against an 11-per-cent chance rate), no instance is a candidate, and no limit derives from them. Separately, crossings towards different targets were cross-matched for recurrence at the same sky frequency – the archival substitute for the ON/OFF cadence this single-pointing survey lacks. Of the 366 star pairs with overlapping searched ranges, none falls within the matching kernel, subject to one bookkeeping caveat (64 crossings of one β Pic window untested) in the appendix.

5. RESULTS

Strict volume ordering (nearest star first; §3) weights these results to the very nearest stars, spanning 1.30–39.63 pc. Per-target values, including each band's searched frequency range, are online (Appendix I).

5.1. *Per-star summary*

Table 24 (online-only) gives the one-line-per-star view complementing the per-window file (Appendix I); two stars of the release are continuum-only. All values come from that file: EIRP values are nominal native-channel thresholds (boxed rule, §4), continuum fluxes primary-beam corrected.

5.2. *Barnard's Star and Wolf 359*

TABLE 3
The survey as frozen for this release. Every count quoted in this paper reconciles with this table; subset numbers are identified as such where used. Thresholds are nominal (boxed rule, §4).

Quantity	Value
Census stars within 40 pc	168
Stars searched; independent systems	88; 82
Star/band datasets	107 of 208 planned (51%)
Spectral windows searched	431
extracted; repeats, defective, withheld	462; 21, 6, 4
fine (<5 MHz) / coarse channels	118 / 313
Execution blocks	102
Distance range	1.30–39.63 pc
Unique frequency union	93.1 GHz (33 intervals, 89.6–495.1 GHz)
Effective after line-exclusion mask	≈87 GHz
Native channel widths	15.3 kHz–31.25 MHz (median 15.625 MHz)
Drift-rate ceilings	1.1–5.9 $\mathrm{kHz\,s^{-1}}$ (12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$)
Per-window on-source time	21–5274 s (median 2177 s)
Exposure×bandwidth	1.4×10^6 s GHz
Nominal $S_{\min}$	0.48 mJy–1.30 Jy (median 6.7 mJy)
Nominal $\mathrm{EIRP}_{5\sigma}$	1.6×10^{13}–2.8×10^{17} W (median 1.4×10^{15})
Measured recovery at threshold, fine/drifting class	42^{+16}_{-14}% (118 windows)
coarse/time-variable class	*not measured* (313 windows)
Hits ($\geq 5\sigma$ at the stellar position)	20 windows
Spatially significant hits	4 windows
circumstellar or foreground CO	3
consistent with the control ensemble	1
Credible technosignatures	0

Barnard's Star and Wolf 359 – the second- and fourth-nearest stellar systems – have not, to our knowledge, previously been searched for technosignatures at ALMA frequencies. Both are non-detections in public Band 6 archival data: $\mathrm{EIRP}_{5\sigma}$ limits of 6.15–6.89×10^{13} W (Barnard's Star, 4 windows) and 7.84–9.10×10^{13} W (Wolf 359), with continuum upper limits of 0.575 and 0.453 mJy.

5.3. *Data-quality exclusions and open items*

Of the 208 planned target/bands, 107 are searched to completion and reported here, the rest queued for later releases bar a few failures excluded from all counts. Four classes within the 107 are absent from the released per-target file (Appendix I) and itemised with forensics in Appendix R: (i) four process-level search failures, each still carrying a continuum non-detection; (ii) one pair excluded for data quality; (iii) windows partially withheld for a noise-handling defect; and (iv) one relabelled entry.

A reproducible defect produces implausibly small noise in a minority of coarse-channel windows; naming the two affected did not generalise as the sample grew, so item (iii) is now caught by a physical test. In a correctly weighted measurement $\sigma\sqrt{t_{\rm on}\,\Delta\nu_{\rm ch}}$ is set by system temperature and collecting area, varying by factors of a few across a heterogeneous archive; we exclude windows more than 100× below the sample median. Six of 431 windows fail, across five stars, a factor 1.5×10^3 below the lowest retained across an empty gap of 3.2 decades (Figure 5); the threshold is not fine-tuned, any value between 1.5×10^1 and 2.2×10^4 selecting the same six. Their nominal thresholds, 2.0–7.8×10^{12} W, would otherwise have set this paper's headline depth; excluding them, the best is 1.6×10^{13} W. 21 of the 462 extracted rows repeat a (star, execution block, frequency range, channel width) key; we keep one per key. Most repeats are identical, but two are not – two reductions of the same τ Ceti Band 6 window differ by 44 per cent in threshold, and the two HR 1010 Band 6 rows straddle the 5σ threshold (5.10 and 4.95) – so the hit count is rule-dependent at the level of one window. The four control-ring exceedance windows are not: no discarded row is both a hit and star-beats-ring, and every row, kept or discarded, ships with its status. The audit is otherwise clean apart from four windows of elevated system temperature, and every census identifier re-resolves independently in SIMBAD and *Gaia* (Appendix R).

Item (iv) is a cautionary example for archival work. The catalogue entry "γ Lupi" resolves not to the naked-eye B2IV star of that name but to the F4V debris-disc host HD 139664, which holds the historical main identifier "* g Lup"; the substitution came from the name lookup, not from any positional error, so a position match against the entry's own coordinates would have passed it. The parallax caught it: inverting the pipeline distance returns the *Gaia* DR3 value for HD 139664, not the B2IV star (Appendix O).

5.4. *Technosignature search*

All 107 star/band datasets yielded at least one narrowband result (process-level failures, §5.3, uncounted), spanning 431 spectral-window measurements after the catalogue audit removed 21 duplicate rows (Appendix I). $\mathrm{EIRP}_{5\sigma}$ limits span 1.6×10^{13}–2.8×10^{17} W, median 1.4×10^{15} W (Figures 8, 2, 1); as expected for a volume-ordered survey the deepest are the nearest systems' (1.6×10^{13} W for the UV Ceti pair at 2.7 pc), the shallowest η Corvi, the first Band 8

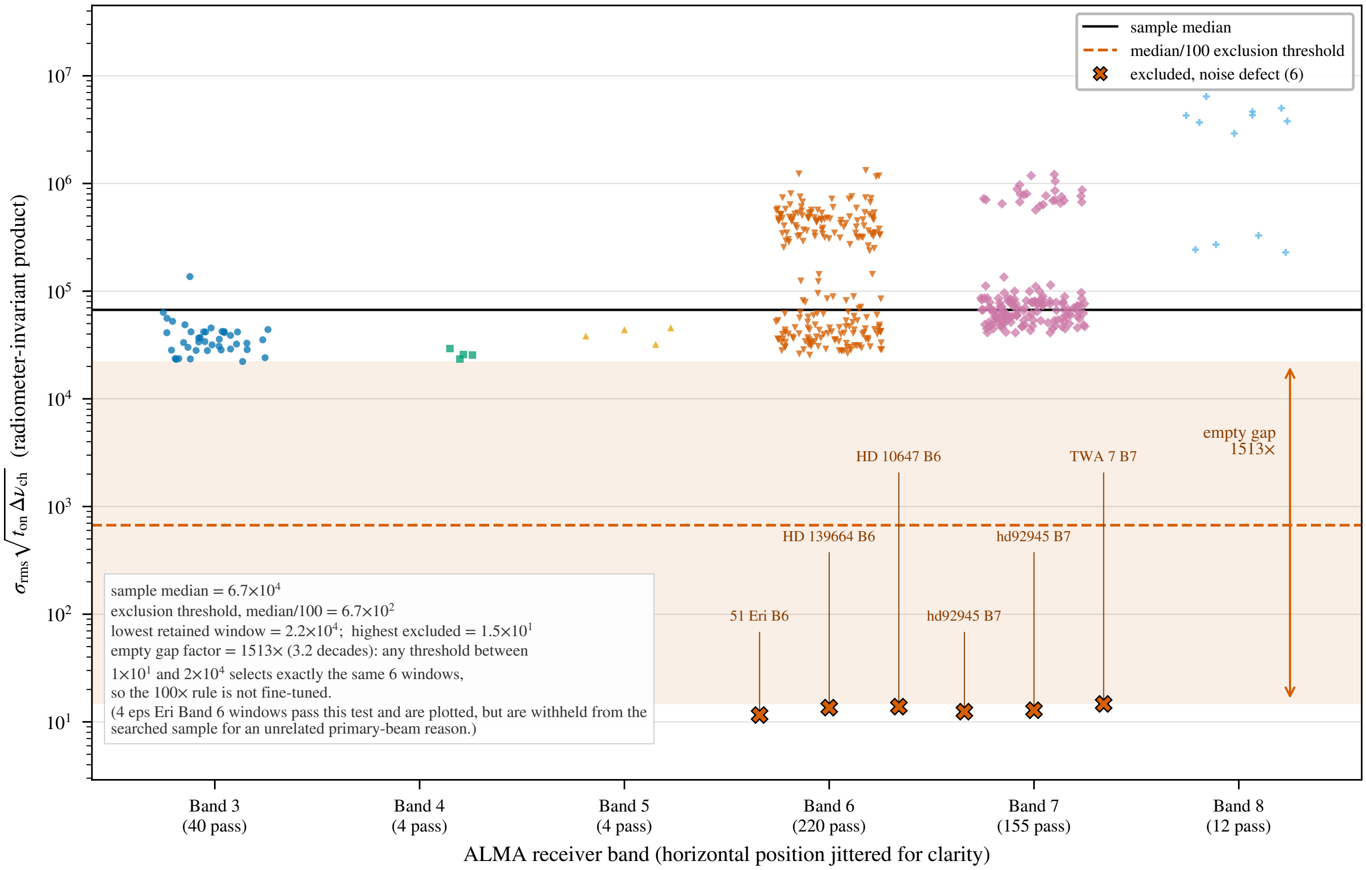

FIG. 5.— Quality control on the noise estimate (§5.3): the radiometer-invariant product $\sigma\sqrt{t_{\rm on}\Delta\nu_{\rm ch}}$, the six excluded windows a factor 1.5×10^3 below the lowest retained across an empty gap.

TABLE 4
THE NARROWBAND DECISION CASCADE OF THIS RELEASE, FOR THE 431 WINDOWS SEARCHED AFTER QUALITY CONTROL (§5.3); VOCABULARY AS IN TABLE 2.

Stage	Count
Channel × drift trial cells (all windows)	4.60×10^7
Control positions compared per window	512
Windows searched after quality control	431
Windows containing ≥1 hit (≥ 5σ at the stellar position)	20
Windows containing a control-ring exceedance	4
attributed to circumstellar or foreground CO	3
indistinguishable from the control ensemble	1
Candidates (surviving every veto)	0

target searched, where distance, coarse channels and receiver performance set the ceiling. Against the benchmarks of §4.3, only the two resolved components of G 272-61 at 2.7 pc reach below the Arecibo-like planetary-radar EIRP, and none approaches the $\sim 4 \times 10^9$ W of unintentional Earth leakage.

Four windows contain a *control-ring exceedance*; none survives vetting, so this release reports *zero candidates*. Table 4 follows the cascade, Table 5 each flagged window and the evidence dispositioning it.

Do the 4 control-ring exceedances exceed this survey's own calibration, or is any distinguishable from the empirical background or known astrophysical emission? No to both.

SURVEY-LEVEL FALSE-ALARM CALIBRATION

Each window-level test compares the peak signal-to-noise ratio at the star's position with the same statistic at 512 controls; the drift grid and thousands of channels apply identically to both, so the per-window trials factor is absorbed empirically. The statistic is now *symmetric*: the on-source quantity is the single propagated stellar position, as each control is. Earlier versions maximised over a small region around it while controls stayed single, an asymmetry that was not benign: under a pure-noise null with exchangeable positions the maximum over $n_{\rm src}$ region positions beats that over $n_{\rm ctrl}$ single controls with probability $n_{\rm src}/(n_{\rm src}+n_{\rm ctrl})$, 21 per cent for a typical window's $n_{\rm src} = 135$ region and 512 controls – the order of the measured region-max star-beats-ring rate, 65 of 431 windows (15.1 per cent; Wilson 95 per cent interval 12.2–19.1). Symmetrically the star ranks first in 5 windows, $\hat{p}_{\rm sbr,sym} = 1.2$ per cent (Wilson 0.5–2.8)

TABLE 5
The four windows containing a control-ring exceedance and the evidence dispositioning each. $T_\star$ is the search statistic at the stellar position (Eq. 1), 'ring max' the largest of that window's 512 control statistics. $\Delta\nu$ and Δv are the same offset of the crossing from the nearest of the eight masked species' catalogued transitions, in frequency and in velocity, topocentric; the $\pm 50\,\mathrm{km\,s^{-1}}$ mask applies in the stellar frame after the conversion chain of Appendix C. Giving both makes the conversion explicit for every row rather than for β Pic alone: at these window frequencies 1 MHz is 2.6, 1.3 and 0.87 $\mathrm{km\,s^{-1}}$ in Bands 3, 6 and 7 respectively. No disposition rests on a single test.

Window	ν (GHz)	$T_\star$	ring max	$\Delta\nu$ (MHz)	Δv ($\mathrm{km\,s^{-1}}$)	Disposition
β Pic B3	115.2605	14.65	7.27	−10.36	−26.9 (CO 1→0)	circumstellar CO
β Pic B6	230.5164	11.68	9.12	−21.87	−28.4 (CO 2→1)	circumstellar CO
HD 48370 B6	230.7305	27.10	26.83	−26.33	−34.2 (CO 2→1)	CO cloud emission
CP−72 2713 B7	345.1152	5.81	5.68	−1526.2	−1326 (CO 3→2)	control-ensemble background

against $\hat{p}_{\mathrm{sbr,asym}} = 15.1$ per cent – a factor 13, never to be interchanged.

That symmetric rate still needs explaining, not being the exchangeable expectation. Under exchangeability one designated position of 513 ranks first with probability exactly 1/513, a rank statement independent of whether the noise is Gaussian or heavy-tailed; earlier versions of this paper wrongly attributed the excess to heavy tails. The expectation is $431/513 = 0.84$ windows, and $P(\geq 5\,|\,0.84) = 1.7\times10^{-3}$ under an independent-Poisson approximation. That is a real excess over the noise null, for an astrophysical rather than statistical reason: in 3 of the 5, genuine celestial line emission sits at the stellar position (Table 5), making the star rank first by construction rather than by chance, so those 3 are not draws from the noise null at all. Removing them leaves 2 rank-first windows against 0.84 expected, $P(\geq 2\,|\,0.84) = 0.21$ – entirely unremarkable: the CP−72 2713 window of §5.4, and one towards Barnard's Star whose stellar peak (4.40 against a ring maximum of 4.38) does not reach the trigger threshold and so is not a flagged window at all.

Three levels stay distinct throughout: an *instrument/noise* null, calibrated by the control ring; an *astrophysical-source* null, addressed by line identification and kinematics; and only then a technosignature interpretation. The chance budget must account for exceedances unattributed after the first two, not every window in which the star happens to be brightest.

One distinction is load-bearing. The statistic of Eq. 1, applied at the single propagated stellar position and identically at each of the 512 single controls, covers *all* 431 windows: it is the operative criterion, and every count here derives from it. The *richer* variant that maximises over the source region at every control is a consistency check on it, not the criterion itself, and exists for only seven locally reprocessed windows (below).

The joint criterion gating a candidate is dominated by its threshold leg: only 20 of 431 windows contain any $\geq 5\sigma$ crossing at the stellar position. Each contributes at most the exchangeable rate $1/(1+n_{\mathrm{ctrl}}) = 1/513$, the $431/513 = 0.84$ expectation above. Four flagged windows exceed that budget ($P(\geq 4\,|\,0.84) = 0.9$ per cent) with no tension: excluding the three astrophysical, it need explain one unattributed crossing against 0.78 expected across the 402 non-β Pictoris windows. This operative estimate absorbs execution-block systematics into the fixed block margins, assuming only that crossing and star-beats-ring status are exchangeable within a block. Secondary bookkeeping models span 1.0–9.2 per cent (Appendix P); the dominant residual uncertainty is the star-beats-ring rate (Wilson 95 per cent interval 12.2–19.1 per cent), giving a 4–19 per cent bracket on $P(F \geq 3)$.

One ensemble limit follows: with 512 controls the smallest resolvable per-window p-value is 1/513, so a Bonferroni threshold over 431 windows (1.1×10^{-4}) lies below what the controls can measure. They can show a crossing unremarkable but cannot certify one significant survey-wide – a burden on the localisation, recurrence and astrophysical attribution that disposition each flag below. Stated in advance for later releases: N windows contribute $\sim N/513$ chance flags, so the full $\sim$2000-window census expects $\sim$4, and one unattributed marginal crossing per release is the *expected* outcome, not an accumulating anomaly. We pre-commit that no crossing in this programme will be advanced as a candidate on rank statistics alone: promotion needs evidence the ensemble cannot supply – recurrence at an independent epoch, or a ring enlarged until the rank probability resolves survey-wide significance ($\gtrsim 10^4$ positions for a Bonferroni threshold over the completed census; §6.2) – and until then every such crossing is reported, dispositioned as consistent with chance, and released with its products. No model in the bracket promotes a flagged window: the three CO dispositions rest on kinematic attribution to emission resolved, mapped or independently published, and the CP−72 2713 standing is a rank statement invariant to the model.

Is the ring exchangeable with the star? The 1/513 expectation holds only if the star and its 512 controls are exchangeable under the null, which spatially varying noise, sidelobes, correlated pixels or phase-centre-localised calibration residuals could spoil; we measure it by stratum (Fig. 6). The median per-window control maximum is flat across bands – 4.46, 4.59, 4.42, 4.52, 4.63 and 4.69 in Bands 3 to 8 – and a Kolmogorov–Smirnov test of the star's rank among its controls against $U(0,1)$ is consistent in every band ($p = 0.14$–0.83) and the coarse-channel stratum ($p = 0.71$). Fine-channel windows sit higher (control maximum 5.76 against 4.42), as expected from their far larger channel×drift trial count; not a violation, the test being conditional on the window.

Those tests are conditional on the stellar rank; for the stronger question – exchangeability of the ensemble itself – we ranked each of a window's 512 controls against the other 512-1 by the statistic and add-one rule applied to the star, pooled over the release. The 220 672 pseudo-star ranks are uniform to the resolution the ensemble can express (mean 0.5010 against 0.5; the Kolmogorov–Smirnov $D = 1/512$ is the discreteness floor of a 512-point rank statistic,

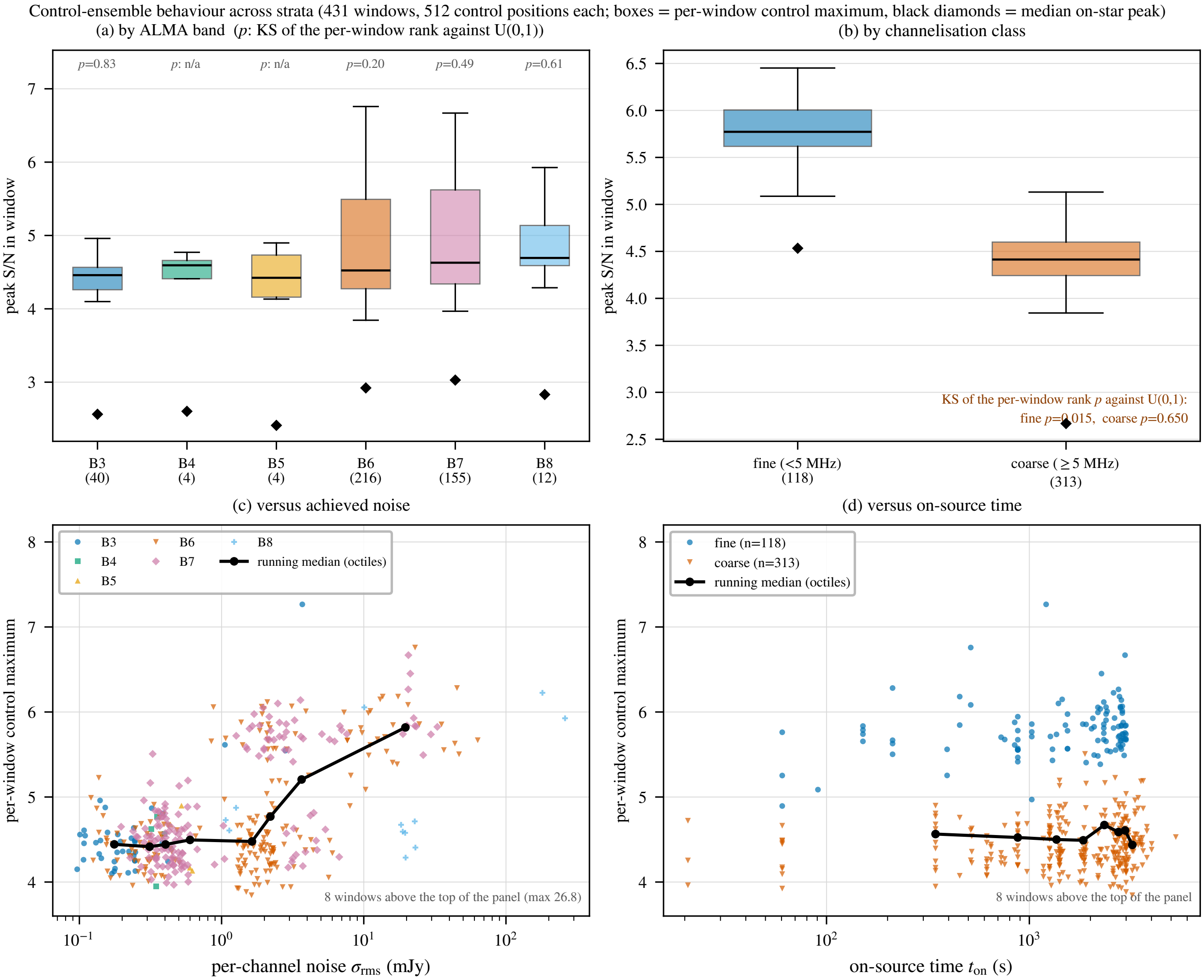

FIG. 6.— Exchangeability of the 512-position control ring by stratum. Boxes: per-window control maximum; diamonds: median on-star peak; p: the Kolmogorov–Smirnov test of the text. The ensemble tracks the on-star statistic in every band and at both channelisations; the one departure is the fine-channel stratum ($p = 0.018$), driven by the four flagged windows of Table 5.

$p = 0.37$), in every band and at both channelisations, and a two-sample comparison of the 431 real stellar ranks with that pooled distribution gives $p = 0.38$: the stellar position behaves like one more position in its own ring – the null calibration the control-ring argument needs, measured not assumed.

One stratum does depart, reported rather than pooled away: the fine-channel windows fail the same $U(0, 1)$ test at $p = 0.018$ ($n = 118$), containing four windows whose on-star peak exceeds their own control maximum against 0.22 expected – precisely the four flagged windows of Table 5. That is astrophysical signal concentrated where channels are finest and narrow lines detectable, not a misbehaving ring; but the departure is genuine, so fine-window rank probabilities are slightly optimistic. Two stratifications the retained products cannot support, storing no per-control radial position or primary-beam gain: with the ring radius small compared with the beam (targets lie a median 0.1 arcsec from the pointing centre) all controls should share the target's response to a good approximation – but we have not measured it, and because the whole false-alarm framework rests on the ring being exchangeable with the star, we record this as a *required* item for the completed survey rather than a refinement: the holography-model gain at a sample of control positions, relative to the target, for representative windows in each band. Remaining robustness checks – trials-count reconciliation under channel adjacency, control-ring geometry – are in Appendix P.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL

The designed principal test applies identical source-region, drift-rate and frequency maximisation at the star's position and at each of the 512 controls under one search rule. The frozen release products cannot meet that, so its domain is the seven locally reprocessed windows – six validation, plus the AU Mic window's first-epoch block – all giving $T_\star = 3.7$–4.9 against control maxima of 6.7–7.7, $p = 0.81$–1.00 (Figure 11, online-only appendix): the star is exchangeable with its ring wherever the test runs at full symmetry. The empirical star-versus-ring calibration

TABLE 6
LINE-ATTRIBUTION AUDIT OF THE THREE LINE-ATTRIBUTED WINDOWS (FIRST THREE ROWS) AND THE COARSE β PICTORIS BAND 6 WINDOW WITHOUT A CANDIDATE SHOWING THE SAME CO COMPONENT, THROUGH THE FRAME CHAIN OF §4.2. TOPOCENTRIC OFFSETS ARE OBSERVED MINUS CATALOGUED REST FREQUENCY (CO J=1→0 115.271202 GHz, CO J=2→1 230.538000 GHz, PICKETT ET AL. 1998); THE BARYCENTRIC CORRECTION IS THE EARTH-EPHEMERIS TERM AT THE EXECUTION-BLOCK MID-TIME (BLOCK UIDS, TABLE 11); SYSTEMIC RADIAL VELOCITIES ARE CATALOGUED (SIMBAD; *Gaia* DR3 FOR β PICTORIS, GAIA COLLABORATION ET AL. 2022; DR2 FOR AU MIC, FOUQUÉ ET AL. 2018).

Flag	UTC start	$f_{\rm obs}$ (GHz)	Transition	Topo. offset (MHz; km s^{-1})	Bary. corr. (km s^{-1})	$v_{\rm sys}$ (km s^{-1})	Stellar-frame offset (km s^{-1})
β Pic B3	2022-03-02 23:14	115.260644	CO(1→0)	−10.56; −27.46	−7.90	+16.84	−2.72
β Pic B6	2019-03-12 00:14	230.516128	CO(2→1)	−21.87; −28.44	−8.15	+16.84	−3.45
AU Mic B6	2014-08-18 04:01	230.825230	CO(2→1)	+287.23; +373.51	−10.02	−4.71	+378.82
β Pic B6†	2013-10-06 06:23	230.528134	CO(2→1)	−9.87; −12.83	+7.63	+16.84	−3.62

†Coarse Band 6 window without a candidate (Appendix I), shown because it independently shows the same CO component as that band's candidate fine window.

above therefore remains the operative release-wide estimate over all 431 windows, the symmetric statistic a validation subset. The machinery ships with the release; the full statement, with the family-wise trials factors of Table 17, is in Appendix P.

DISPOSITIONS OF THE FOUR FLAGGED WINDOWS

Two of the four are towards β Pictoris, a debris disc with reported CO emission (Matrà et al. 2017): one in Band 3 at 115.2606 GHz, 10.56 MHz below the CO(J=1→0) rest frequency, the other in Band 6 at 230.5161 GHz, 21.87 MHz below CO(J=2→1). Neither coincidence is exact, so the automated check missed them. Through the frame chain (Table 6; Fig. 7) the offsets are one kinematic component – −2.7 and −3.5 km s^{-1} from stellar systemic, 0.74 km s^{-1} apart – in the two lowest CO rotational transitions of a system with detected circumstellar CO; the unflagged coarse Band 6 window, from a third epoch six years earlier, independently shows CO at −3.6 km s^{-1} in the same frame. All three epochs and both transitions land within 0.9 km s^{-1} of each other and 3.6 km s^{-1} of systemic, deep inside the ±50 km s^{-1} mask, though raw topocentric offsets are −27.5 and −28.4 km s^{-1}: we attribute both to this kinematically offset CO. They are also the survey's empirical positive control – two transitions of one species at the same stellar-frame velocity, recovered at separate epochs – showing that the pipeline does detect narrow spectral emission localised on a target when it is there.

Their empirical significance under the release background is nonetheless extreme, so the attribution is no dismissal: the Band 3 region peak of 20.3 exceeds all 431 per-window control maxima (add-one $p < 0.0044$), the Band 6 peak of 12.8 all but one – that window's own CO-filled control ring, reaching 15.2, by far the release maximum ($p \approx 0.0088$). The alternative needs a transmitter coincidentally placed at matched offsets from two CO transitions in the same star.

The third, towards HD 48370 in Band 6, is the release's largest excess in absolute and smallest in relative terms: $T_\star = 27.10$ against a control-ring maximum of 26.83, the star exceeding its ring by one per cent of the statistic. Star and ring rising together is what emission filling the primary beam looks like, and the window sits −34.2 km s^{-1} topocentric from CO(J=2→1). The attribution is independently published: Cataldi et al. (2023) report prominent CO(2→1) and ^{13}CO(2→1) emission towards HD 48370 at $v_{\rm bary} \approx 41$ km s^{-1}, well separated from its 23.6 km s^{-1} systemic velocity and attributed by them to foreground cloud contamination; our window recovers it. It is also a stated limitation of the spatial test: this window is identified by the velocity coincidence and near-equality of star and ring, not the rank statistic.

The fourth, towards CP−72 2713 in Band 7, is the only flagged window with no astrophysical attribution, and is marginal: $T_\star = 5.81$ against a ring maximum of 5.68, a margin of 0.13 attaining only the rank-probability floor $p = 1/513$, the smallest value 512 controls can express and not a measurement of significance. That margin is smaller than the scatter across this star's own windows, whose other fine-channel control maxima are 5.16, 5.41, 5.67 and 5.85: its rings reach 5.81 routinely where the star does not cross threshold. The nearest transition of the eight masked species lies 1326 km s^{-1} away; since that mask is an automated contamination cut, not a complete line catalogue, and a chemically rich or circumstellar environment can present many further ALMA-band transitions, we re-queried a wider catalogue, whose closest entry is H^{13}CN(J=4→3) at −195 km s^{-1}, still far outside any circumstellar tolerance. The defensible reading is "no plausible known transition identified", a statement about the catalogues consulted, not a proof that none exists; and with a single execution block at this frequency no recurrence test is possible. Five facts fix the disposition, none decisive alone: a single occurrence; the 0.13 separation from its own control maximum; a rank probability on the 1/513 floor; no catalogued transition nearby; no recurrence test available. We record it as an unclassified statistical or astrophysical feature, not candidate evidence – and as the one window here a small, well-defined observation would settle: a targeted ALMA re-observation at 345.1152 GHz would test recurrence directly, as no further analysis of this single archival window can.

LINE EXCLUSION: DEFINITION, COST, AND LIMITS

The searches used an automated line check matching crossings to catalogued rest frequencies within one spectral channel at zero drift. The β Pictoris windows showed that inadequate: both fell outside its one-channel tolerance (by 10.6 and 21.9 MHz) while lying, after the frame chain, within 3.6 km s^{-1} of the stellar systemic of a resolved, mapped disc

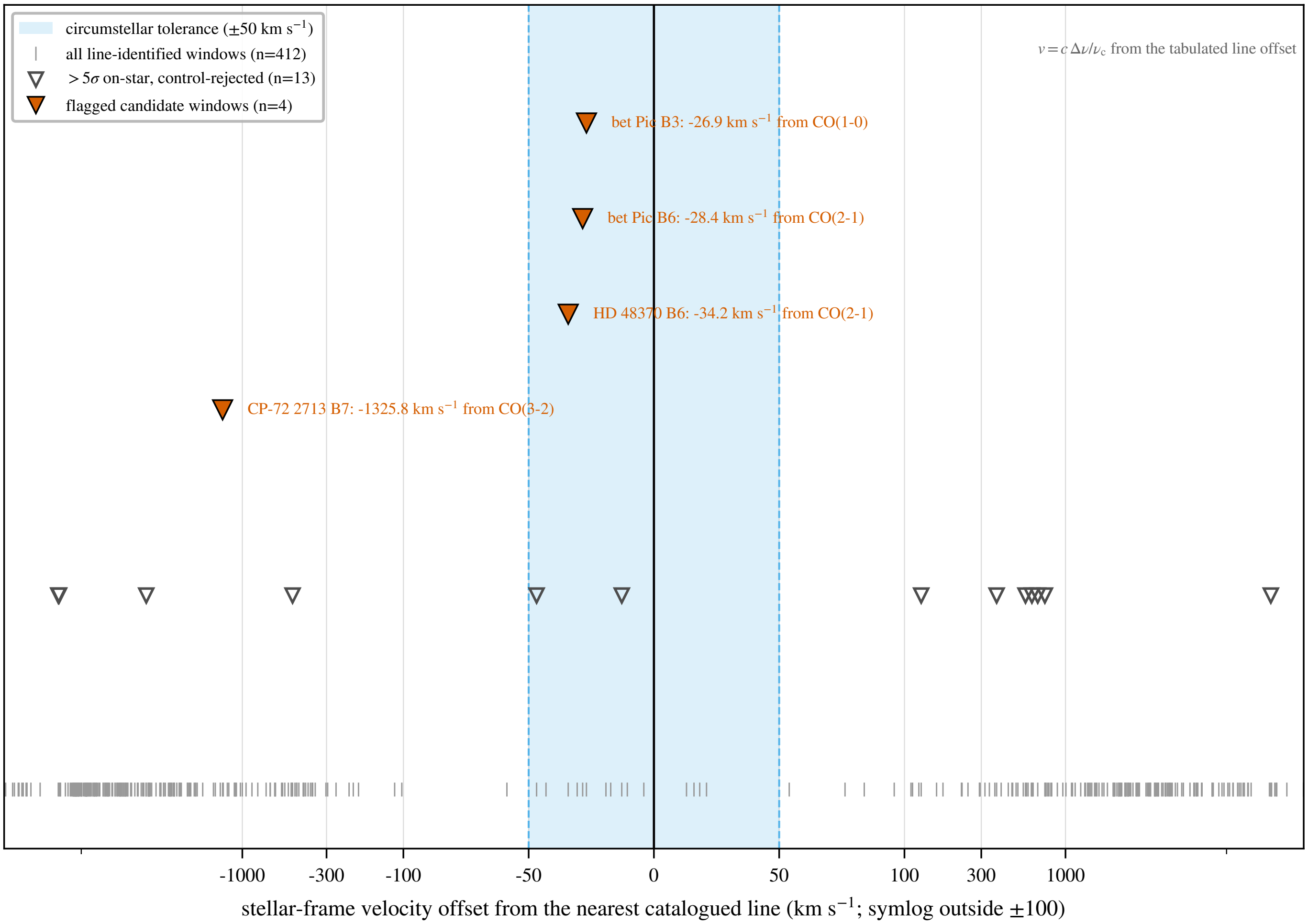

Fig. 7.— Stellar-frame velocity offsets of the flagged windows, through the frame chain (topocentric → barycentric → stellar; Table 6). Shaded band: the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line exclusion mask. *Top:* the two β Pictoris windows fall well inside the mask as one kinematic component, resolved in the inset. *Bottom:* the flagged window towards CP−72 2713 and the AU Mic crossing the symmetric statistic does not flag (§5.5) lie far outside any circumstellar tolerance, dispositioned by their spatial behaviour, not the mask.

– exactly the regime a channel-width tolerance is blind to. Every crossing was therefore reclassified before publication with a velocity-space mask ($\pm 50\,\mathrm{km\,s^{-1}}$ in the stellar frame around each catalogued transition), the pipeline's line-exclusion definition from this release on. Its kinematic justification, empirical conservatism, that reclassification and the "Schelling point" counter-argument that a communicator might deliberately choose these frequencies are in Appendix J; retrospectively the mask rejects exactly the two β Pictoris windows, changing no other disposition.

The masking cost uses one definition: every JPL catalogued transition of the eight species (276), displaced by each system's radial velocity, $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame half-width, intersected with each window and merged. The release effectively searches 87.0 GHz of unique sky frequency – the disjoint union of searched windows minus the mask inside it (Figure 2) – down from 93.1 GHz, a 5.7 per cent reduction; over the 431 windows it removes 41.4 GHz of 700.8 GHz gross bandwidth (5.9 per cent; breakdown in Table 23), the per-window usable fraction having median 0.960 (10th percentile 0.790). Union bandwidth is exposure-blind, so we also give the summed per-star union, 371.9 GHz, and the summed product of on-source time and searched bandwidth, 1.4×10^6 s GHz (on-source time 12.1–5274 s per window, median 2570 s). The mask governs dispositions, not the noise-derived thresholds, so the quoted $\mathrm{EIRP}_{5\sigma}$ values are unchanged; no signal was found outside the excluded velocity regions in any window.

The mask is a design selection effect: a signal coincident with a molecular line is not thereby "false" but *excluded from the technosignature search domain*, the astrophysical prior there being overwhelming and this pipeline unable to separate the two. We place no constraint on transmitters in the excluded regions – 5.9 per cent of the searched bandwidth – and a search for deliberately line-adjacent carriers would need a complementary strategy. Nor is the exposure confined to the flagged windows: 131 of 431 windows (58 per cent) contain an in-band catalogued transition of the eight species, as do all four flagged windows, yet in none does a crossing fall within one native channel of a rest frequency – precisely the regime the velocity-space criterion covers.

5.5. *Changing the detection statistic, and the AU Mic crossing*

The initial region-max statistic maximised over an $n_{\mathrm{src}} = 135$-position region centred on the star against 512 *single*-position controls; under it a crossing towards AU Mic in Band 6 at 230.83 GHz was carried as a candidate. A

TABLE 7
The AU Mic crossing frequency (230.8252 GHz) under the three treatments that decide it: the release's frozen products (region-max statistic), the fully symmetric reprocessing of the same first-epoch block (2014 August 18, observatory calibration recipe verbatim) and the second-epoch block of the same programme (2014 June 5, identical search treatment).

Statistic	Release (region-max)	Symmetric reprocess.	Second epoch (recurrence)
Star statistic	5.97	4.93	4.77
Control maximum (512)	5.85	7.70	6.78
Star rank p (add-one)	—	0.81	0.97
Single-position star statistic	5.22	—	—
Per-channel S/N (zero drift)	—	2.46	2.22
Peak drift ($\mathrm{Hz\,s^{-1}}$)	1418.5	1593	1281

TABLE 8
Every window flagged by the original region-max statistic, shown under both statistics, so the effect of the revision can be audited across all of them rather than taken on trust for one target. $S_{\rm reg}$ is the maximum of Eq. 1 over the $n_{\rm src} = 135$-position source region, 'ctrl max' the largest of the 512 single-position controls it was compared with; $T_\star$ is the symmetric single-position statistic at the propagated stellar position and 'ring max' the largest of the same window's 512 control statistics. ν is the window centre. The symmetric criterion flags a strict subset, four of the seven. In the three that drop out the stellar position never exceeded its own control ring ($T_\star$ of 5.22, 5.09 and 5.22 against ring maxima 5.85, 5.63 and 5.74): the region-max statistic cleared the ring only by maximising over the source region. For CP−72 2713 the region maximum already lay on the star, so the two statistics coincide exactly and nothing about it changes.

			region-max statistic		symmetric statistic		
Star	Band	ν (GHz)	$S_{\rm reg}$	ctrl max	$T_\star$	ring max	Symmetric outcome
HD 48370	6	230.7305	27.95	26.83	27.10	26.83	flagged
β Pic	3	115.2605	20.31	7.27	14.65	7.27	flagged
β Pic	6	230.5164	12.78	9.12	11.68	9.12	flagged
AU Mic	6	230.5384	5.97	5.85	5.22	5.85	not flagged
ALMA J1537−3319	6	217.0183	5.83	5.63	5.09	5.63	not flagged
CP−72 2713	7	345.1152	5.81	5.68	5.81	5.68	flagged
HD 14055	7	345.1377	5.80	5.74	5.22	5.74	not flagged

criterion revised after inspecting the object it must judge invites distrust, so we set out the sequence explicitly. The bias argument below is algebraic, holds for any window, and was derived before any crossing was re-examined – and Table 8 audits the revision across every window the original statistic flagged, not one. For the completed survey the statistic will be pre-registered, frozen before any crossing is inspected, so the objection cannot arise again.

The defect belongs to the statistic: the region-max asymmetry raises the star-beats-ring probability under a pure-noise null to 20.9 per cent (§5.4) against $1/513 = 0.19$ per cent for a symmetric test, and the measured release-wide region-max rate, 15.1 per cent, is of that order. We replaced it with the symmetric single-position form of Eq. 1, applied identically to star and controls, and re-ran the release.

The change is a bias correction, not a loss of sensitivity, and it is checkable. Across the 431 searched windows the symmetric criterion flags a strict *subset* of the region-max set: seven before, four now, the three dropping out (ALMA J1537−3319, AU Mic, HD 14055) all with the *stellar position never exceeding its own control ring* – star statistics of 5.09, 5.22 and 5.22 against ring maxima of 5.63, 5.85 and 5.74. No window in which the star dominated its controls was removed – the only configuration a localised transmitter can produce, since a point source at the stellar position raises $T_\star$, the quantity the new test uses. Consistently, the injection campaign recovers signals through the unmodified pipeline on the star position (§4.4), so its completeness is a property of the symmetric statistic as reported.

Under the symmetric criterion the AU Mic crossing is not flagged: 10 of its 512 controls reach or exceed the stellar statistic, an empirical $p = 0.021$ far from the 1/513 floor. It also fails two pre-named tests, localisation and recurrence (Table 7).

Localisation, recurrence and statistical scale are in Appendix K, with the numbers of Table 7. Neither the reprocessed first epoch nor the one further public Band 6 block covering 230.83 GHz shows a localised excess at the stellar position, and that second epoch is a clean non-recurrence – rejecting a continuous or repeating source at this frequency and these epochs, though not intermittent transmission. The archival epochs also bound what this release can say about duty cycle: each star was observed in one to three execution blocks, searched independently with no stacking, so the non-detection concerns the epochs observed, not permanent activity. The surplus of available over searched blocks is the recurrence pool the second-epoch test drew on, and §6 turns the epoch structure into an explicit duty-cycle constraint.

5.6. *Ancillary results*

Three ancillary results are reported in full online.

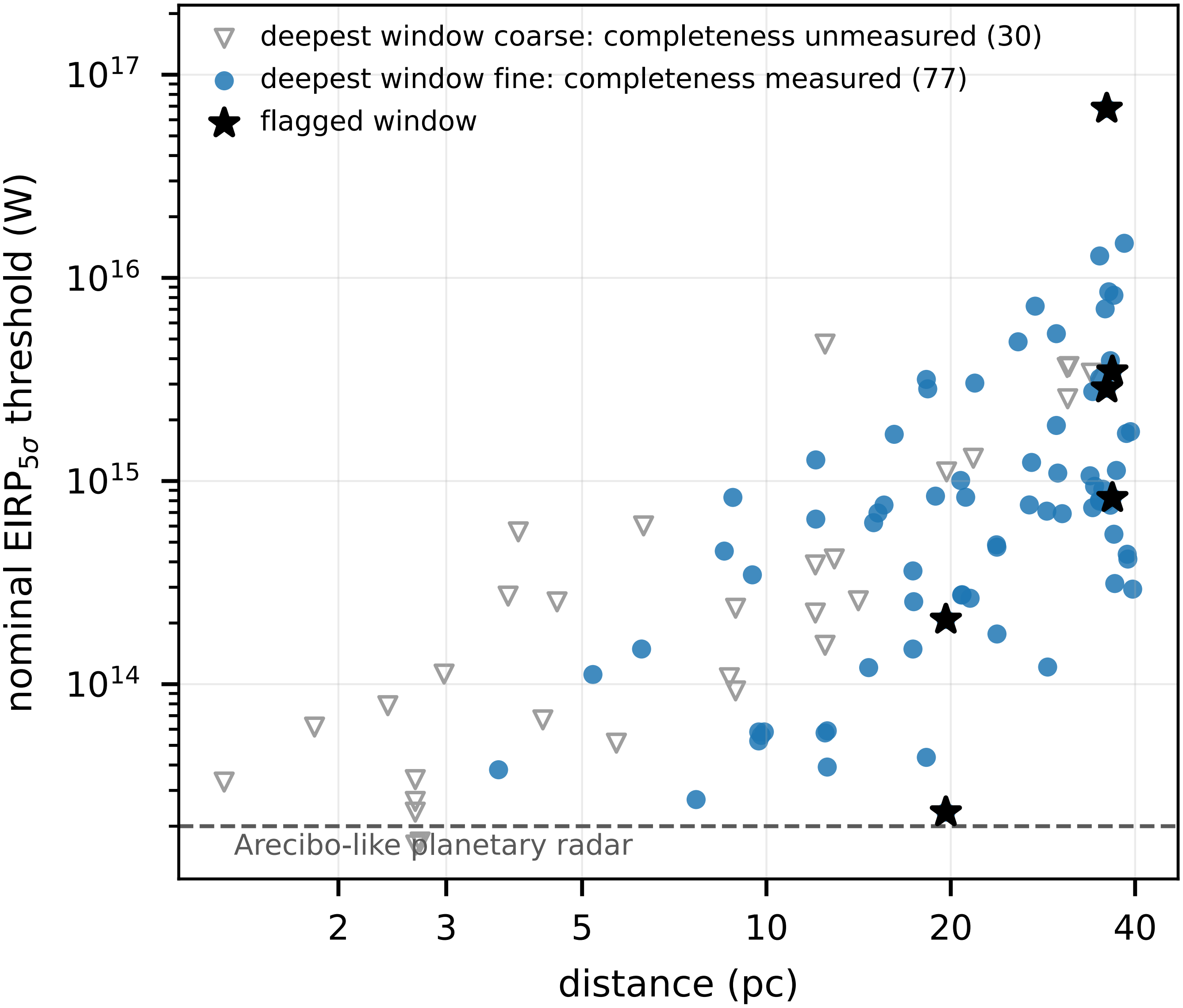

FIG. 8.— Nominal EIRP$_{5\sigma}$ trigger thresholds (boxed rule, §4) against distance, one deepest window per star and band (107 datasets, 88 stars). Symbols mark whether that window is fine-channel, where the injection campaign measures a recovery fraction, or coarse-channel, where completeness for the time-variable class those windows search is unmeasured; survey-wide the split is 118 fine against 313 coarse windows. Every point is an upper limit. Stars mark the four windows with a control-ring exceedance (Table 5). The dashed line is an Arecibo-like planetary radar, 2×10^{13} W, a scale for human transmitter power, not a claim that the survey reaches Earth-technology detectability (§4.3).

Integration-time audit (Appendix Q). On-source integrations were reconstructed from pipeline bookkeeping, not retained natively, so all 227 window-level records were audited against the archive's exposure accounting. Two of the 58 star–execution-block groups over-count – AT Mic by a factor 1.43, an internally diagnostic dump-bookkeeping artefact already corrected to the archive's 877 s in Table 22, and CD−38 10980 by 7 per cent – and the rest are conservative. No EIRP threshold depends on it, thresholds deriving from measured per-channel rms rather than integration time; the audit is conservative for every window of both flagged hosts, so no disposition changes; and the exposure-weighted coverage of §5.4 is at worst optimistic by under 1 per cent. The class was found by chance in one target, hence the direct per-execution-block export committed for the next release.

Continuum search (Appendix M). An exploratory by-product, run over the 57 target/bands processed far enough for a usable measurement – a subset of the narrowband sample, and the one count here referring to fewer than the 431 searched windows. Seventeen show a $\geq 5\sigma$ detection, forty are non-detections reaching 0.036–3.6 mJy (median 0.52 mJy); every detection is astrophysically unsurprising, the auroral emitter LSR J1835+3259 meriting deeper follow-up. Its statistical power is unquantified at this depth and no technosignature disposition rests on it.

Serendipitous molecular-line catalogue (Appendix N). The line-exclusion step identifies every $\geq 5\sigma$ outlier channel as a by-product; across the subset scanned, 18 groups were found, six plausibly real, all reported openly, including the two we think probably spurious.

6. DISCUSSION

This first stand-alone processed release is preliminary by construction: 107 of 208 planned target/bands (51 per cent) are complete, and that subset is sensitivity-ranked, not an unbiased draw; the full sample will reach larger distances and shallower faint-end limits. No credible technosignature has been found; the $\mathrm{EIRP}_{5\sigma}$ thresholds of Appendix I bound any transmitter above them. Two flagged windows towards β Pic are attributable to CO disc emission, underlining Mason et al. (2024)'s point that mm-band crossings need checking against known molecular transitions; that check is automated (§4) and now applies a velocity-space match retroactively to every window (§5.4). The one window flagged without a line attribution, CP−72 2713, is not statistically distinguishable from the empirical background; it is recorded for re-examination, not dismissed. The AU Mic crossing is unflagged under the operative criterion and not a candidate (§5.5).

The searched space uses one definition throughout (§5.4, Figure 2). Of the 41 planets known around the 18 planet-hosting targets only TRAPPIST-1 b reaches the drift ceiling, in its edge-on worst case at $\sim$13 Hz s^{-1} GHz^{-1}; GJ 581 e, 61 Vir b, and TRAPPIST-1 c reach roughly half, the remaining 33 well inside the searched grid.[5]

Table 3 (§5) consolidates the axes on which a transmitter is or is not detectable; Table 21 (online-only appendix) decomposes the release by ALMA band.

Coverage eligibility and search completeness differ. The census of public usable ALMA data (§3) sets eligibility; every held execution block in scope was searched, so no EB-level selection lies behind the 102. Availability is measured directly, not inherited from release bookkeeping: a proper-motion-matched ALMA archive census (2026 September 9; public rows, all bands, pointing within 30 arcsec of the propagated stellar position at the observation epoch) counts 107 unique public on-star EBs over the subset run so far, re-finding all but one searched EB – TRAPPIST-1's Band 7 execution A002/Xcccc19/X5564, which `obscore` carries only through its calibrator-field row, an archive-representation artefact biasing that star's census low by one. The selection funnel is Figure 10 (online-only appendix); the ledger `v323_star_coverage.json` gives per-star distance, availability, scope and searched frequency intervals.

Figure 14 (online-only appendix) shows the drift ceiling as an explicit selection function of planet period and host mass: the search covers the drift of a transmitter on any known planet in the sample bar the innermost TRAPPIST-1 planets, while a faster drift falls outside the searched grid, sacrificed by design (Appendix G recovers some non-linear morphologies). The host-mass dependence is weak ($a \propto M_\star^{1/3}$ at fixed period), so the ceiling is effectively a function of orbital period alone; and it constrains only transmitters co-located with *known* planets – one on an undiscovered planet, or on a platform accelerating as it chooses, is covered only if its drift falls in the searched grid.

The release sits against Wright et al. (2018)'s published haystack axis by axis. On frequency (10 MHz–115 GHz) only our Band 3 falls inside (20.6 GHz of 115 GHz, 0.18 of the axis, and one their Hz-resolution surveys reach far more finely); the remaining 66.6 GHz (Bands 4–8, 125–495 GHz) lies wholly outside – the genuinely new territory. On transmitted bandwidth (0–20 MHz), a native-channel search probes a fraction set by channel width (window-mean 0.63, median 0.78). On sensitivity, our median threshold (7.3 mJy across a 15.625-MHz channel, $\phi \approx 1.1 \times 10^{-21}$ W m^{-2}) detects a 10^{13} W transmitter only within $\sim$0.9 pc. The sole prior ALMA-archival search, Mason et al. (2024), reaches 28 bycatch targets at $\geq$1.01 kpc, deepest quoted $\mathrm{EIRP}_{\min}$ 6.91×10^{17} W; this release's 82 systems span 1.30–39.63 pc, 41 of them inside 20 pc, at median per-target thresholds $\sim$1250$\times$ deeper. We do not multiply these fractions into a combined coverage: the axes are strongly coupled (distance and sensitivity above all), so a product of marginals would misstate the overlap by orders of magnitude.

6.1. *Occurrence limits*

What a zero-detection result bounds is fully conditioned. The per-system detection probability factorises into four terms,

$$C_i(P) \;=\; F_i\; R_i(P)\; D_i\; C_{\mathrm{morph},i}, \tag{3}$$

F_i: *frequency occupancy* – the transmitter's sky frequency lies in system i's searched intervals;
$R_i(P)$: *amplitude recovery* – a transmitter of EIRP P would have crossed system i's threshold had it emitted there;
D_i: *duty cycle* – it emitted during the analysed archival interval;
$C_{\mathrm{morph},i}$: *morphology match* – the measured recovery of a carrier of the searched drift and temporal morphology (Appendix E).

The bound below sets $F_i = D_i = 1$ *by conditioning, not by measurement*: neither factor is estimated anywhere in this release. With that convention the probability of zero credible detections if a fraction f of systems hosts one is

$$P(N_{\mathrm{det}} = 0 \,|\, f) = \prod_{i=1}^{N_{\mathrm{sys}}} (1 - f\, C_i)\,, \tag{4}$$

and the one-sided 95% upper limit f_{95} solves $P(N_{\mathrm{det}} = 0 \,|\, f_{95}) = 0.05$.

> *Reading rule.* Among the ALMA-observed systems, fewer than $f_{95}(P)$ per cent host a transmitter of EIRP $\geq P$ emitting during the archival observations, at a sky frequency in the particular spectral intervals searched for that system, with drift and temporal morphology in the searched class.

[5] Star–planet relative accelerations $GM_\star/r^2$ from NASA Exoplanet Archive planetary-companion parameters (Mission Star/Planet table, accessed 2026 September), maximised over orbital phase and inclination.

TABLE 9
The 95 per cent upper limit $f_{95}(P)$ on the occurrence of transmitters above EIRP P, conditional on emission inside the searched windows and throughout the archival epochs. n is the number of systems with non-zero completeness at that P; the *measured* column is this paper's result. 'Vacuous': no system has non-zero completeness, so no bound exists.

P (W)	thresholded (unit C)		measured C *(result)*	
	n	f_{95}	n	f_{95}
3×10^{13}	4	52.7%	2	vacuous
10^{14}	16	17.1%	11	47.1%
10^{15}	57	5.1%	45	10.1%
10^{16}	81	3.6%	59	**6.2%**
10^{17}	82	3.7%	57	6.2%

What is and is not being inferred. Fig. 2 shows which spectral intervals the frequency condition refers to. The proposition differs from star to star – for system i, "hosts a transmitter within frequency set i" – so we call f not a prevalence but an *effective conditional in-band transmitter fraction given occupancy of each target's own observed frequencies*, defined on the 82 systems processed here, not the 168-star ALMA-covered census and still less the 40 pc stellar population.

That conditioning can be measured: under the simplest transparent transmitter-frequency prior, a uniform $p(\nu)$ over ALMA's tuning range, $F_i = \int_{\Delta\nu_i} p(\nu)\,\mathrm{d}\nu$ rather than one. The per-system fine-channel frequency union has median 1.73 GHz, so F_i averages 4.5×10^{-3} over Bands 3–8 and 2.2×10^{-3} over Bands 3–10, and the product likelihood excludes nothing whatever: $\sum_i C_i = 0.27$ and 0.13 respectively, so one detection would need $f \approx 3.8$ or 7.8 transmitters per system. Under any frequency prior spread across ALMA's tuning range, then, this release constrains nothing, and the conditioning supplies the entire content of the numbers below.

We evaluate Eq. 4 on the 82 independently observed systems (88 narrowband-searched stars; six bound pairs share observations and count once each) in two forms (Table 9). The *thresholded* form is the step function the retained products support: $R_i(P) = 1$ when system i's deepest threshold lies below P, else 0. The *measured* form integrates the injection-recovery curve of Appendix E over each system's searched bandwidth, $R_i(P) = \sum_w \Delta\nu_w\,\mathrm{rec}(5P/\mathrm{EIRP}_{5\sigma,w})/\sum_w \Delta\nu_w$, folding in the $17 \to 83$ per cent recovery of the searched class measured between four and ten times the noise; defined on the 59 systems with a fine-channel window, coarse-channel-only ones lacking measured recovery for the drifting class: a 59-system bound.

How far the curve should be carried. The drifting-class recovery curve is measured on one fine-channel configuration, then applied to all 118 fine windows, which span Bands 3–8, on-source times of 21–5274 s and a range of primary-beam offsets. Achieved noise, integration time and drift-grid density should shift the curve mainly in *position* – the amplitude at which a given recovery is reached – since they set the ratio of injected signal to the statistic's own noise; channelisation and the median-subtraction dwell fraction control its *shape*, and both are common to every fine window by construction. None of this has been measured, so the transfer is an assumption, and its error is not bounded by the $^{+16}_{-14}$ percentage-point binomial interval, which describes only sampling noise within the one configuration run.

Why this is a pilot number. The retained products preserve the campaign's pooled recovery fractions but not its trial-level records, so their denominators come from the design, not a log – weaker provenance than a headline number should have: completeness is measured on six configurations and three stars, the drifting-class curve rests effectively on one fine-channel configuration, and the coarse-window searched class has no calibrated completeness at all. The conditional in-band transmitter fraction is therefore a *pilot, configuration-limited* estimate, not a survey-grade measurement, and secondary to the non-detection and threshold range. Trial-level logging – identifier, dataset, amplitude, frequency phase, drift, morphology and recovered flag per injected signal – is committed for the next release.

Which number is which. The measured form is this paper's observational result, the only one whose recovery fraction is measured rather than assumed: at EIRP $\geq 10^{16}$ W, $f_{95} = 6.2$ per cent. The thresholded form (3.6 per cent) assumes unit recovery and a uniform $C = 0.42$ gives 8.6 per cent, both bracketing calculations not to be quoted as the survey's constraint. Nor is the measured form simply the weakest of the three: where fine-channel thresholds lie well below P the measured recovery rises above 0.42 towards its 0.83 plateau, so it falls between the two idealisations. A Bayesian treatment with a uniform prior on f, $p(f\,|\,\mathrm{data}) \propto \prod_i (1 - f\,C_i)$, gives a 95 per cent credible upper bound of 6.1 per cent for the same model – agreeing to better than the last quoted digit, so nothing turns on the frequentist construction.

Three caveats bound these numbers. Even the measured form fixes $F_i = D_i = 1$ and the recovery at configuration level – a frequency-integrated completeness for the searched class, not a per-system calibration; the population is heterogeneous in threshold, so each point applies to transmitters above that P, not one well-defined class; and systems are treated as independent trials in a randomly chosen archival interval, with duty-cycle dependence quantified below. These are, to our knowledge, the first ALMA-side numbers of their kind; removing the remaining idealisations needs the per-system $C_i(P)$ the retained campaign supports.

The archival epoch structure gives a duty-cycle constraint directly. Write D_i for the probability that a transmitter of duty cycle D emitted during system i's analysed intervals ($D_i = D$ to the accuracy the epochs support; Table 22 gives per-star epochs and integrations). The same product form gives the $f_{95}(D)$ ladder of Table 10 and Figure 9: continuous emitters give Table 9's limits; at $D = 0.5$ the measured-form limit degrades from 6.2 to 12.3 per cent; by $D = 0.1$ it is 62 per cent, carrying essentially no information; below that no bound of this form exists, and such

TABLE 10
Duty-cycle dependence of the 95 per cent upper limit at $P \geq 10^{16}$ W, from Eq. 4 with $D_i = D$: thresholded form (82 systems, unit recovery) and the measured-completeness form (59 fine-calibrated systems), this paper's result. 'Vacuous': no bound exists.

Duty cycle D	f_{95} thresholded	f_{95} measured
1 (continuous)	3.6%	6.2%
0.5	7.5%	12.3%
0.1	37.7%	65.0%
0.01	vacuous	vacuous

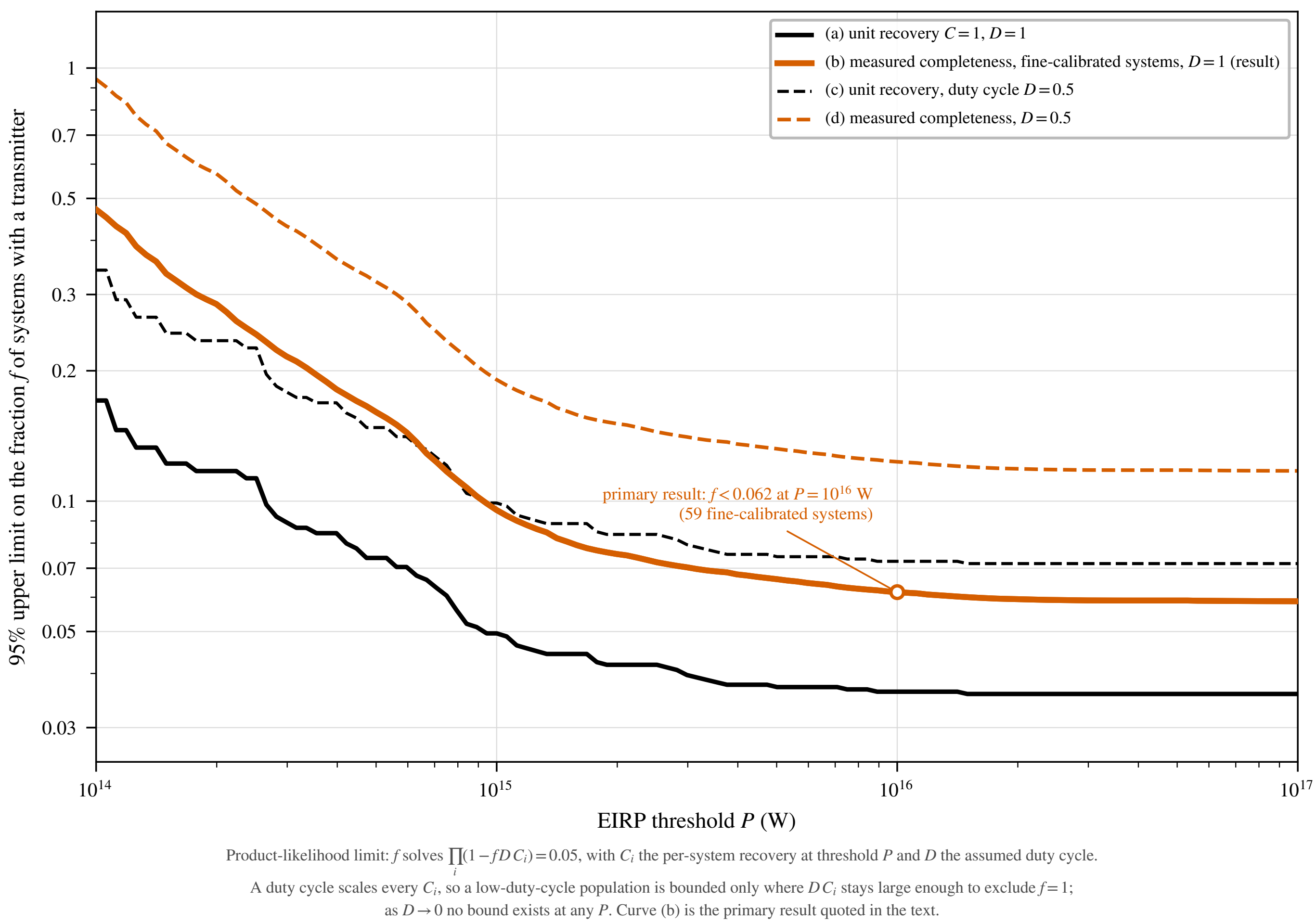

Fig. 9.— 95 per cent upper limits on the fraction f of systems hosting a transmitter above EIRP P, from $\prod_i (1 - f D C_i) = 0.05$. Curve (b) is this paper's result, the injection-measured recovery integrated over each system's fine-channel bandwidth; curves (a) and (c) assume unit recovery and are bracketing only. Shaded: where the surviving systems cannot exclude $f = 1$ and no bound exists; it grows to cover every P as $D \to 0$.

transmitters are reachable only by future releases' recurrence-pool reprocessing. An intermittent transmitter is at least as plausible *a priori* as a continuous one, so the $D = 1$ numbers should never be quoted without this dependence; the non-recurrence logic of §5.5 points the same way.

A cumulative look-elsewhere effect accompanies multi-release use: the probability that some unremarkable crossing is reported somewhere grows toward unity as releases accumulate, even if each is individually clean. The guard is the per-release procedural pre-commitment of §5.4, not aggregated significance.

The data-quality exclusions of §5.3 carry two open engineering items: the continuum-imaging step still lacks the narrowband step's pointing-offset guard, and the window-level noise-handling defect affecting HD 10647 and HD 139664 is excluded from results but not yet fixed at source. Both are corrected, and the affected datasets reprocessed, in the single corrected public release (Data Availability); Appendix O gives general lessons for archival mining. The four supplementary analyses of Table 15 are infrastructure investments, not sources of results at this stage: the Type-I error budget is carried by the primary lane alone (§5.4): the screens gate nothing in it, are correlated with it by construction, and disposition none of the 4 control-ring exceedance windows, so their non-detections must not be multiplied into a stronger constraint than the primary lane's.

Because selection is on ALMA archival coverage alone, useful subsamples fall out without compromising the census framing, subject to the unresolved-pair caveat of §3. First, 18 of the 88 processed stars (41 known planets) are confirmed exoplanet hosts (Proxima Centauri, τ Ceti, ϵ Eridani, β Pictoris and the seven-planet TRAPPIST-1 system among them), extractable as a sub-survey from Appendix I; HN Lib and LHS 1140, each with a habitable-zone planet, now carry an ALMA technosignature non-detection beside their planet-search literature. Second, four searched stars are white dwarfs (Gentile Fusillo et al. 2021) (Sirius B, Van Maanen's Star, Wolf 219, CD−38 10980), all continuum non-detections with $\mathrm{EIRP}_{5\sigma}$ limits of 2.3×10^{13}–8.4×10^{14} W – among the few mm/submm SETI constraints for white dwarfs (Huang et al. 2026), a channel motivated by post-main-sequence survival and the compact debris discs of metal-polluted ones.

6.2. *Priorities for the next release*

Four changes would enlarge what this experiment can say, ranked by how much signal space they buy.

First, and by a wide margin: a branch preserving phase-steady carriers. The per-channel temporal-median subtraction that suppresses stationary ground-based interference also removes a stable narrowband beacon – a canonical technosignature morphology – at every amplitude injected: a whole signal class the pipeline cannot see. An interferometer need not subtract the temporal median to reject terrestrial interference: sky localisation, baseline-dependent phase, delay and fringe-rate signatures and multi-block consistency each separate a celestial source from a ground-fixed emitter without touching time dependence.

Second, use visibilities rather than extracted spectra. A genuine source at the star should show on every baseline the geometric phase behaviour $V_{ij}(\nu, t) = A(\nu, t) \exp[-2\pi i(u_{ij}l + v_{ij}m)]$ expected at that sky position, and a likelihood ratio between a celestial point source there, a common-mode terrestrial interferer and image-plane noise is strictly more powerful than ranking a peak against a control ring – the distinctive opportunity of ALMA SETI, unavailable to single-dish or beamformed searches.

Third, enlarge the control ensemble for interesting windows. The 1/513 rank floor (§5.4) cannot certify significance across a census of thousands of windows. A two-stage scheme – 512 controls routinely, then 10^4–10^5 local null evaluations for any window that passes – would settle candidates statistically rather than hand each to astrophysical vetting.

Fourth, extend the injection campaign and the polarisation axis. The completeness curve should be measured per band, channel width, integration time, antenna count and primary-beam position rather than transferred by construction, and pushed past 10σ until it asymptotes, with EIRP_{90} unlocated. The analysis is also Stokes I only; a polarised carrier discriminates against weakly polarised astrophysical backgrounds, and polarisation-resolved reprocessing costs no sensitivity.

A boundary on drift linearity. The search assumes drift linear over a track. Curvature matters when $\frac{1}{2}\ddot{\nu}T^2$ exceeds half a native channel; for a circular orbit of period P this is $P \lesssim 2\pi(\nu/c)\, a_{\max} T^2/\Delta\nu_{\mathrm{ch}}$, i.e. $P \lesssim 1.6$ d for a typical fine-channel window (230 GHz, 488 kHz channels, $T \approx 2000$ s) and about an hour for a coarse-channel one. Only the shortest-period orbits reach it: TRAPPIST-1 b, at 1.51 d, sits just inside the boundary and is also the case exceeding the drift ceiling. Elsewhere the constant-drift model is adequate, and the chirp-search extension (Appendix G) covers the remainder.

What this survey does not constrain. *Nothing in this paper bounds the prevalence of:* (i) *phase-steady carriers* – a stable beacon dwelling in one channel for more than roughly 40 per cent of a track is removed by the per-channel temporal-median subtraction, by construction and at every amplitude tested (§4.4); (ii) transmitters emitting *outside the searched spectral windows*, which are 33 narrow islands totalling 93.1 GHz (Fig. 2), not ALMA's frequency range; (iii) transmitters active *outside the archival epochs* observed – the limits assume continuous emission and are already weak at duty cycle 0.5 and vacuous below ∼0.1 (Table 10); (iv) drift rates *above* the searched ceiling $|a| = 3.6\,\mathrm{m\,s^{-2}}$, which excludes transmitters on the most close-in orbits, including a TRAPPIST-1 b-like one (Fig. 14); (v) carriers deliberately placed *inside* the $\pm 50\,\mathrm{km\,s^{-1}}$ molecular-line mask, which is excluded search space rather than searched-and-empty space; (vi) emission substantially *broader* than one native channel, for which the quoted thresholds must be rescaled; (vii) anything below the nominal threshold, at which measured recovery is 42^{+16}_{-14} per cent, not 95 per cent.

---

7. CONCLUSIONS

We have described the design, methodology and processed release of a technosignature search of the 168 stars within 40 pc with public ALMA archival coverage: a complete census of the ALMA-covered subset, with no astrophysical selection of its own and every entry independently audited for position and identity against SIMBAD and *Gaia* (§5.3), but covering only ∼1% of the catalogued stars there and inheriting the archive's non-uniform targeting, so the contrast with interest-selected samples (§3) is one of degree, not kind. It processes 107 of 208 planned target/bands: 88 of the 168 census stars, 82 independent systems.

The release yields $\mathrm{EIRP}_{5\sigma}$ limits of 1.6×10^{13}–2.8×10^{17} W (unchanged if the primary-beam-affected Sirius B datasets are also excluded) and forty non-detections in the ancillary continuum screen of unquantified statistical power (median

5σ limit 0.52 mJy). No credible technosignature was identified: nothing of the searched morphology rose above the nominal instrumental threshold during the analysed intervals. Of the 4 windows with a control-ring exceedance, 3 are attributed to CO – two towards β Pictoris, one towards HD 48370, the last independently reported by Cataldi et al. (2023) – and 1 towards CP−72 2713 is a marginal crossing consistent with the survey's own quantified false-flag rate, against 0.78 expected over the non-β Pictoris windows ($P(\geq 1) = 54$ per cent). None survives vetting, so this release reports 0 candidates. The operative criterion, the symmetric single-position statistic, is applied uniformly to all 431 windows (Figure 11).

The result is a constraint, not a bare non-detection: the searched windows cover 87.0 GHz of unique effective frequency space after molecular-line masking, at nominal thresholds constraining transmitters of the searched spectral and drift morphology during those intervals, read under the boxed rules of §4 and §6. These limits are not narrow-band parity with Hz-resolution centimetre-wave surveys: the sensitivity comes from proximity, not spectral resolution (Figure 2); the contribution is bounding total transmitter power for the nearest stars in a virtually unexplored regime. Epoch coverage bounds it likewise: with one to three execution blocks per star (median 46 minutes on source), transmitters active only outside the sampled epochs are unconstrained. Conditional on a transmitter occupying the frequencies already observed toward its own host and emitting throughout the archival epochs, zero credible detections bound the in-band fraction of the 59 systems the pilot recovery model applies to at < 6.2 per cent for EIRP $\gtrsim 10^{16}$ W, weakening to 12.3 per cent at duty cycle 0.5 and 62 per cent at 0.1 (idealised brackets: 3.6 per cent at unit recovery, 8.6 per cent at a uniform 0.42). Drop the frequency condition – adopt any transmitter prior spread across ALMA's tuning range – and the bound vanishes entirely. It is a pilot-grade worked example, not a prevalence measurement.

The release includes the first reported ALMA technosignature searches of Barnard's Star and Wolf 359 (§5.2) and, to our knowledge, the first technosignature-survey use of a closure-phase diagnostic – demonstrated but with no discriminating power at these flux densities – a population-level spectral-type-normalised continuum anomaly check, a chirped-drift and periodicity extension re-using retained spectra, and a cross-target frequency-occupancy check for sky-frequency-recurrent interference, all four shipping with the pipeline at the two grades of Table 15, gaining power as the survey proceeds. A serendipitous molecular-line catalogue and the exoplanet-host and white-dwarf subsamples are census by-products. This release is a fixed, citable partial result: the pipeline is frozen and every number tied to a named snapshot, so the completed survey will supersede it row by row without invalidating it. The remaining 208−107 planned datasets are queued behind archive retrieval and reprocessing; the full-sample paper follows once they are searched, and we anticipate extending beyond 40 pc once the 40 pc sample is complete.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 102 execution blocks searched (project codes and epochs in Appendix A) should be cited from that list in the standard JAO.ALMA#XXXX.X.NNNNN.S form. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), MOST and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia*, processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

COMPETING INTERESTS

G.J.W. declares no competing interests. R.D. is an employee of VBRL Holdings Inc., a private software company with no astronomy, archival-tooling or SETI-related product line; the collaboration arose independently of that employment, and no proprietary VBRL tooling was used at any stage of the pipeline; VBRL has no financial or intellectual stake in ALMA archival technosignature work, provided no funding for this study, and had no role in its design, analysis, or decision to publish. The authors therefore declare no competing interest.

AUTHOR CONTRIBUTIONS (CREDIT)

Glenn J. White: Conceptualisation, Methodology, Supervision, Writing – review & editing.
Robin Dey: Software, Investigation, Formal analysis, Data curation, Visualisation, Writing – original draft.

DATA AVAILABILITY

The reduced data products underlying this paper, with the full target-selection and analysis pipeline, are at `https://github.com/Tilanthi/SETI`, live at submission. All numerical values here come from the *submitted-analysis snapshot*: the repository commit tagged at submission freezes pipeline code and input data as run, is retained unchanged, and reproduces every number, table and figure via the regeneration scripts shipped with it. The *public release* is built from that snapshot with the two pipeline defects of §5.3 corrected and the affected datasets reprocessed *before* freezing, so it carries one set of products, not a submitted/corrected pair. Those datasets are exactly those excluded from this paper's results by §5.3 (the Giclas 9−38 pair and the two defective Band 6 windows); the Sirius B and ε Eri holography-beam recomputation committed in §4 joins the same freeze, its outcome bounded by the systematic quoted there. No value printed here is silently altered: a change-log names every affected product, table and row with its reason. The machine-readable per-window table carries archive-consistent execution times: where the printed integration-time audit differs (the CD−38 10980 row is optimistic by 7%, Appendix Q), it gives the archive's. Public-release per-window hit lists are uncapped (the interim product capped them at 50 frequencies; Appendix H).

TABLE 11

Execution blocks searched in this release, ordered by epoch. Epochs (UTC start, to the minute) and proposal codes are from the ALMA Science Archive `ivoa.obscore` service at `almascience.eso.org`, queried by execution-block identifier on 2026 September 8; MJD is the archive `t_min`.

Execution block	Proposal	UTC date	MJD
uid://A002/X6f1341/X1484	2012.1.00142.S	2013-10-06 06:23	56571.26616
uid://A002/X74deb6/Xff1	2012.1.00268.S	2013-12-02 07:15	56628.30252
uid://A002/X877e41/X14c1	2012.1.00993.S	2014-05-22 11:24	56799.47525
uid://A002/X899169/X1838	2012.1.00198.S	2014-08-18 04:01	56887.16777
uid://A002/X99c183/X4f1e	2013.1.00645.S	2015-01-17 00:35	57039.02475
uid://A002/X9f2ff8/X25b0	2013.1.01214.S	2015-04-26 12:19	57138.51354
uid://A002/Xb0be8b/X785f	2015.1.00307.S	2016-03-22 23:25	57469.97599
uid://A002/Xb25e1a/Xed74	2015.1.00783.S	2016-05-01 17:17	57509.72060
uid://A002/Xb2e94e/X361a	2015.1.00307.S	2016-05-13 04:19	57521.18017
uid://A002/Xb39557/X3789	2015.1.00307.S	2016-05-26 11:30	57534.47920
uid://A002/Xb9a2bb/X1899	2016.1.00817.S	2016-10-22 02:07	57683.08829
uid://A002/Xb9cc97/X150a	2016.1.00803.S	2016-10-25 02:14	57686.09356
uid://A002/Xba3ea7/X2ab4	2016.1.00817.S	2016-11-06 20:01	57698.83471
uid://A002/Xbe7502/X2047	2016.1.00637.S	2017-03-28 05:22	57840.22390
uid://A002/Xc067f7/X822	2016.2.00015.S	2017-05-17 01:51	57890.07740
uid://A002/Xc19d6f/X19ce	2016.2.00015.S	2017-07-02 08:43	57936.36365
uid://A002/Xc6141c/X3af0	2017.1.00786.S	2017-10-27 03:04	58053.12832
uid://A002/Xc66ce1/Xaa9	2017.1.00786.S	2017-11-04 03:17	58061.13698
uid://A002/Xc74b5b/X7c16	2017.1.01644.S	2017-11-29 07:29	58086.31237
uid://A002/Xc8b2b0/X9c35	2017.1.01644.S	2018-01-05 00:25	58123.01783
uid://A002/Xc9831a/X443	2017.1.00986.S	2018-01-22 20:39	58140.86104
uid://A002/Xc99ad7/X348	2017.1.00698.S	2018-01-24 23:51	58142.99391
uid://A002/Xcbb583/Xb78	2017.1.00704.S	2018-04-06 23:22	58214.97419
uid://A002/Xcc626d/X579	2017.1.00698.S	2018-04-21 21:13	58229.88426
uid://A002/Xcccc19/X5564	2017.1.00215.S	2018-05-03 11:42	58241.48813
uid://A002/Xcd97a1/X3f1e	2017.1.01583.S	2018-05-19 07:56	58257.33086
uid://A002/Xcda49e/Xdebb	2017.1.00561.S	2018-05-21 09:50	58259.41000
uid://A002/Xcf0c6b/X6513	2017.1.00167.S	2018-06-27 10:14	58296.42698
uid://A002/Xcf196f/Xabc5	2017.1.00200.S	2018-06-29 09:32	58298.39773
uid://A002/Xd0fb35/X90dc	2017.1.00698.S	2018-08-22 15:32	58352.64755
uid://A002/Xd16a1e/X8d4	2017.1.01644.S	2018-08-31 22:26	58361.93490
uid://A002/Xd248b5/X5186	2017.1.00698.S	2018-09-22 11:12	58383.46702
uid://A002/Xd28a9e/Xbcdf	2017.1.00698.S	2018-09-28 10:57	58389.45629
uid://A002/Xd6ebab/X41a3	2018.1.01149.S	2018-12-20 09:49	58472.40910
uid://A002/Xd86f11/X584b	2018.1.01833.S	2019-01-20 19:05	58503.79517
uid://A002/Xd95042/X2d99	2018.1.01149.S	2019-03-09 06:13	58551.25945
uid://A002/Xd9668b/X3a90	2018.1.00072.S	2019-03-12 00:14	58554.01019
uid://A002/Xd98580/X3bf6	2018.1.01833.S	2019-03-15 20:39	58557.86052
uid://A002/Xd9d7c7/X6fd2	2018.1.01833.S	2019-03-21 15:03	58563.62709
uid://A002/Xda2c82/X765d	2018.1.01833.S	2019-03-28 21:57	58570.91480
uid://A002/Xdc621b/X2bb0	2018.1.01833.S	2019-05-20 04:28	58623.18665
uid://A002/Xe5808e/X80a1	2019.1.01681.S	2019-12-20 17:03	58837.71056
uid://A002/Xf5d76d/X32f1	2021.1.00485.S	2022-03-02 23:14	59640.96830
uid://A002/Xfdb69b/X667a	2021.1.01209.S	2022-08-30 16:22	59821.68235
uid://A002/Xfdf45f/X478e	2021.1.01209.S	2022-09-05 02:32	59827.10618
uid://A002/Xfeb5e6/X13ee	2021.1.01209.S	2022-09-22 01:39	59844.06896
uid://A002/Xff294f/X2db6	2022.1.00338.L	2022-10-04 04:49	59856.20118
uid://A002/X106a1f1/X6a0c	2022.1.01163.S	2023-05-06 03:35	60070.14972
uid://A002/X107fa96/Xce38	2022.1.01163.S	2023-06-02 02:51	60097.11926
uid://A002/X10a341d/X82d2	2022.1.00134.S	2023-07-17 18:28	60142.76964
uid://A002/X10b0d7d/X323f	2022.1.00134.S	2023-08-06 09:20	60162.38930
uid://A002/X11e10bc/X12662	2023.1.00390.S	2024-09-28 16:03	60581.66907
uid://A002/X1207c39/X9417	2024.1.00722.S	2024-11-15 08:17	60629.34515
uid://A002/X12277ed/X22e53	2024.1.00165.S	2024-12-22 10:09	60666.42301
uid://A002/X1237e23/Xdb0	2024.1.01095.S	2025-01-06 18:51	60681.78602
uid://A002/X12925a3/Xa4b1	2024.1.01095.S	2025-05-17 08:00	60812.33402
uid://A002/X12a8fc8/X10b8	2024.A.00040.S	2025-06-11 08:49	60837.36781

The complete release will also be deposited on Zenodo, with a DOI minted on final acceptance and inserted here; it comprises (i) per-window search products for all 431 windows (spectral extracts, per-drift-trial peak statistics, the machine-readable line-exclusion mask of §5.4); (ii) control-ring calibration products at all 512 positions; (iii) closure-phase and population-anomaly outputs; (iv) the full injection-recovery campaign products (all 1200 trials of Appendix E, including the 24-tone pilot); (v) the frozen pipeline code and input data snapshot; (vi) the audit tables of Appendix Q, machine-readable; (vii) the molecular-line catalogue of Appendix N; (viii) the proper-motion-matched execution-block availability census of §6 with its per-star ledger. Data products carry the Creative Commons Attribution 4.0 International licence (CC BY 4.0), pipeline code the MIT licence. Raw ALMA visibility data are public from the ALMA Science Archive at `https://almascience.org`.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; calibration, imaging and the local reprocessing of §§5.4–5.5), Python 3 with astropy, numpy and scipy (spectral extraction and statistics), and the ALMA Science Archive and SIMBAD TAP interfaces; all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

EXECUTION-BLOCK METADATA

Table 11 lists the 57 execution blocks for which archive metadata had been recovered when the search products were frozen, out of the 102 searched in this release; the metadata for the remainder are being retrieved and accompany the machine-readable release. They were recovered by execution-block identifier after freezing and are not part of the frozen products. The per-window linkage between execution block and searched window, with per-observation antenna configurations, synthesised beam sizes and polarisation setup, will accompany the machine-readable table of the corrected tagged release (Data Availability). The epochs span 2013 October 6 to 2025 June 11 (MJD 56571–60837) across 36 unique proposal codes; the AU Mic crossing window of §5.4 comes from the 2014 August 18 execution block of project 2012.1.00198.S. One further public execution block of that project (uid://A002/X7d80c7/X15c6, executed 2014 June 5, 1633 s of science exposure, public since 2015 April 8) also covers the candidate 230.83 GHz window but was not processed; it is the second epoch of the recurrence test of §5.4.

GLOSSARY OF SURVEY VOCABULARY

Quantitative definitions are in the sections cited.

FREQUENCY CONVENTIONS, LINEWIDTH READING, AND SMEARING

REFERENCE FRAMES

The line-exclusion mask's chain – topocentric → barycentric (Earth ephemeris at the execution-block mid-time) → stellar rest frame (catalogued systemic radial velocity; JPL catalogue rest frequencies, Appendix N) – has as residual error the catalogue radial-velocity uncertainty alone (at most a few $\mathrm{km\,s^{-1}}$ within 40 pc); second-order relativistic terms at the $50\,\mathrm{km\,s^{-1}}$ scale are $\lesssim 5$ kHz at 350 GHz, below the finest native channel searched (15.3 kHz). The β Pictoris and AU Mic kinematic attributions (§5.4, Fig. 7) use the same chain, so their offsets are comparable to the exclusion tolerance. Enlarging the drift-rate grid needs no further frame machinery: the searched rates are topocentric and depend only on elapsed time and channel width, not sky position.

TABLE 12
Survey vocabulary.

Term	Meaning in this paper
Census	every star within 40 pc with public ALMA archival coverage (168 entries; §3)
Bycatch	census stars observed by ALMA for unrelated science goals
Target/band dataset	one star in one ALMA band; the unit of the 208-dataset plan (107 complete)
Spectral window	one correlator spw of one execution block, after deduplication; the unit of search and bookkeeping (431)
Execution block (EB)	one contiguous ALMA observing session (102 searched; Appendix A)
MOUS	member observation unit set: the ALMA data hierarchy level at which pipeline products are delivered (one calibration recipe per MOUS)
QA2	the ALMA observatory pipeline's second-stage quality-assurance calibration and scoring, whose products this survey consumes
`obscore`	the IVOA table protocol of the ALMA Science Archive metadata service from which coverage and EB metadata are queried
Native channel	the correlator channel width of the window, 15.3 kHz–31.25 MHz here
Threshold crossing ("hit")	a $\geq 5\sigma$ spectral feature at the star's position in a searched window
Candidate	a control-ring exceedance that survives every astrophysical and instrumental veto (0 in this release)
Control ring	512 off-source test positions per window, used for empirical false-alarm calibration
Star-beats-ring	a window whose star statistic exceeds all 512 control statistics (5/431)
Drift trial	one trial Doppler drift rate of the stack grid; 2–1944 per window
Nominal threshold	noise-derived 5σ detection threshold, *not* a calibrated completeness limit (boxed rule, §4)
$\mathrm{EIRP}_{5\sigma}$	equivalent isotropic radiated power at the nominal threshold; $4\pi d^2 S_{\min}\Delta\nu_{\mathrm{ch}}$
t_{on} / T_{span}	summed on-source integration / elapsed block span per star (Table 22)
Molecular-line exclusion mask	the $\pm 50\,\mathrm{km\,s^{-1}}$ stellar-frame region around catalogued transitions where crossings are excluded (§4.2)
Closure phase	interferometric phase triplet test discriminating point-source structure (Appendix F)
Stellar frame	topocentric $\rightarrow$ barycentric $\rightarrow$ systemic-RV frame chain (§4.2)
Funnel	the candidate-narrowing cascade from channels searched to vetted candidates (Table 4)

CHANNEL WIDTHS AND THEIR VERIFICATION

Channel widths come from the measurement sets' own spectral window tables (CHAN_WIDTH). For the local reprocessing set we verified these are the *effective* resolution: the tables carry no EFFECTIVE_BANDWIDTH column (no online spectral averaging) and the full-resolution windows are named FULL_RES, so the 15.625-MHz coarse and 488.28125-kHz fine channelisations are the noise bandwidths the thresholds mean.

LINEWIDTH CONVENTION

The detection statistic is the flux density in a single channel after drift correction, so the quoted $\mathrm{EIRP}_{5\sigma}$ applies to any transmitter narrower than the native channel, flat in the assumed linewidth. Two sub-channel qualifications: flatness is exact for a tone centred in a channel (a boundary-straddling tone splits its power between two, at worst a factor of two; measured in Appendix E); and the transition between adjacent channels follows the correlator's passband shape (Hanning weighting or spectral averaging included), which varies with correlator generation and is not retained in the frozen products, making the flat-in-linewidth statement a channel-centred idealisation accurate at the tens-of-per-cent level. A robustness test with injected tones at randomised sub-channel positions is committed with the validation items of §6. For the deepest window (UV Ceti Band 3) the quoted limit is 1.6×10^{13} W at a 15.625-MHz channel and the minimum detectable spectral power $1.0 \times 10^{6}\,\mathrm{W\,Hz^{-1}}$: a 1-Hz carrier there needs the full 1.6×10^{13} W. Channelised at 1 Hz from the start, the same observation would give $P_{\min} \propto \sqrt{\Delta\nu}$, falling to 4×10^{9} W; archival data cannot be re-channelised below native resolution, so the ratio $\sqrt{\Delta\nu_{\mathrm{ch}}/\Delta\nu'}$ ($\sim$4000 here) is the unavoidable price of archival channelisation.

Worked example: from flux to EIRP. Take UV Ceti (G 272-61B) Band 3. The searched window's per-channel rms is 0.244 mJy, so $S_{\min} = 5 \times 0.244 = 1.22$ mJy in one 15.625-MHz native channel. At 2.67 pc that is $1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$, and $\mathrm{EIRP}_{5\sigma} = 4\pi d^2 S_{\min}\Delta\nu = 4\pi(2.67\,\mathrm{pc})^2 \times 1.22 \times 10^{-29}\,\mathrm{W\,m^{-2}\,Hz^{-1}} \times 15.625\,\mathrm{MHz} = 1.6 \times 10^{13}$ W. A wider transmitter needs $S_{\min}$ per channel.

INTRA-INTEGRATION SMEARING

The frequency excursion at the maximum searched drift rate across one integration, $\dot{\nu}_{\max} t_{\mathrm{int}}$, is evaluated for every window: at most 1.10 channels across the 431 windows (median 0.001, 90th percentile 0.06). A linearly drifting signal smeared by Δ channels within an integration retains a fraction $\eta_{\mathrm{smear}} = \mathrm{sinc}(\pi\Delta/2)$ of its peak-channel amplitude, so a worst-drift transmitter's effective threshold is $\mathrm{EIRP}_{5\sigma}/\eta_{\mathrm{smear}}$; the three affected windows of 431 are those of Table 13, the median window essentially unsmeared. A signal drifting midway between adjacent trial rates accumulates at most 0.03 channels of residual drift across the grid as realised, so no significant sensitivity is lost between trial rates.

TABLE 13

The only windows with non-negligible intra-integration smearing at the maximum searched drift rate ($\eta_{\rm smear} < 0.99$; the other 222 have $\eta_{\rm smear} \geq 0.99$, nominal and effective thresholds agreeing to better than 1%). Δ is the excursion in channels across one integration, $t_{\rm int}$ the per-integration on-source time, and $\mathrm{EIRP}_{5\sigma,\rm eff}$ the smeared threshold $\mathrm{EIRP}_{5\sigma}/\eta_{\rm smear}$.

Window	$\Delta\nu_{\rm ch}$ (kHz)	$t_{\rm int}$ (s)	Δ (ch)	$\eta_{\rm smear}$	$\mathrm{EIRP}_{5\sigma,\rm eff}$ (10^{13} W)
β Pic B6	15.3	6.05	1.096	0.574	4.08
η Crv B8	61.0	6.05	0.585	0.865	365.8
η Crv B7	61.0	6.05	0.411	0.932	4.68

TABLE 14

Selected prior technosignature programmes below 10 GHz and the sole prior ALMA search, as cited in the text.

Programme / study	Facility	Scale	Frequency	Note as stated by the cited work
Project Ozma (Drake 1960)	—	2 stars	1.42 GHz	τ Cet and ϵ Eri
Project Phoenix (Backus & Project Phoenix Team 2002)	multi-site	—	—	pioneered multi-site RFI verification
Allen Telescope Array (Tarter 2001)	ATA	—	—	first purpose-built commensal SETI instrument
ATA TRAPPIST-1 (Tusay et al. 2024)	ATA	—	—	—
ATA ISO survey (Sheikh et al. 2025b)	ATA	—	—	—
BL first data release (Enriquez et al. 2017)	GBT	692 stars	1.1–1.9 GHz	EIRP $> 10^{13}$ W
BL 1327-star search (Price et al. 2020)	—	1327 stars	1.10–3.45 GHz	—
BL Earth Transit Zone (Gajjar et al. 2019)	GBT	—	3.95–8.00 GHz	—
Margot et al. (Margot et al. 2023)	GBT	11,680 stars	1.15–1.73 GHz	—
ML vetting (Ma et al. 2023)	GBT	820 stars	—	β-CVAE cuts human-vetting load by $\sim$100×
turboSETI correction (Choza et al. 2024)	—	—	—	S/N mischaracterisation; retrospective corrections
FAST commensal ISOs (Zhang et al. 2020; Li et al. 2026)	FAST	—	—	interstellar-object targets
Occultation timing (Tusay et al. 2025)	—	—	—	eclipsing/occultation strategy
COSMIC (Tremblay et al. 2024)	VLA	—	—	commensal system
MeerKAT dragnet (Czech et al. 2021)	MeerKAT	—	—	—
MWA back-ends (Morrison et al. 2023)	MWA	—	—	—
ALMA stellar bycatch (Mason et al. 2024)	ALMA B3	28 stars	90.642/93.151 GHz	first ALMA technosignature search; $\mathrm{EIRP}_{\rm min} = 6.9 \times 10^{17}$ W (nearest star)

PRIOR LOW-FREQUENCY TECHNOSIGNATURE SURVEYS

Table 14 summarises the surveys cited in §2, limited to what their abstracts state; a dash marks quantities the cited work does not quote. Conventions differ by construction, so the table is context, not a sensitivity ranking (Figure 1).

TABLE 15
Scope of the analyses in this release.

Analysis	Applied to	Validation status	In headline limits?
Primary narrowband search	431 windows / 88 stars (82 systems)	Injection-calibrated for the searched class on one configuration (1200 trials; transfer unmeasured, Appendix E); false-alarm rate empirical (§5.4)	Yes
Closure-phase vetting	4 flagged windows	Simulation-validated; sensitivity-limited on sky (Appendix F)	Disposition only
Injection-recovery validation	6 configurations / 3 stars	This release; scope limits stated (Appendix E)	Completeness only
Chirp/periodicity screen	122 retained spectra	Demonstration-grade (Appendix G)	No
Frequency-occupancy check	366 overlapping star pairs	Executed (Appendix H)	Disposition only

TABLE 16
Selection function of the searched sample against a volume-limited reference census. Reference census: Gaia DR3 sources with $\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$ and $1/\varpi \leq 40$ pc (17566 objects; 8594, 49 per cent, carry a `teff_gspphot` estimate). Sample: the 88 stars with at least one searched window in this release. Distances for the sample are the values used by the search itself and are complete; effective temperatures are available for 51 of the 88 stars (all 87 were matched to the 168-entry ALMA-covered census, 58 of them by exact name; 51 of those matches carry a temperature), so the temperature panel is normalised within each column over classified objects only and $T_{\rm eff}$ is used as a proxy for spectral class. "Ratio" is the sample column fraction divided by the census column fraction, so 1 means the sample tracks the reference population and >1 means over-representation.

Property	Census	Searched	Ratio
Distance (all 88 searched stars; 17566 census objects)			
$d \leq 5$ pc	48 (0.3%)	12 (13.6%)	49.9
5–10 pc	240 (1.4%)	13 (14.8%)	10.8
10–20 pc	2069 (11.8%)	18 (20.5%)	1.7
20–30 pc	5391 (30.7%)	18 (20.5%)	0.7
30–40 pc	9818 (55.9%)	27 (30.7%)	0.5
$T_{\rm eff}$ class[a] (51 searched; 8594 census)			
A (≥ 7500 K)	24 (0.3%)	1 (2.0%)	7.0
F (6000–7500 K)	402 (4.7%)	8 (15.7%)	3.4
G (5300–6000 K)	860 (10.0%)	16 (31.4%)	3.1
K (3900–5300 K)	1400 (16.3%)	5 (9.8%)	0.6
M (< 3900 K)	5908 (68.7%)	21 (41.2%)	0.6
Other axes			
Known planet hosts[b]	20 (11.9%)	18 (20.5%)	1.7
Searched stars / systems	–	88 / 82	–

[a]Bins are temperature bins, not MK types: A ≥ 7500, F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and the earlier-type ratios are correspondingly upper limits. White dwarfs are not separable in this parameterisation and fall wherever their (unreliable) photometric temperature places them.

[b]The Gaia reference census carries no planet information; the census column here is instead the 168-entry ALMA-covered census, of which 20 entries are known planet hosts. The 88 searched stars include 18 hosts of 41 known planets.

SUPPLEMENTARY MATERIAL (ONLINE ONLY)

These online-only appendices are not part of the printed article but are carried in the preprint source, and each – with each relocated figure and table – is cross-referenced where the main text uses it.

Index of the online-only appendices.

INJECTION-RECOVERY SENSITIVITY VALIDATION

The quoted limits assume a 5σ threshold behaves as idealised Gaussian noise predicts. We test that by injecting synthetic narrowband signals of known amplitude into real calibrated visibilities through the extraction code's own

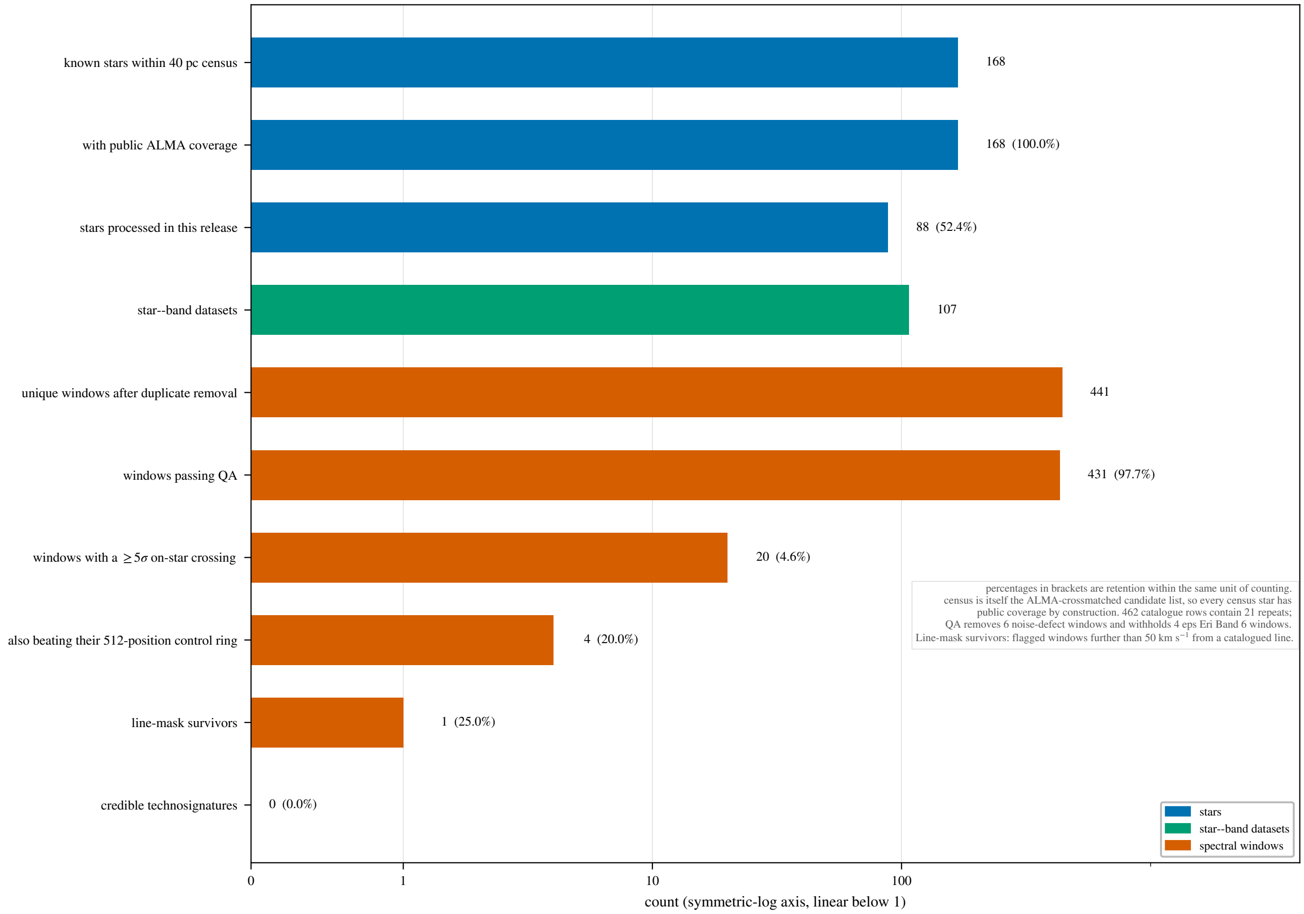

Fig. 10.— The survey's selection funnel, from archive availability to the zero-candidate outcome. Top: star selection. Bottom: archive-availability accounting on the ≤20 pc subset – these ancillary checks are scoped that way because they need dense overlapping coverage between targets, which only the inner sample has, not because they inherit an earlier release; the headline false-alarm calibration of §5.4 is computed on the full 431-window release – quoting its counts – public on-star execution blocks from the proper-motion-matched archive census, the star–execution-block pairs in scope, the window-level records with their correct unit (∼4 spectral-window rows per pair), and the outcome. The searched release spans 102 execution blocks; the difference between availability and scope is the recurrence pool drawn on by the AU Mic second-epoch test (§5.4).

TABLE 17

Consolidated family-wise trials accounting for this release; the bookkeeping ladder is in Appendix P.

Trials factor	Unit	Count
Channels per window (median; maximum)	window	118; 3770
Drift trials per window (median; maximum)	window	4; 1944
Channel × drift trials	release	4.60×10^7
Effectively independent channel cells[b]	release	8.5×10^4–1.1×10^5
Control positions / $N_{\rm eff}$	window	512 / ∼30–43
Windows (star-level tests)	release	431
Windows containing an on-star hit	release	20
Gaussian-expected chance crossings	release	11.4
Cross-target frequency pairs tested	release	946
Star-beats-ring rate $\hat{p}_{\rm sbr}$	window/release	65/431 (15.1%)
Execution blocks spanned	release	102
Principal false alarm: EB-block permutation	release	$P(F{\geq}3) = 6.1\%$ ($E[F]$=1.04)
Expected flags under the symmetric null (N/513)	snapshot	0.84
Wilson-rate bracket on $P(F{\geq}3)$	release	4–19%

[b]From the measured lag-one channel autocorrelation of the drift-tracked statistic, $\rho_{\rm lag1} = 0.989$–0.991.

mechanism – as far upstream of the statistic as retained products allow – and measuring what the unmodified pipeline recovers. An early 24-tone, zero-drift Proxima Cen pilot put 50 per cent recovery near 6σ; the campaign below supersedes it, reproducing that as threshold marginality.

Six configurations span three bands and both resolution classes: three coarse Band 7 spws of AT Mic, one coarse Band 3 and one coarse Band 6 spw of AU Mic, and AU Mic's 488-kHz flagband window (49-trial drift grid). Into each we place a complex tone at the star position at amplitudes 4–10× the statistic's own noise, drift fractions $f_{\rm drift} = 0$, 0.25,

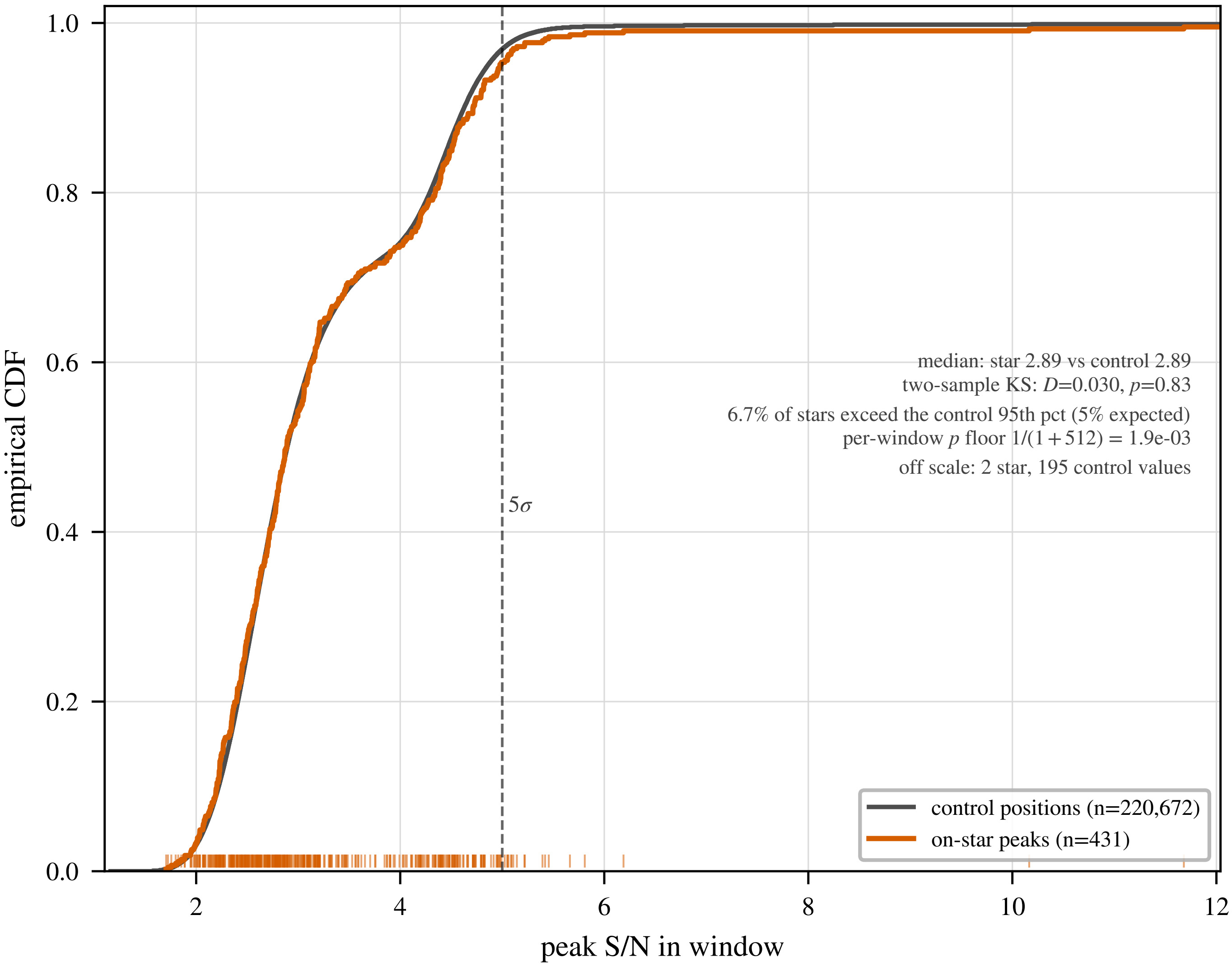

Fig. 11.— The symmetric star-versus-control null on the seven locally reprocessed windows. Each window's p is the add-one rank of $T_\star$ against the 512 control statistics under identical treatment, Uniform(0,1) under the null. All seven lie at $p \geq 0.81$; with $n = 7$ the 95 per cent Kolmogorov band is wide, so the content is the absence of star-side excess, not a precise distribution.

0.5, 0.75, 1.0 of the window's searched ceiling, on- and half-grid rates, and channel-centred and boundary placements: 200 trials per configuration, 1200 in all, under a frozen recovery criterion ($T \geq 5$, peak channel within 3 of the injection, peak drift within one grid step; Fig. 12).

The first result is structural. The per-channel time-median subtraction removing each position's continuum also absorbs any phase-steady carrier dwelling in one channel for more than roughly 40 per cent of the track. On coarse windows this is total – every in-grid drift is sub-channel over the track, so *every* in-grid carrier is static in this sense – and the pipeline-identical variant recovers 0 of 500 injections at every amplitude from 4σ to 10σ. Coarse-window $\mathrm{EIRP}_{5\sigma}$ limits therefore bound transmitters time-variable on track timescales, amplitude-, phase- or duty-cycle-modulated below the $\sim$50 per cent dwell threshold, and say nothing about phase-steady carriers. The fine window measures the same boundary in drift space: carriers with $f_{\mathrm{drift}} \leq 0.25$ dwell $\geq$40 per cent of the track per channel and are absorbed, those with $f_{\mathrm{drift}} \geq 0.5$ surviving to constitute the searched class.

The second result is the completeness curve. On the fine window, drifting carriers ($f_{\mathrm{drift}} \geq 0.5$, pooled over grid phase and sub-channel placement) are recovered 17, 42, 58, 75 and 83 per cent of the time at 4, 5, 6, 8 and 10σ, saturating by twice threshold power (Fig. 4). The machinery alone (tone injected after the subtraction) is amplitude-faithful to within $-9/+14$ per cent for channel-centred tones at every drift fraction; boundary-straddling placement costs a factor 0.6–0.8 in the statistic and recovers 0 per cent at 5σ against 83 per cent for centred ones (67 and 100 per cent at 10σ), its 50-per-cent crossing near 1.4× threshold amplitude and full recovery at $\sim$1.6× threshold power. Survey completeness uses the placement-pooled curve, conservative for the boundary-straddling case. For that class $\mathrm{EIRP}_{50} = 1.10\,\mathrm{EIRP}_{5\sigma}$ (the interpolated half-recovery point, 5.5σ), while $\mathrm{EIRP}_{90} > 2.0\,\mathrm{EIRP}_{5\sigma}$ is a bound only, the curve reaching 83 per cent at the largest amplitude injected. Coarse windows carry no searched-class EIRP_{50} by construction; Table 18 collects the per-class reading. Band 8 windows share the coarse-channel construction, so the static-carrier result holds there by the same mechanism, but their drifting-class completeness is transferred rather than measured (§6). Not

TABLE 18
Per-class reading of the quoted limits. $C_{5\sigma}$ is the measured recovery of the class at the nominal threshold.

Window class	Searched class	$C_{5\sigma}$	EIRP_{50}	Static-carrier recovery
Fine ($\leq$1 MHz)	drifting	0.42	1.10 $\mathrm{EIRP}_{5\sigma}$	0/20 (absorbed)
Coarse (15.625 MHz)	time-variable	not calibrated[a]	...	0/500 (by construction)

[a]The injected tones were all phase-steady on a coarse window (see the appendix text), so recovery of the time-variable modulation defining the coarse searched class is unmeasured. Static carriers are suppressed on both classes.

spanned: other target environments, primary-beam-offset positions, integration durations and edge-phased drift grids; campaigns on Band 8 data and on the β Pictoris windows are infeasible with the measurement sets retained locally.

How far the curve travels. Survey completeness is measured on one configuration – AU Mic's 488-kHz Band 6 window – and applied to all 118 fine windows, spanning Bands 3–8, the release's full range of on-source time and drift-grid density (Table 3) and a range of primary-beam offsets. On physical grounds the factors differing between windows – achieved noise, integration time, drift-grid density – should move the curve's *position* along the amplitude axis, the statistic already being in units of each window's own noise and the trial count setting how much of the drift axis is sampled; its *shape*, set by the temporal-median subtraction interacting with sub-channel drift and placement phase, should be the more portable part. Neither expectation is measured, so the transfer is an assumption whose size is *not* contained in the quoted 42^{+16}_{-14} per cent interval, the binomial spread of one configuration's trials and nothing more; bounding it needs the multi-configuration campaign committed in §6.2.

CLOSURE-PHASE POINT-SOURCE VETTING: SUMMARY AND FULL SPECIFICATION

Threshold crossings face a further automated test, one single-dish and beamformed surveys lack because it needs multiple independent baselines. At these flux densities it has no discriminating power, so a flagged window's formal point-source consistency carries no evidential weight (§4.4).

The closure phase $\arg(V_{ij}) + \arg(V_{jk}) - \arg(V_{ik})$ around a baseline triangle i-j-k vanishes for an unresolved point source anywhere on sky – its visibility phase is linear in (u, v) and cancels around any loop – and is immune to per-station calibration errors. It asks only whether the signal is a coherent point source at all: a coherent far-sidelobe interferer (satellite, aircraft) would itself produce near-zero closure phase and pass.

Consistency with zero is scored by a z-score, the mean drift-tracked closure phase over its standard error across triangles, with three properties handled explicitly. Triangles of one observation are not independent – any two sharing a baseline share its thermal noise – so independent closure quantities are fewer than triangles (of order 10^4 for a 40-antenna configuration) and are not treated as independent trials. Closure phases wrap at $\pm\pi$, so the statistic is circular (unit-phasor averaging); its null is calibrated empirically on the observation's own integrations rather than assumed Gaussian, absorbing the shared-baseline covariance too.

Validation used synthetic visibilities: zero closure phase for a noiseless point source, consistency with zero for a noisy one, inconsistent scatter for an RFI-like non-point source. As positive control we drove continuum detections through the machinery at their own frequencies on four fields of the local reprocessing set – AT Mic Band 7 (the known ∼2-arcsec binary, as resolving control), AU Mic Band 6 and Band 3 continuum, and the AU Mic Band 6 frequency 230.8252 GHz – taking triangle closure phases from time- and channel-averaged unflagged cross visibilities with σ_{CP} from per-baseline scatter between integrations. Per-baseline S/N is 0.5–0.7 at the median and never exceeds 5.1 (0.02 per cent of triangles have all three baselines above S/N = 5), so every field's closure phases are indistinguishable from noise – including the resolved binary, whose ∼0.4 mJy total flux sits below the per-baseline noise of these short integrations. The test triggers only on threshold crossing, at negligible cost.

CHIRPED-DRIFT AND PERIODICITY SCREEN: SUMMARY AND FULL SPECIFICATION

Stacking at constant linear drift rate makes the primary search blind by design to a transmitter whose drift rate changes during the observation, and to pulsed or periodically-modulated carriers. We implemented a chirped (quadratic-drift) extension of the same stacking, on a coarser grid of 21 linear-drift × 5 chirp-rate trials, and a per-channel time-domain periodicity search – a per-position FFT along each channel's time axis, its peak power against a robust per-channel noise estimate. Both vet the star's dynamic spectrum against 8 retained control positions, as the primary search does its ring; those 8 are a strict subset of that window's 512-position ring, not an independent ensemble (§4). On the 122 retained spectra, with a 9-member ensemble and coarser grids, this is a demonstration rather than survey-grade, establishing only that the extension runs and behaves correctly under the null, which it does. The exceedance rates (10.7% chirp, 13.1% periodicity, against the ∼11% chance rate of a star topping such an ensemble) are as expected, no instance is treated as a candidate, and no limit derives from it. Extending both to the full 512-position ring, at negligible cost since the spectra are retained, would promote the screen to survey grade, and is deferred to the completed survey.

CROSS-TARGET FREQUENCY-OCCUPANCY CHECK: SUMMARY, KERNEL, COORDINATES, AND SCOPE

This single-pointing archival survey has no ON/OFF cadence against radio-frequency interference; its analogue is the many unrelated stars sharing similar correlator setups, where a narrowband feature recurring at the same observed *sky frequency* is almost certainly RFI or instrumental. We cross-match crossings towards different targets falling within

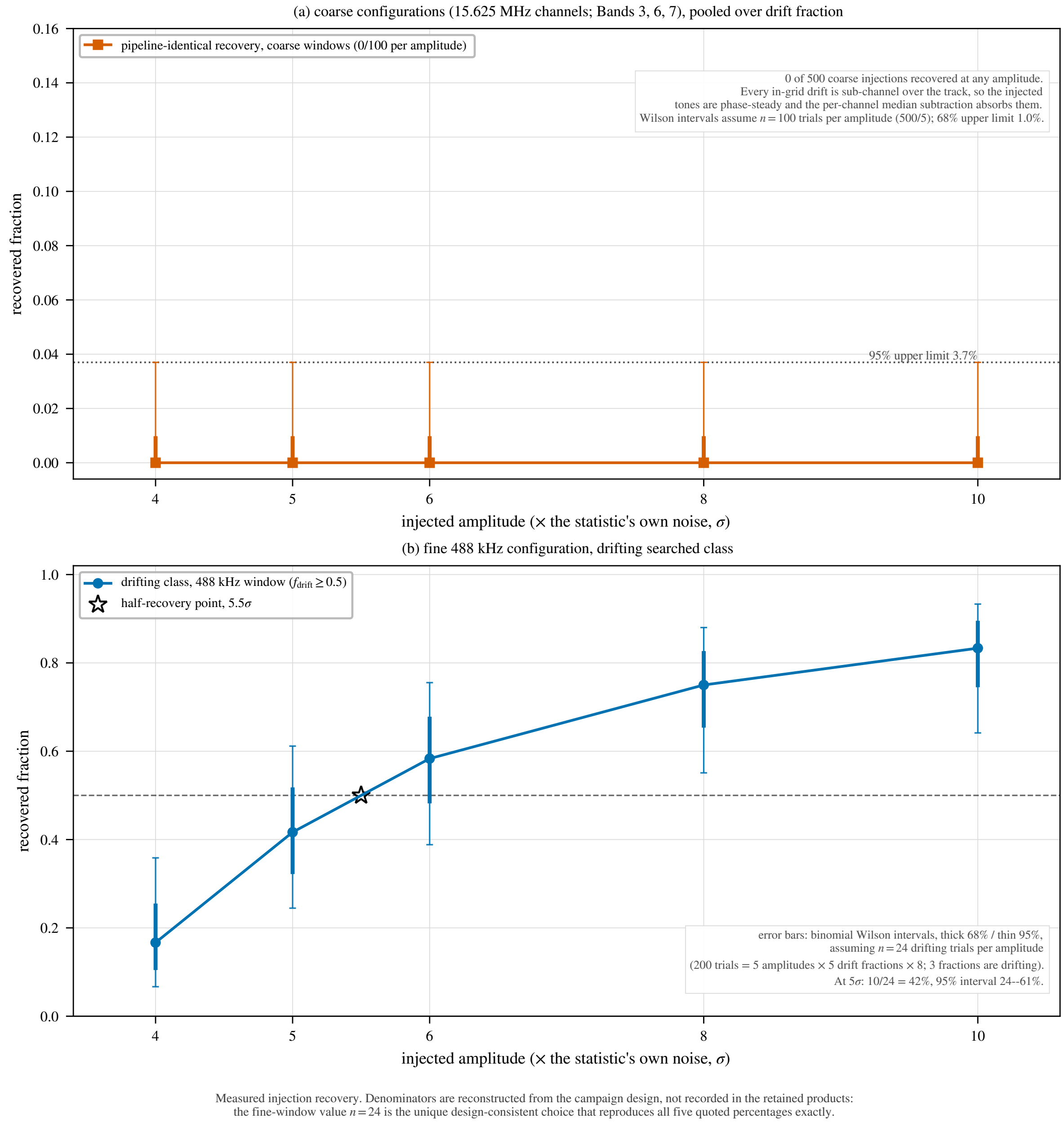

FIG. 12.— Measured injection recovery on real ALMA data (1200 trials, this appendix). Top: the five coarse-channel configurations, pooled over drift fraction and grid phase – the pipeline-identical variant (solid) recovers nothing by construction, leaving only the machinery's detection fraction (dashed) meaningful. Bottom: the fine-channel configuration, where the drift dimension is real and pipeline-identical recovery $C(A, f_{\rm drift})$ rises with amplitude for drifting carriers, static and near-static ones being absorbed. Boundary-straddling tones can fail the frozen criterion's peak-drift clause at high drift while still detected – hence the non-monotonic drift dependence near the ceiling.

$k = 3$ native channels of the coarser width. Across the 431 windows and 107 tabulated crossings, no pair falls within the kernel, or within 0.5 MHz. The kernel (∼46 kHz) is narrower than the β Pictoris line offsets of 10.6 and 21.9 MHz, so those windows were dispositioned by line-catalogue inspection, not this check.

Matching kernel. Across every target/spw result, the check cross-matches both the best-fit-channel frequency (whether or not it formally crossed the threshold) and the list of formal $\geq 5\sigma$ crossings, pairing two crossings at frequencies ν_1, ν_2 towards different targets when $|\nu_1 - \nu_2| < \max(k\,\Delta\nu_1,\ k\,\Delta\nu_2)$. The 5 MHz tolerance of the original formulation corresponds to k between ∼0.2 and ∼330 across this release's native channel widths.

The three frequency coordinates. The *sky frequency* of a hit, reconstructed from archive spectral-window metadata in the observation's recorded frame; the *topocentric frequency*, differing from a barycentric convention by up to the $\pm 30\,\mathrm{km\,s^{-1}}$ barycentric term (∼23 MHz at 230 GHz, far wider than the kernel); and the *correlator channel index* of the native spectral window. Our cross-match probes the first; a frame-convention inconsistency between two observations, or an artefact fixed in the second, would defeat it, so the frame-convention disclosure queued with the symmetric-null recomputation (§5.4) feeds this check too. Artefacts fixed in the correlator frame – birdies, local-oscillator and quantization spurs, bandpass-edge ripple – recur at the same intermediate frequency or channel number, not the same

TABLE 19
Target pairs with the deepest searched-frequency overlap.

Target pair	Overlap (GHz)
γ Lep – γ Vir	14.25
γ Vir – TRAPPIST-1	10.78
γ Lep – TRAPPIST-1	10.77
HD 10647 – HR 1010	10.13
AU Mic – β Pic	8.18
GL 3379 – HD 285968	7.32
GJ 849 – LHS 1140	7.32
TRAPPIST-1 – Wolf 359	7.31
Proxima Cen – Wolf 359	7.31
Proxima Cen – TRAPPIST-1	7.31

TABLE 20
The full crossing funnel: every one of the 20 windows, of 431, containing a $\geq 5\sigma$ crossing at the propagated stellar position, ordered by $T_\star$. 'Ring max' is the largest of that window's 512 control statistics, and a window is control-ring exceedance (bold, four of the 20) when $T_\star$ exceeds it. $\Delta\nu$ and Δv are the offsets of the crossing from the nearest catalogued transition of the eight masked species, topocentric; the $\pm 50\,\mathrm{km\,s^{-1}}$ mask applies in the stellar frame after the conversion chain of Appendix C. ν is the window centre, not the crossing frequency, so it is not $\Delta\nu$ from the transition. A dash means the released products record no nearest masked transition for that window. ALMA J1537−3319 is ALMA J153702653-33192492 in the released table.

Star	Band	ν (GHz)	$T_\star$	ring max	Spatial	Nearest masked transition	$\Delta\nu$ (MHz)	Δv ($\mathrm{km\,s^{-1}}$)
HD 48370	6	230.7305	**27.10**	26.83	**yes**	CO 2→1	−26.33	−34.2
β Pic	3	115.2605	**14.65**	7.27	**yes**	CO 1→0	−10.36	−26.9
β Pic	6	230.5164	**11.68**	9.12	**yes**	CO 2→1	−21.87	−28.4
β Pic	6	230.5439	10.16	15.17	—	CO 2→1	−9.87	−12.8
HD 285968	6	229.9574	6.19	10.85	—	CO 2→1	−35.92	−46.8
CP−72 2713	7	345.1152	**5.81**	5.68	**yes**	CO 3→2	−1526.2	−1326
HD 139084B 805632	7	346.0074	5.66	17.65	—	CO 3→2	28.06	24.3
HIP10679	6	220.4135	5.46	6.76	—	^{13}CO 2→1	19.51	26.5
β Pic	6	241.7688	5.42	5.81	—	CS 5→4	−3184.8	−3949
HD 10647	7	331.2709	5.39	5.93	—	—	—	—
AU Mic	6	230.5384	5.22	5.85	—	CO 2→1	287.2	374
HD 14055	7	345.1377	5.22	5.74	—	CO 3→2	−561.3	−488
HD 170773	7	331.2950	5.14	5.84	—	CO 3→2	−15185.3	−13741
CD−57 1054	7	345.9857	5.11	5.71	—	CO 3→2	713.9	619
ALMA J1537−3319	6	217.0183	5.09	5.63	—	SiO 5→4	486.8	672
ALMA J1537−3319	6	231.0349	5.09	5.67	—	CO 2→1	435.0	564
HD 14055	7	331.3048	5.06	5.84	—	CO 3→2	−15349.4	−13889
HD 48370	8	492.1173	5.05	6.23	—	CO 4→3	30908.1	18829
LHS 1140	6	229.9985	5.05	5.81	—	CO 2→1	97.46	127
ALMA J1537−3319	6	219.2683	5.02	6.28	—	H_2CO	544.6	744

sky frequency. A cross-match on the third coordinate (same channel number, identical correlator setup) is possible with the retained products and also returns zero pairs; a full intermediate-frequency reconstruction, pairing artefacts across tunings, is not recoverable from the retained metadata and stays with the completed survey.

Exposure and exclusions. The null's power is the count of target pairs sharing frequency coverage. On the $\leq$20 pc subset this check has run over, 366 of the 946 star pairs (39%) overlap in searched frequency by at least 0.01 GHz, 359 (38%) by 0.5 GHz and 345 (36%) by 1 GHz; of those, 109 involve one of the five stars carrying formal crossings, so every crossing frequency was tested against genuinely overlapping independent targets (Table 19). Two execution blocks are shared between catalogue entries – the UV Ceti binary's (one field, the G 272-61A/B entries) and the AT Mic pair's (AT Mic B and AT Mic AB; §3) – whose intra-pair "coincidences" are identities by construction, excluded from the cross-target test.

A bookkeeping caveat bounds the null on that subset: the per-window hit lists of the interim retained product are capped at 50 frequencies, so the true count over the subset is 171 crossings against 107 tabulated. The difference is one window, β Pic Band 6 ultra-fine, with 114 formal crossings of which the 50 strongest are tabulated; those 50 span 0.748 MHz, and the four crossings towards HD 285968, the only other star with crossings there, lie 13.7 MHz from that extent but inside the window's full 54-MHz searched span, so the 64 untabulated crossings could not be tested. The null is demonstrated for the tabulated set, unresolved for the remainder – a limitation of the retained product, not of the criterion, which the public release removes by shipping uncapped hit lists (Data Availability).

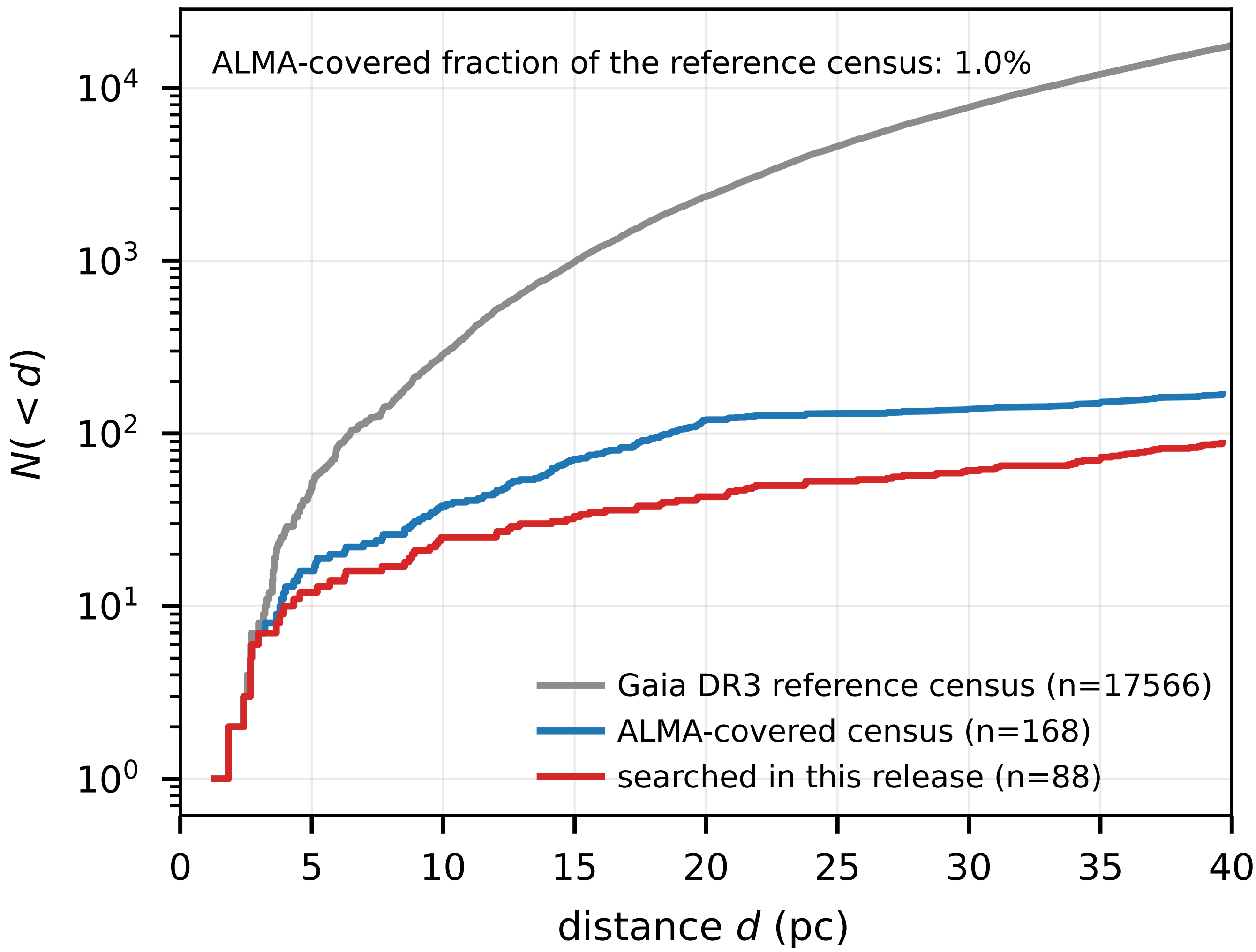

Fig. 13.— The ALMA-covered subset versus the stellar population within 40 pc: cumulative $N(< d)$. Grey: the reference Gaia DR3 catalogue (17 566 stars passing the parallax-quality cut of Table 16); blue: this paper's census sample, ∼1 per cent of the total, of which 88 are searched here. The curves need not track one another: the archive strongly over-represents the nearest stars (§3).

PER-TARGET RESULTS

At this scope a typeset table would run to many pages, so these results – data rather than argument – ship with the manuscript source as `per_target_results_v3.31.csv`, summarised here. One row per extracted spectral window (462 at the snapshot epoch, each with its quality-control status) carries star name and distance, ALMA band and execution block, searched frequency range and native channel width, nominal $\mathrm{EIRP}_{5\sigma}$ threshold, line-excluded continuum rms, stellar and control-ring peak S/N, the count of controls at or above the star, the empirical p-value of §5.4, drift ceiling, on-source time, and nearest catalogued transition with velocity offset.

$\mathrm{EIRP}_{5\sigma}$, the nominal trigger threshold, is the instrumental power at a 5σ spectral amplitude before morphology-dependent recovery is folded in, $\mathrm{EIRP}_{5\sigma} = 4\pi d^2(5\sigma_{\mathrm{comb}})\Delta\nu$, on the same threshold and primary-beam correction as the continuum column, so the row's two limits are comparable. The per-channel flux-density threshold follows as $S_{\mathrm{min}} = \mathrm{EIRP}_{5\sigma}/(4\pi d^2\,\Delta\nu_{\mathrm{ch}})$, the narrowband per-channel noise and *not* the tabulated rms, that target's line-excluded full-bandwidth continuum noise. Bands 6 and 7 supply 83 per cent of the windows, an archive imprint rather than a design choice; Band 8 windows are an order of magnitude less sensitive in EIRP at fixed distance, and Bands 4 and 5 have one star each, so neither has weight in the aggregate limits.

THE VELOCITY-SPACE LINE MASK: JUSTIFICATION AND RETROSPECTIVE APPLICATION

The one-channel line check failed the β Pictoris windows (§5.4): a native channel spans a fraction of a $\mathrm{km\,s^{-1}}$, while circumstellar kinematics shift real emission by many channels. The replacement mask takes its tolerance from the kinematic budget of circumstellar gas – Keplerian motion where these molecules survive ($\gtrsim$10 au) $\lesssim$13 $\mathrm{km\,s^{-1}}$; cold-gas linewidths $\lesssim$1 $\mathrm{km\,s^{-1}}$; the eight masked species cold dense-gas tracers, no bulk-wind term for the dwarfs dominating the sample; the uncorrected barycentric term bounded by $\pm$30 $\mathrm{km\,s^{-1}}$ – and is empirically conservative: the largest attributed circumstellar component (β Pic, -2.7 to -3.6 $\mathrm{km\,s^{-1}}$) sits at 7 per cent of it, and *zero* crossing frequencies lie

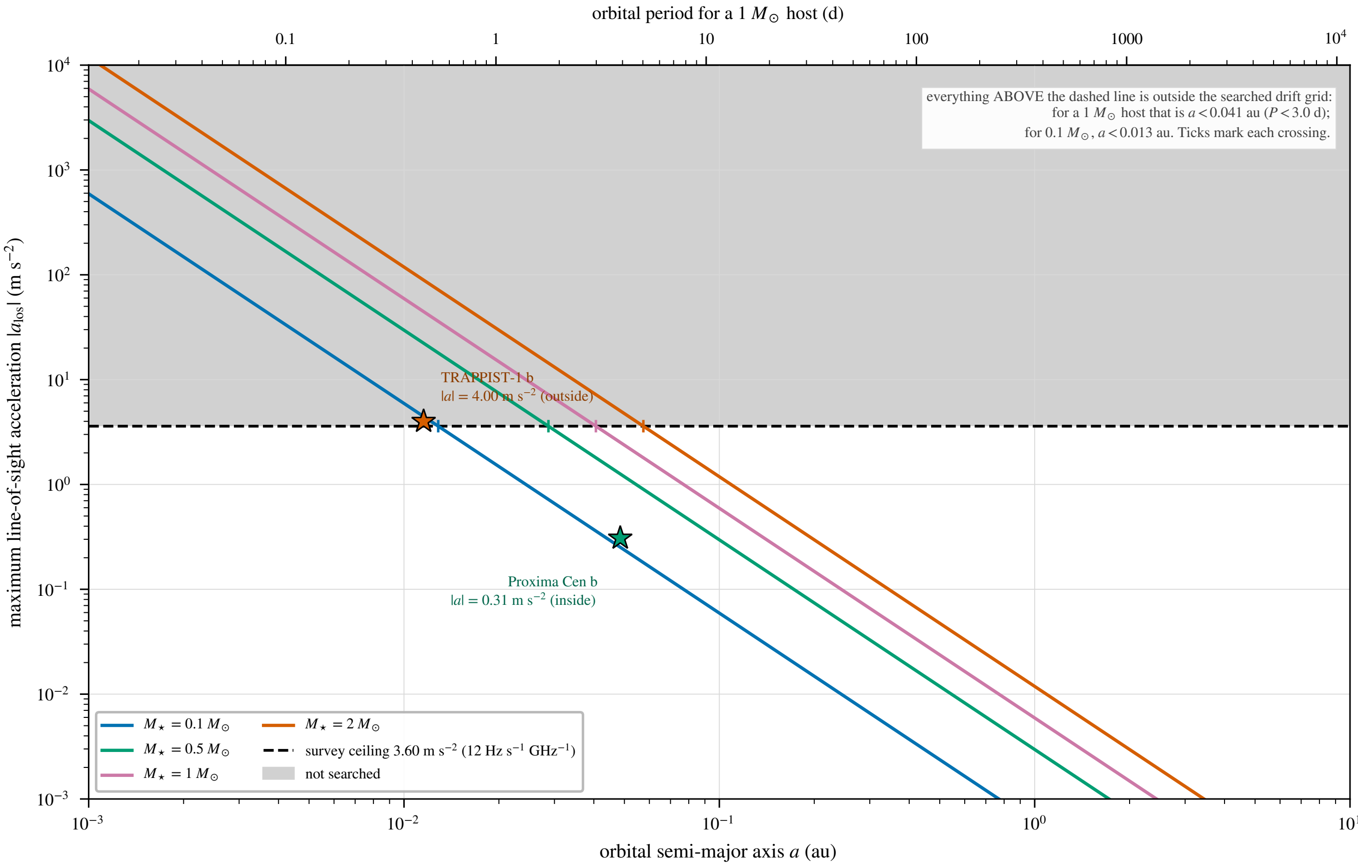

FIG. 14.— The drift-rate ceiling as a physical selection function. Maximum star–planet relative acceleration, edge-on and maximised over orbital phase, as the carrier drift rate it induces (Hz s^{-1} per GHz of carrier frequency), against orbital period, for host-star masses 0.09–1.8 $M_\odot$ (lines) and the known planets of the planet-hosting targets (points). Shaded band: the survey's drift-search ceiling; above it a transmitter lies outside the searched grid. Only TRAPPIST-1 b reaches it (§6).

TABLE 21

PER-BAND SURVEY CHARACTERISATION OF THE 431 SEARCHED WINDOWS. "SYS." COUNTS UNIQUE STELLAR SYSTEMS SEARCHED IN THE BAND, SO A SYSTEM OBSERVED IN TWO BANDS IS COUNTED IN BOTH ROWS (82 SYSTEMS, 88 STARS IN TOTAL); "EBS" THE EXECUTION BLOCKS CONTRIBUTING, ATTRIBUTED BY THE PROVENANCE OF THE DE-DUPLICATED WINDOWS; "WIN." THE SEARCHED WINDOWS; $\Delta\nu_{\rm ch}$ THE NATIVE CHANNEL WIDTH IN MHz (MEDIAN, WITH THE RANGE IN BRACKETS); "UNION" THE UNIQUE SKY FREQUENCY SEARCHED IN THE BAND, AFTER MERGING OVERLAPPING WINDOWS; EIRP$_{5\sigma}$ THE MEDIAN NOMINAL THRESHOLD ACROSS THE BAND'S WINDOWS; $\dot{\nu}_{\rm ceil}$ THE RANGE OF DRIFT-RATE CEILINGS IN Hz s^{-1} PER GHz OF CARRIER FREQUENCY. THE "ALL" ROW IS COMPUTED OVER THE WHOLE SAMPLE, NOT SUMMED DOWN THE COLUMNS: EBS AND SYSTEMS RECUR ACROSS BANDS AND THE FREQUENCY UNIONS ARE DISJOINT BY CONSTRUCTION.

Band	Sys.	EBs	Win.	$\Delta\nu_{\rm ch}$ (MHz)	Union (GHz)	Med. EIRP$_{5\sigma}$ (W)	$\dot{\nu}_{\rm ceil}$
3	8	9	40	15.63 (0.244–15.63)	20.6	1.1×10^{14}	12.0–13.3
4	1	1	4	15.63	6.9	2.6×10^{13}	12.0
5	1	1	4	15.63	6.9	4.4×10^{13}	12.0
6	54	54	216	15.63 (0.015–31.25)	29.4	1.3×10^{15}	12.0–13.3
7	34	34	155	15.63 (0.061–15.63)	22.6	2.9×10^{15}	12.0–13.3
8	3	3	12	15.63 (0.061–15.63)	6.6	6.1×10^{16}	12.0
All	82	102	431	15.63 (0.015–31.25)	93.1	1.4×10^{15}	12.0–13.3

13–50 km s^{-1} from their nearest transition, so a $\pm$13 km s^{-1} Keplerian-only mask would have suppressed and released exactly the same windows: the wider tolerance costs no yield and buys robustness against radial-velocity catalogue error.

Two of the six remaining crossing windows fall inside the mask (the unflagged β Pic coarse window at -3.6 km s^{-1}; HD 285968 at $+8.7$ km s^{-1}, already rejected by its control ring, maximum 10.9 against a star peak of 6.5); the rest lie far outside any circumstellar tolerance (AU Mic +379, LHS 1140 +115 km s^{-1}). Of the 20 windows with an on-star hit, 4 also beat their control ring, three accounted for by the mask. But the transitions that make astrophysical false positives cheap are also natural "Schelling points" for a deliberate communicator (Cocconi & Morrison 1959), so the released products keep a known-line-coincident list – every on-source 5σ crossing inside the masked intervals, with its offset from each nearby transition – for a future search with independent discrimination to revisit.

TABLE 22

Per-star observing duty cycle, ordered by distance, for the subset with compiled epoch accounting. The two spatially unresolved binaries share every observation – separate catalogue rows, single trials – giving 82 independent systems. $N_{\rm win}$: searched windows after deduplication; $N_{\rm EB}$: distinct EBs searched; $t_{\rm on}$: summed on-source integration in minutes, each block's median across its windows; $T_{\rm span}$: summed elapsed span of those blocks in hours, including calibration and overheads, so exceeding $t_{\rm on}$. Not retained by the frozen products, these were reconstructed from the ALMA Science Archive (queried 2026 September 8) and audited against its exposure accounting (Appendix Q), whose two exception groups are carried at archive-consistent exposures. Dates and MJDs are in Appendix A; per-window values accompany the machine-readable table.

Target	$N_{\rm win}$	$N_{\rm EB}$	$t_{\rm on}$ (min)	$T_{\rm span}$ (h)
Proxima Cen	4	1	50.0	1.05
Barnard's Star	4	1	50.0	1.01
Wolf 359	4	1	50.0	1.10
Sirius B	12	3	17.2	0.27
G 272-61B	4	1	31.0	0.62
G 272-61A	4	1	31.0	0.62
CD-23 14742	4	1	50.0	1.10
eps Eri	4	1	42.8	0.89
tau Cet	1	1	23.2	0.47
BD+05 1668	4	1	23.7	0.46
HD 33793	1	1	14.1	0.26
Wolf 28	1	1	48.4	0.98
GJ 674	4	1	50.0	1.04
GL 3379	4	1	45.9	0.91
LSR J1835+3259	4	1	43.1	0.80
HN Lib	4	1	24.2	0.49
GJ 581	4	1	50.0	1.03
lp 876-10	4	1	46.5	0.74
61 Vir	4	1	37.8	0.95
chi01 Ori	4	1	46.4	0.92
GJ 849	4	1	44.4	0.88
gam Lep	8	2	63.7	1.23
HD 285968	4	1	46.9	0.93
AU Mic	12	3	136.1	2.93
AT Mic B	4	1	14.6	0.32
AT Mic AB	4	1	14.6	0.32
gam Vir	8	2	67.7	1.28
HR 1010	8	2	70.1	2.02
TRAPPIST-1	12	3	168.4	2.91
HD 69830	8	2	92.4	2.04
CD-38 10980	4	1	9.6	0.18
LP 349-25	4	1	44.2	0.82
GJ14	4	1	51.1	0.90
LHS 1140	4	1	43.3	0.95
HD 38858	4	1	0.3	0.03
HD 207129	4	1	47.9	1.04
HD 23484	4	1	45.9	0.91
HD 10647	8	3	107.2	1.98
HD 139664	3	1	8.7	0.12
eta Crv	8	2	98.6	2.04
HD 53143	4	1	41.8	0.84
Wolf 219	4	1	1.0	0.02
bet Pic	15	3	93.7	1.81
eta Cru	4	1	30.2	0.59

AU MIC: THE THREE TESTS IN FULL

Localisation. The first-epoch block was reprocessed from the raw science data model with the observatory pipeline's own calibration recipe – retaining 310 integrations against the release pipeline's 249, on a finer drift grid – and the symmetric star-versus-control statistic of §5.4 run on the crossing window: the star does not beat its control ring. The frozen products agree only asymmetrically, the source-region peak exceeding the 512-control maximum by 0.12, below that statistic's window-to-window scatter, and the crossing fails the single-position variant too.

Recurrence. The one further public Band 6 block covering 230.83 GHz, the same programme executed 2014 June 5, was searched identically: no channel within ±10 of the first-epoch flagged channel exceeds 3.1, and nothing is localised to the stellar position. Its band rms (1.48 mJy per 488-kHz channel) matches the first epoch's 1.9 mJy, so a comparable excess would have been recovered, and the frequency recurs at no other target. Intermittent or precisely-scheduled transmissions outside the sampled windows remain unconstrained, a caveat quantified by the duty-cycle analysis of §6.

Statistical scale. The on-source peak (5.97) sits at the 92.8th percentile of the control-maxima distribution over all 431 windows (median 4.55, 90th percentile 5.84; Fig. 15): thirty (7.2 per cent) match or exceed it, add-one empirical

TABLE 23

Masked bandwidth (GHz) by species and ALMA band on the operative mask. Columns B3–B8 and 'Total' are window-summed (denominator: the gross searched bandwidth); 'Union' merges the same intervals across windows and stars (denominator: the 93.1 GHz unique-frequency union). Species rows overlap where two masks share a channel, so they exceed the all-species totals. Per-star intervals are in the released veto-mask export.

Species	B3	B4	B5	B6	B7	B8	Total	Union
CO	0.12	0.00	0.00	1.51	0.86	0.00	2.49	0.33
^{13}CO	0.35	0.05	0.07	1.71	0.51	0.00	2.69	0.74
C^{18}O	0.04	0.00	0.00	0.00	0.00	0.00	0.04	0.04
HCN	0.00	0.00	0.00	0.00	0.12	0.00	0.12	0.12
HCO^{+}	0.10	0.00	0.07	0.00	0.36	0.00	0.53	0.50
CS	0.00	0.00	0.07	0.57	0.11	0.00	0.75	0.29
CN	0.87	0.00	0.00	11.04	2.59	0.00	14.50	2.69
SiO	0.00	0.00	0.00	0.27	0.81	0.00	1.08	0.26
All species (merged)	1.48	0.05	0.21	15.10	5.36	0.00	22.20	4.96

TABLE 24

One row per searched star, ordered by distance; the table is split into two side-by-side blocks that read down the left block and then down the right. d is the distance used by the search; "Bands" lists the ALMA bands in which that star has at least one searched window; $N_{\rm w}$ is the number of searched windows; EIRP$_{5\sigma}$ is the deepest (smallest) nominal 5σ threshold of any of those windows, in W. Star names are the identifiers used in the released machine-readable table (per_target_results_v3.31.csv) and cross-reference it directly. 88 stars, 82 independent systems, 431 searched windows; de-duplicated, noise-defect and withheld windows are excluded here but are present, and labelled, in the released file.

Star	d (pc)	Bands	$N_{\rm w}$	EIRP$_{5\sigma}$	Star	d (pc)	Bands	$N_{\rm w}$	EIRP$_{5\sigma}$
Proxima Cen	1.30	6	4	3.30e+13	V star TX PsA	20.83	7	4	2.75e+14
NAME Barnards star	1.83	6	4	6.15e+13	V star WW PsA	20.87	7	4	2.76e+14
Wolf 359	2.41	6	4	7.84e+13	j1256-1257	21.15	7	4	8.32e+14
alf CMa B	2.67	3,4,5	12	2.34e+13	hd92945	21.51	7	4	2.65e+14
G 272-61B	2.67	3	4	1.62e+13	WD2039-202	21.77	6	4	1.29e+15
G 272-61A	2.72	3	4	1.67e+13	HD 45184	21.89	6	4	3.03e+15
CD-23 14742	2.98	6	4	1.12e+14	LP 476-207 783296	23.75	7	4	4.85e+14
tau Cet	3.65	6	1	3.79e+13	LP 476-207 384128	23.79	7	4	4.73e+14
BD05 1668	3.79	6	4	2.71e+14	HD202628	23.79	6	4	1.77e+14
HD 33793	3.93	6	1	5.62e+14	HD 31392	25.76	6	4	4.85e+15
Wolf 28	4.31	6	1	6.67e+13	CD-57 1054	26.87	7	4	7.63e+14
GJ 674	4.55	6	4	2.54e+14	LP 353-51	27.10	7	4	1.23e+15
GL 3379 Gaia DR3 3316364602541746048	5.21	6	4	1.12e+14	hd107146	27.47	7	4	7.26e+15
LSR J18353259	5.69	3	4	5.12e+13	L 836-122	28.70	7	4	7.11e+14
HN Lib	6.25	6	4	1.49e+14	HD172555	28.79	6	4	1.21e+14
GJ 581	6.30	6	4	6.01e+14	HD161868	29.75	6,7	8	1.88e+15
lp 876-10 Gaia DR3 6623351805412369024	7.68	7	4	2.70e+13	51 Eri	29.91	6	4	1.09e+15
61 Vir	8.53	7	4	4.52e+14	HD 155555C	30.41	7	4	6.91e+14
chi01 Ori	8.70	3	4	1.08e+14	2MASS J05241914-1601153 551040	30.99	7	3	3.62e+15
GJ 849	8.81	6	4	8.30e+14	WD0346-011	31.04	6	4	2.54e+15
gam Lep	8.91	6,7	8	9.27e+13	2MASS J05241914-1601153 717696	31.19	7	3	3.66e+15
HD 285968	9.49	6	4	3.45e+14	UCAC2 20312880	33.77	7	4	1.06e+15
AU Mic	9.71	3,6	12	5.26e+13	HR 2562	33.93	7	4	3.39e+15
V AT Mic B	9.81	7	4	5.60e+13	TWA 7	34.10	7	4	7.39e+14
NAME AT Mic AB Gaia DR3 6792436799475128960	9.92	7	4	5.82e+13	WD 0407179	34.11	6	4	2.76e+15
gam Vir	12.02	6,7	8	2.24e+14	UCAC3 176-23654	34.35	7	4	9.42e+14
HR 1010 Gaia DR3 4722135642226902656	12.04	6,7	8	6.50e+14	GJ 2006B	34.98	7	4	8.04e+14
TRAPPIST-1	12.47	3,6,7	12	5.76e+13	ALMA J153702653-33192492	35.01	6,7	9	3.20e+15
HD 69830	12.57	6,7	8	3.91e+13	GJ 2006A	35.03	7	4	7.95e+14
CD-38 10980	12.91	6	4	4.14e+14	TYC 2211-1309-1	35.42	7	4	9.12e+14
LP 349-25	14.13	3	4	2.57e+14	HD14055	35.74	6,7	8	7.85e+14
GJ14 Gaia DR3 383426372059603712	14.69	6	4	1.21e+14	HD 48370	35.95	6,8	8	2.86e+15
LHS 1140	14.96	6	4	6.24e+14	HD 89452	36.23	6	4	8.54e+15
HD 38858 Gaia DR3 3023711269067191296	15.21	6	5	6.96e+14	HD61005	36.45	6,8	8	7.60e+14
HD 207129 Gaia DR3 6564091190988411520	15.56	6	4	7.62e+14	CP-72 2713	36.72	6,7	8	8.26e+14
HD 23484 Gaia DR3 4856592239127350400	16.17	6	4	1.70e+15	HD170773	36.93	6,7	8	5.47e+14
HD 10647	17.35	6,7	8	1.49e+14	TWA 3A 696000	37.05	6	4	3.13e+14
HD 139664	17.40	6	3	2.55e+14	Barta 161 12	37.28	7	4	1.13e+15
eta Crv	18.24	7,8	8	4.36e+13	HD377	38.40	6	4	1.48e+16
HD53143 Gaia DR3 5479222240596469632	18.34	6	4	2.84e+15	HD 139084B 921024	38.72	7	4	1.71e+15
Wolf 219	18.88	6	4	8.43e+14	HIP 490	38.83	6	4	4.36e+14
bet Pic	19.63	3,6	15	2.34e+13	HD110411	38.92	6	6	4.13e+14
eta Cru	19.69	6	4	1.11e+15	HD 139084B 805632	39.31	7	4	1.75e+15
PM J034331958	20.76	6	4	1.01e+15	HIP10679	39.63	6	4	2.94e+14

$p = 0.074$, before any correction across 431 windows. It matches no catalogued molecular transition (287 MHz from CO $J{=}2{\to}1$); closure-phase vetting finds it point-source consistent, one-sided evidence only (Appendix F).

Two astrophysical contexts are recorded for future re-examination; neither grounds the rejection. AU Mic is an exceptionally active flare star, but the block's own data test that: its broadband continuum sits at a quiescent 0.15 mJy, the two flares of MacGregor et al. (2020) (16.8 and 6.1 mJy, optically thin, GHz-wide) belong to a later epoch, and an 11 mJy narrowband event over it would be a factor ~70 the continuum does not show. A flare origin needs a spectrally narrow coherent process leaving the continuum untouched, which the data neither require nor establish; the same-tuning second-epoch twin shows no burst domination. Second, the controls share the primary-beam footprint, so smooth structure such as AU Mic's debris belt raises star and controls alike and cannot alone produce a localised excess.

THE POPULATION-LEVEL CONTINUUM ANOMALY SCREEN: SUMMARY AND FULL SPECIFICATION

Being a census, not an interest-selected list, the sample supports an exploratory by-product screen: is a star's continuum flux anomalous against its own photospheric expectation (Planck function with Gaia DR3 parameters, Pecaut & Mamajek fallback radius where GSP-Phot is unavailable) and against same-$T_{\rm eff}$ comparisons in coarse bins of at least five? It is no headline constraint: the samples are small (57 measurements: 17 detections, 40 limits; bins of 4–7 stars) and a millimetre excess is at best an indirect, model-dependent observable of waste-heat technology. One outlier results, χ^1 Ori at Band 3 (excess ratio 2.52, largest of the 17 detections; resampled tail probability 0.53 in its six-star bin, a ranked outlier of a small sample, not a significant departure), consistent with a spectroscopic G0V binary's unresolved companion or chromosphere (Appendix M).

Radius sourcing. Where a Gaia GSP-Phot radius is unavailable – close to half the sample by construction – the pipeline substitutes a coarse Pecaut & Mamajek main-sequence $T_{\rm eff}$–radius relation, whose scatter propagates into the excess-ratio distribution; stars can also depart from a mean relation through age, metallicity or unresolved multiplicity. Rerunning with those parameters imposed on every star tests that scatter against the outlier threshold: the sole flag survives, the M and G/F bin scales unchanged (already fallback-dominated), the A/F-hot median moving from 4.3 to 3.9.

Error model. Evaluated before any flag, it is $\sigma^2_{\rm tot} = \sigma^2_{\rm map} + (f_{\rm cal}\, S_\nu)^2 + \sigma^2_{\rm model}$: $\sigma_{\rm map}$ the image rms, $f_{\rm cal}$ the band-dependent absolute-flux systematic (5 per cent in Bands 3–6, 10 per cent in Bands 7–8), $\sigma_{\rm model}$ the photospheric-model uncertainty propagated from the Gaia temperature and radius errors and the main-sequence-relation scatter for fallback stars. The per-target file's $\sigma_{\rm tot}$ column carries the map and calibration terms only; the model term enters the population-level excess-ratio screen, not the tabulated per-star errors. Read excess ratios near unity with both in mind: the model term dominates for fallback stars, the calibration term for the brightest detections.

Binning and the outlier statistic. Excess ratios are compared against similar-$T_{\rm eff}$ stars in coarse equal-width bins through a robust median/MAD outlier statistic, the bin width a fixed pipeline constant published with the release, chosen so populated bins meet the five-comparison-star minimum. At that minimum the bin median rests on two or three values, the MAD scale is small-sample, and repeating the comparison for every star and bin brings a small expected number of chance flags: the screen flags for inspection, not per-star hypothesis tests. Raising the minimum to ten awaits the completed survey, which will reach it in the occupied bins.

Exact status. The screen is live: stellar parameters were ingested for every continuum-measured target (Gaia GSP-Phot for 16), detections binned M/G/F/A-hot (7/6/4 members of the three populated bins), with per-bin medians and log-scatters shipping in the machine-readable release. The one measurement crossing the outlier threshold ($z = 4.1$) is dispositioned above, with the two committed robustness reruns (fallback-only parameters; leave-one-out bin scatter).

CONTINUUM SEARCH (EXPERIMENTAL ANCILLARY ANALYSIS)

The continuum lane is an exploratory by-product, not a co-equal search (Appendix L). Its seventeen detections and forty non-detections are plotted in Figure 8 (right panel); an "upper limit" is 5× the map rms at the star's position, primary-beam corrected like the EIRP thresholds, and the per-target rms of Appendix I rescales it to any confidence level. The deepest non-detection is towards TRAPPIST-1 in Band 3. Under the combined error model sixteen of the seventeen stay above 5σ, the exception (η Corvi Band 7, 4.7σ combined) being marginal. Bright peaks far from the target (ϵ Eri Band 6 at 10.2σ, 18 arcsec off the phase centre; GJ 674 and CD−23 14742 at 10.7 and 9.9σ) fall outside the 3-arcsec tolerance and are correctly recorded as non-detections; the retained products cannot decide whether the ϵ Eri peak is real off-target emission or a primary-beam artefact at that offset (§4).

Every detection is astrophysically unsurprising: the G 272-61A/B pair is the unresolved UV Ceti binary (0.85 mJy, attributable to neither component); six are young active systems with debris discs or chromospheric activity (τ Cet, β Pic B3/B6, AU Mic, AT Mic A/B); six show the photospheric flux expected of their spectral types (γ Lep and γ Vir in both bands, χ^1 Ori, η Cru; η Corvi's Band 8 limit is likewise photosphere-consistent); and two are late-M dwarfs of documented strong magnetic activity, LP 349−25 (10.5σ, 0.071 mJy) and the auroral emitter LSR J1835+3259 (6.7σ, 0.078 mJy; Hallinan et al. 2007), which merits deeper follow-up. Closure phase cannot discriminate structure for any of the seventeen at these flux densities (Appendix F), and the anomaly screen's single outlier is astrophysically dispositioned (Appendix L).

A SERENDIPITOUS MOLECULAR-LINE CATALOGUE

The continuum search's line-exclusion step (§4) flags every $\geq 5\sigma$ outlier channel as a by-product. No existing mm/submm line survey of the solar neighbourhood rests on a distance-limited census of archival coverage, so even a modest catalogue has value beyond its SETI motivation. Twelve of the outlier groups scanned to date are single- or few-channel features with no known transition within 300 MHz, most plausibly statistical; six are plausibly real: CO(J=1→0) and CO(J=2→1) towards β Pictoris (the stellar-frame component of Table 6); a marginal CO(J=2→1)-consistent feature recurring across two epochs towards HD 285968, tentative only (1–2 channels of significance per epoch); and a single-channel, 195-MHz-offset feature near SiO(J=5→4) towards HD 53143, more likely statistical than real. All are reported openly; the machine-readable catalogue (per-group target, band, channel coordinates, significance, nearest transition) ships with the tagged public release; the completed survey's full-sample re-scan will extend it.

LESSONS FOR ARCHIVAL PIPELINES

Three of the four audit items of §5.3 are failure modes generic to automated archive mining. First, every lane needs its own geometric guard: the narrowband step aborts on item (ii)'s pointing mismatch while the continuum-imaging step had none. Second, window-level bookkeeping anomalies yield to cross-target pattern recognition, not per-target sanity checks: item (iii) recurred with matching correlator parameters across two unrelated targets. Third, name resolution is not position resolution: the γ Lupi entry was misidentified through the SIMBAD identifier list, not by positional error – HD 139664 carries the historical main identifier "* g Lup", so a name lookup returns it, not the B2IV star seven times farther off (parallax 7.75 mas, 3.7° away). The cheap check that caught it should be standard: the star's Gaia DR3 parallax, 57.476 mas, recovered digit-for-digit by inverting the pipeline distance, with $T_{\rm eff} \approx 6600$ K and cross-identifiers HR 5825, HIP 76829 and GJ 594 consistent with F4V.

FALSE-ALARM ACCOUNTING UNDER THE SYMMETRIC CRITERION

Earlier drafts carried a four-model bookkeeping ladder (execution-block permutation, hypergeometric, block bootstrap and a Poisson rate heuristic, spanning $P(F{\geq}3) = 6.1$–9.2 per cent) to calibrate a false-alarm rate for an *asymmetric* statistic with no closed-form per-window null rate. The symmetric statistic of §5.4 removes that need, giving a per-window rate of 1/513 and a survey-level expectation of $431/513 = 0.84$ flags, so the ladder is retired, not updated.

The 1/513 rank floor of §5.4 can be lowered only by enlarging the control ring, which the primary beam bounds; and windows are not fully independent, grouped into 102 EBs sharing weather, calibration and correlator setup, so $N/513$ is an expectation over a mildly correlated sample, not N independent trials. Both limits are conservative for the conclusion drawn – that the flagged windows are consistent with chance – widening rather than narrowing the plausible null range.

THE SYMMETRIC STAR-VERSUS-CONTROL NULL IN FULL

The per-window p-value of the test of §5.4 is $p = (1 + \#\{T_i \geq T_\star\})/(N+1)$, and its seven measured windows are the six validation windows of the local reprocessing set plus the AU Mic crossing's reprocessed first-epoch window (§5.5). It is the designated replacement for the empirical star-versus-ring calibration as the remaining sets are re-run.

Table 17 consolidates the family-wise trials factors: every per-window factor applies identically at the 512 controls, making the control ring and the empirical star-beats-ring rate built on it the operative calibration, not any analytic correction.

ROBUSTNESS OF THE EMPIRICAL CALIBRATION

The release evaluates 3.97×10^7 channel×drift cells; a Gaussian one-sided 5σ tail (2.87×10^{-7}) over that count predicts 11.4 chance crossings, against the 18 windows with an on-star crossing. Raw counts run higher because the drift-tracked statistic is strongly correlated between adjacent channels ($\rho_{\rm lag1} = 0.989$–0.991, i.e. 8.5×10^4–1.1×10^5 effectively independent cells, not 3.97×10^7), so crossings arrive in contiguous runs. Collapsed to clusters (kernel of 3 native channels) they leave far fewer independent events – the count the trials expectation must be compared with; the two agree. The per-crossing census for the extended sample is in the released products.

No single stratum drives the star-beats-ring rate. The stratification was run on the $\leq$20 pc subset, whose counts it quotes. By band: 20 per cent (B3, $n = 40$), 10.9 per cent (B6, $n = 128$), 17.8 per cent (B7, $n = 45$), zero in the sparsely sampled B4/B5/B8 ($n = 4$ each). By channelisation: 17.0 per cent fine ($n = 47$) against 12.4 per cent coarse ($n = 178$). By primary-beam offset: 15.7 per cent above the median offset (0.10 arcsec) against 11.0 per cent below. The disc-hosting caveat inverts empirically: disc- or planet-hosting stars contribute every crossing window and every control-ring exceedance but a *lower* star-beats-ring rate (9.8 per cent) than non-hosts (18.3 per cent), so circumstellar emission shared by star and ring does not drive the statistic. Excluding the fifteen β Pictoris windows leaves one unattributed crossing among 402, against 0.78 expected ($P(\geq 1) = 54$ per cent).

Beyond the shared primary-beam gain of §5.4, neighbouring control positions are correlated below the synthesised-beam scale, so the independent spatial trials fall from 512 to $N_{\rm eff} \simeq 30$–43, as the symmetric reprocessing runs measure directly – a floor to which the empirical 15.1 per cent rate, measured on the correlated ensembles themselves, is immune by construction. A continuum source in the ring raises the narrowband statistic only if it also emits a narrow line in-window, and Appendix H covers recurring frequencies.

TABLE 25
Exceptions of the integration-time audit (§5.6). The 53 remaining star–EB groups all have pipeline on-source time at or below the archive science exposure, the comparison conservative for them; three more (Wolf 28, the narrow AT Mic window, G 272-61) agree to within 1%.

Star / block	Pipeline (s)	Archive (s)	Ratio
AT Mic (A002_Xcd97a1_X3f1e, wide windows)	1258	877	1.43
CD−38 10980 (A002_Xe5808e_X80a1)	617	575	1.07

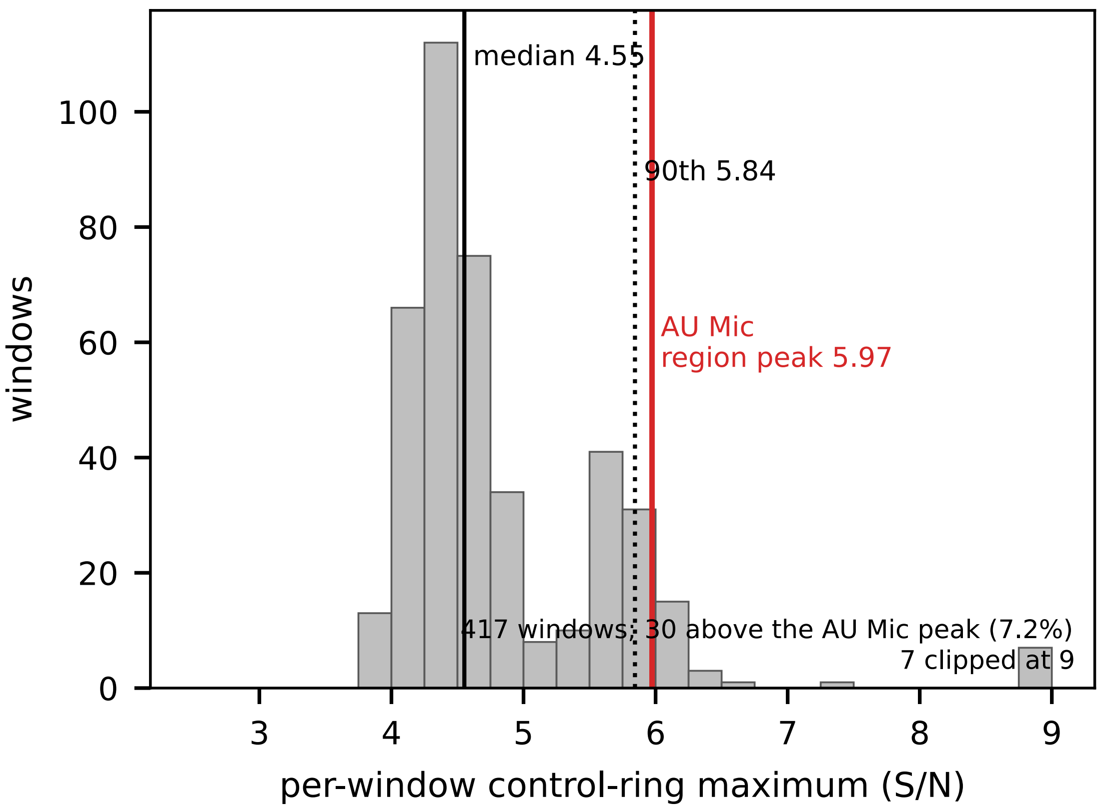

Fig. 15.— Diagnostic distribution for the marginal AU Mic crossing: control-ring peak S/N of all 431 windows (histogram) – the maxima expected from noise plus background structure alone. Dashed: the formal 5σ crossing threshold. Solid: the crossing window's on-source peak; dotted: its own control maximum, both at the values quoted in the text. Three windows lie beyond the plotted range, marked at upper right: HD 285968 Band 6 at 10.9, and β Pictoris Band 6 at 9.1 and 15.2, the latter the CO-filled control ring of the text.

INTEGRATION-TIME AUDIT: METHOD AND EXCEPTIONS

The archive cross-check of §5.6 uses the ALMA Science Archive's `ivoa.obscore` service (queried 2026 September 8), whose rows give the science-intent exposure `t_exptime` per observing unit set and pointed science source, summed over its executions. For each of the 57 searched EBs the pointed science source came from the archive's field list, aliases resolved by hand (Sirius B enters through the Sirius A pointings, the UV Ceti and G 272-61 pairs share one field each, AT Mic's block points at the pair), and its on-source time compared with the archive exposure for that source and set. `obscore` counts science-intent scans only, hiding target time in calibrator-intent scans, so the archive figure is a lower bound and any pipeline over-count conservative; per-execution durations are not exposed, so a multi-execution set is compared through its aggregate. It covers every window-level record of the subset, by star and EB: the 59 level-1 bookkeeping pairs, with AT Mic's narrow flagband window kept separate from its wide ones. It is conservative for both flagged hosts' windows (§5.4): β Pictoris's (1824–1845 s) sit inside an archive science exposure of 47 444 s and AU Mic's inside 109 334 s, with no disposition change, and the archive values lower the exposure-weighted coverage total of §5.4 by less than 1 per cent.

The AT Mic block shows the signature of a dump-counting defect: three windows report 208 integrations where 145 exist, and more on-source time than the block's own 1140 s elapsed span. The duty table uses the archive-consistent value for both AT Mic entries; the over-counted values survive only in per-window machine-readable fields, to be corrected next release. The CD−38 10980 excess is smaller than the intent-filter's conservative margin and its limit is unaffected, thresholds deriving from measured rms rather than integration time. A Wolf 359 discrepancy in the raw comparison is an audit-side artefact – an over-aggressive calibrator-source exclusion in the query – and disappears once the star's science-source exposure (9463 s, ample against the 2999 s searched) is included.

EXCLUSION AND DEFECT FORENSICS

THE FOUR CLASSES OF EXCLUDED ENTRY, ITEMISED

The process-level failures of §5.3 are item (i) below. The six windows failing the noise-defect test lie across five stars (51 Eri, HD 10647, HD 139664, TWA 7, HD 92945 twice). The supporting audit is clean across all 431 windows bar the four χ^1 Ori Band 3 windows (elevated system temperature; limits weakened, not invalidated), and all 168 census identifiers re-resolve through SIMBAD and *Gaia* to 168 distinct sources inside 40 pc.

(i) Narrowband process failures. Four target/bands (Barnard's Star [B7], Wolf 359 [B7], SCR J1845−6357, Wolf 358) have no narrowband result after process-level failures; each keeps a valid continuum non-detection and carries no EIRP threshold.

(ii) Giclas 9−38 pair. Giclas 9−38A and Giclas 9−38B [B3] are excluded entirely: the pair's only matched MOUS points ∼52 arcmin from either star, 35× its own primary-beam FWHM. The narrowband step correctly aborts; the continuum-imaging step lacks that guard, so the target/band waits on one (Appendix O).

(iii) Window-level noise defect. One spectral window in each of two Band 6 targets with similar correlator setups (HD 10647, HD 139664) reports 12.1 s of on-source integration against 393–520 s for its siblings in the same block, yet a combined noise estimate ∼1000× deeper than those siblings and than the survey's best continuum limit − the opposite of radiometer-noise scaling for a ∼40× shorter integration. The 12.1 s and the noise estimate come from the same machine-written per-window product carrying every other window's values, and are defective together. That window's narrowband result is excluded and the combined continuum measurement withheld for both target/bands, being built from all configured windows; their other narrowband windows are retained.

(iv) The "γ Lupi" resolution. The object observed under the catalogue entry "γ Lupi" is the F4V debris-disc host HD 139664 at 17.40 pc (Gaia DR3 5989102478619519616), not the naked-eye γ Lupi (§5.3, Appendix O). All such entries are relabelled; no measured value changes and item (iii)'s defect status is unaffected.

REFERENCES

Backus P. R., Project Phoenix Team, 2002, in Bulletin of the American Astronomical Society. p. 1173
Cataldi G., et al., 2023, ApJ, 951, 111
Choza C., et al., 2024, AJ, 168, 254
Cocconi G., Morrison P., 1959, Nature, 184, 844
Czech D., et al., 2021, PASP, 133, 064502
Czech D. J., et al., 2026, arXiv:2607.23651
Drake F. D., 1960, Physics Today, 13, 40
Drake F. D., 1974, in Sagan C., ed., Communication with Extraterrestrial Intelligence (CETI). MIT Press
Enriquez J. E., et al., 2017, ApJ, 849, 104
Fouqué P., et al., 2018, MNRAS, 475, 1960
Gaia Collaboration, Vallenari A., Brown A. G. A., et al., 2022, arXiv:2206.05595 (*Gaia* DR3 summary)
Gajjar V., et al., 2019, in Bulletin of the American Astronomical Society. AAS Meeting #233
Garrett M. A., et al., 2026, MNRAS, accepted (arXiv:2605.10212)
Gentile Fusillo N. P., et al., 2021, MNRAS, 508, 3877
Hallinan G., et al., 2007, ApJ, 663, L25
Huang B.-L., Tao Z.-Z., Zhang T.-J., 2026, ApJ, 1006, 9
Isaacson H., et al., 2017, PASP, 129, 054501
Jacobson-Bell K., et al., 2025, AJ, 169, 233
Johnson O. A., et al., 2023, AJ, 166, 193
Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, arXiv:2603.19023
Ma P. X., et al., 2023, Nature Astronomy, 7, 492
MacGregor M. A., Osten R. A., Hughes A. M., 2020, ApJ, 891, 80
Margot J.-L., et al., 2023, AJ, 166, 206
Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2024, MNRAS, 536, 2127
Matrà L., Dent W. R. F., Wyatt M. C., et al., 2017, ApJ, 842, 9
Morrison I. S., et al., 2023, PASA, 40, e038
Ng C., et al., 2022, AJ, 164, 205
Poznanski D., et al., 2025, AJ, in press (arXiv:2505.03927)
Price D. C., et al., 2020, AJ, 159, 86
Sheikh S. Z., 2020, International Journal of Astrobiology, 19, 237
Sheikh S. Z., et al., 2025, AJ, 169, 108
Sheikh S. Z., et al., 2025, arXiv:2512.18142 ("A Search for Radio Technosignatures from Interstellar Object 3I/ATLAS with the Allen Telescope Array")
Siemion A. P. V., et al., 2015, arXiv:1412.4867
Tarter J., 2001, ARA&A, 39, 511
Tremblay C. D., et al., 2024, PASA, 41, e004
Tremblay C. D., et al., 2026, in Advancing Astrophysics with the SKA II (AASKAII), arXiv:2606.27565
Tristan I. I., Osten R. A., Notsu Y., Kowalski A. F., Brown A., White G. L., Grady C. A., Henry T. J., Vrijmoet E. H., 2025, ApJ (arXiv:2503.14624)
Tusay N., Sheikh S. Z., Sneed E. L., Farah W., Pollak A. W., Cruz L. F., Siemion A., DeBoer D. R., Wright J. T., 2024, AJ, 168, 283
Tusay N., et al., 2025, arXiv:2506.13459
Valente G., et al., 2025, Acta Astronautica, in press
Vidal C., et al., 2026, arXiv:2605.21093 ("The Search for Technosignatures: a Review of Possibilities")
Pickett H. M., Poynter R. L., Cohen E. A., Delitsky M. L., Pearson J. C., Müller H. S. P., 1998, J. Quant. Spectrosc. Radiat. Transfer, 60, 883
White G. J., 2026, MNRAS, submitted
Wright J. T., Kanodia S., Lubar E. G., 2018, AJ, 156, 260
Zhang Z.-S., et al., 2020, ApJ, 891, 174